

DROPBOX

A Book on Computer Education with Artificial Intelligence



Name

Class Section



8

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(M.Tech)

New Edition

Printed By :

Bhumika Industry, Mawana (Delhi/NCR)

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Preface

In the context of the National Education Policy 2020, Computer Science education must undergo a paradigm shift where our children transition from being 'users' to 'developer's at a young age through mastery in coding. Newer technologies such as AI, Natural Language Processing, Data Science, Computer Vision and Neural Networks must be Introduced.

As per CBSE directives, the series stresses upon sustainable development goals while also ensuring that the right attitudes and values are inculcated through of cyber Ethios.

Special Features :

- Carefully selected and graded content
- Stepwise explanation of programs with clear screen shots.
- Inculcation of in depth problem solving and programming.
- Sufficient Lab Activities to ensure practical leaning.
- A section on computers and Health including tips to minimize health issues while working on computers.
- A section on Olympiad Bot containing practice questions for Natural Cyber olympiad.

—Author and Publisher

INCORPORATES NEP 2020

Created with new dimension of joyful learning along with completely mapped parameters of National Education Policy, 2020





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INSIDE THE BOOK

Outline includes the topics to be learnt in the chapter.



Outline

- Computer : An Electronic Machine
- Features of a Computer
- Types of a Computer

Info Zone include some facts related to chapter to enhance the knowledge of the students.



The first version of **MS Paint** was introduced with the first version of windows in **1985**.

Smart Tip provide keyboard shortcuts for menu commands, to help users save time while performing routine operations.



Smart Tip

You can also open **Paint** by typing **Paint** in the **Search box** next to the **Start** button. Then, click on the **Paint** button.

Computer Etiquette presents computer ethics and manners to be followed while working on a computer.



COMPUTER ETIQUETTES

While working on a computer, keep your neck relaxed. Practice this exercise to avoid any strain on your neck.

Let's Implement contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.



Let's Implement

A Tick (✓) the correct option.

1. A monitor is also called:
 - (a) LCD ☐
 - (b) VDU ☐
2. Which part of a computer is called the brain of a computer?
 - (a) CPU ☐
 - (b) VDU ☐
3. It is used to give the output of our work on paper.
 - (a) Speakers ☐
 - (b) Printer ☐

Refreshment Corner includes interesting exercises for enhancing the skills with fun.



Refreshment Corner

A Colour the alphabet keys that spell your name with yellow.



Practical Session for practical hands-on exercises to strong then the concepts learned by doing.



Practical Session

Integrated Learning

- Visit your computer lab and open **Paint**. Draw and colour the following.



Racking Your Brain includes the life skills and values questions.

RACKING YOUR BRAIN

Life Skills and Values

Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

Just Have a Discussion stimulates a small discussion for the topics being covered in the chapter which in turn improves the communication and logical skills.

Just Have a Discussion

Communication

Discuss the different components of Scratch in class in detail.

Teacher's Note provides important information and suggestions on creative approaches to a chapter or a topic.



TEACHER'S NOTE

- Give enough exercises to students so that they get practice of using the tools of **Paint** as well as using the mouse.



COMPUTER NETWORKING



Outlines

- What is a Network
- Computer Network Types
- Networking Hardware
- Wireless Transmission
- Advantages of Network
- Computer Network Topologies
- Transmission Media

What is a Network

A network consists of two or more computers that are linked in order to share resources (such as printers), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites or infrared light beams.

Advantages of Network

Speed : Networks provide a very fast method for sharing and transferring files. Without a network, files are shared by copying them to pen drive, then carrying or sending from one computer to another, which is very time consuming.

Cost : The network version of most software programs are available at considerable prices when compared to buying individually licensed copies.

Centralised Software Management : One of the greatest benefits of installing a network at a school, is the fact that all the softwares can be loaded on one computer (server). This removes the need to spend time and energy installing updates and tracking files on independent computers.

Resource Sharing : Sharing resources is another area in which a network exceeds stand-alone computers. Most users cannot afford enough laser printers, fax machines, modems, scanners and CD-ROM players for each computer. However, if these or similar peripherals are added to a network, they can be shared by many users.

Flexible Access : School networks allow the students to access their files from computers outside the school also. Students can begin an assignment in their classroom, save part of it on a public access area of the network, then go to the media center after the school to finish their work.

Computer Network Types

Generally, networks are distinguished based on their geographical span. A network can be as small as distance between your mobile phone and its Bluetooth headphone and as large as the internet itself, covering the whole geographical world.

Personal Area Network

A Personal Area Network (PAN) is the smallest network which is very personal to a user. This may include Bluetooth-enabled devices or infra-red enabled devices. PAN has connectivity range up to 10 meters. PAN may include wireless computer keyboard and mouse, Bluetooth-enabled headphones, wireless printers and TV remotes.



Piconet is Bluetooth-enabled Personal Area Network, which may contain up to 8 devices connected together in a master-slave fashion.



Local Area Network

A computer network spanned inside a building and operated under single administrative system is generally termed as Local Area Network (LAN). Usually, LAN covers an organisation, offices, schools, colleges or universities. Number of systems connected in LAN may vary from at least as two to as much as 16 million. LAN provides a useful way of sharing the

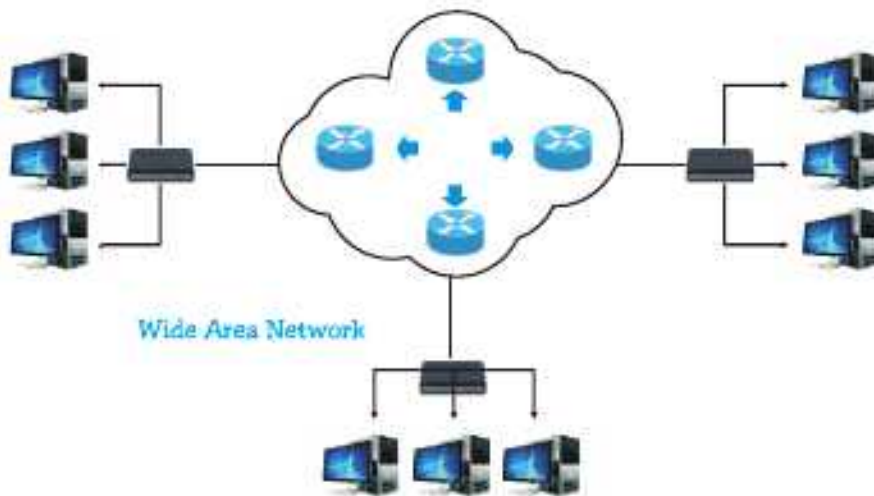
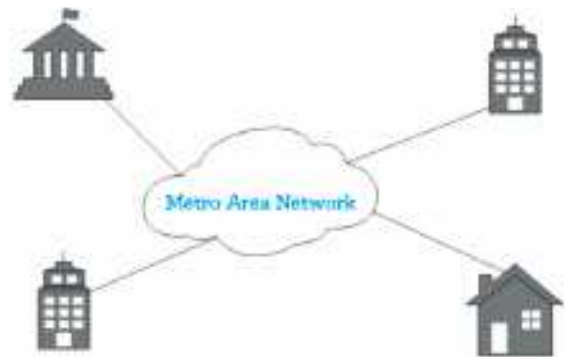
resources between end users. The resources such as printers, file servers, scanners and internet are easily sharable among computers. LAN can be wired, wireless or in both forms at once.



Ethernet is most widely employed LAN technology and uses Star topology.

Metropolitan Area Network

The Metropolitan Area Network (MAN) generally expands throughout a city such as cable TV network. It can be in the form of Ethernet, Token-ring, ATM or Fiber Distributed Data Interface (FDDI). MAN works in between Local Area Network and Wide Area Network. MAN provides uplink for LANs to WANs or internet.



Wide Area Network

Wide Area Network (WAN) covers a wide area which may span across provinces and even a whole country. Generally, telecommunication networks are Wide Area Networks. These networks provide connectivity to MANs and LANs. Since they are equipped with very high speed backbone, WANs use very expensive network equipment.

WAN may use advanced technologies such as Asynchronous Transfer Mode (ATM), Frame Relay and Synchronous Optical Network (SONET).

Internetwork

A network of networks is called an Internetwork or simply the internet. It is the largest network in existence on this planet. The internet hugely connects all WANs and it can have connection to LANs and Home networks.

Internet enables its users to share and access large amount of information worldwide. It uses WWW, FTP, email services, audio and video streaming, etc. At huge level, internet works on Client-Server model.

Internet is serving many purposes and is involved in many aspects of life. Some of them are:

- Websites
- Blogging
- Networking
- Audio and Video Streaming
- E-mail
- Social Media
- Resource Sharing
- Instant Messaging
- Marketing

Computer Network Topologies

A Network Topology is the arrangement with which computer systems or network devices are connected to each other. Topologies may define both physical and logical aspects which could be same or different in a same network.

Point-to-Point

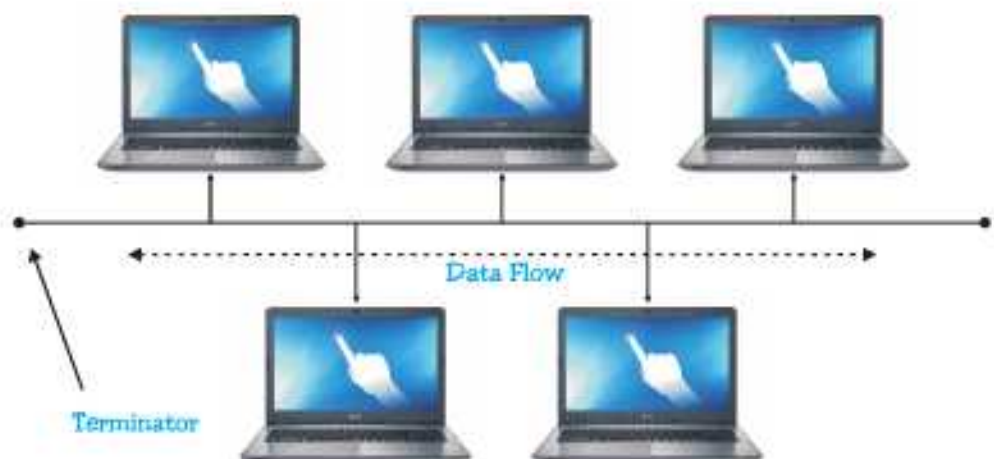
Point-to-point network contains exactly two hosts such as computer, switches or routers, servers connected back to back using a single piece of cable. Often, the receiving end of one host is connected to the sending end of the other and vice-versa.



If the hosts are connected point-to-point logically, they may have multiple intermediate devices. But the end hosts are unaware of underlying network and see each other as if they are connected directly.

Bus Topology

In Bus topology, all devices share single communication line or cable. Bus topology may have problem while multiple hosts sending data at the same time. It is one of the simple forms of networking where the failure of a device does not affect the other devices. But failure of the shared communication line can make all other devices stop functioning.

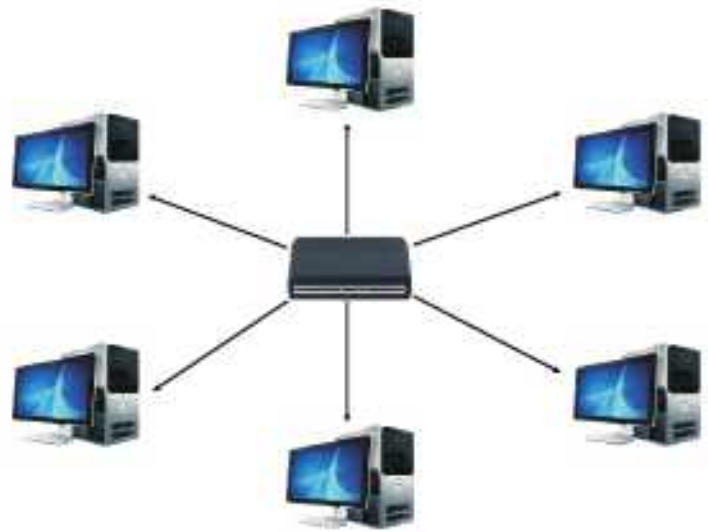


The data is sent in only one direction and as soon as it reaches the extreme end, the terminator removes the data from the line.

Star Topology

All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.

Every communication between hosts, takes place through only the hub. Star topology is not expensive as to connect one more host, only one cable is required and configuration is simple.



Ring Topology

In ring topology, each host machine connects to exactly two other machines, creating a circular network structure. When one host tries to communicate or send message to a host which is not adjacent to it, the data travels through all intermediate hosts.

Failure of any host results in failure of the whole ring. There are methods which employ one more backup ring.



Mesh Topology

In this type of topology, a host is connected to one or multiple hosts. This topology has hosts in point-to-point connection with every other host or to few hosts only.



Tree Topology

Also known as Hierarchical Topology, this is the most common form of network topology in use presently. This topology imitates as extended Star topology and inherits properties of bus topology.

Daisy Chain

This topology connects all the hosts in a linear fashion. Similar to Ring topology, all hosts are connected to two hosts only, except the end hosts. Means, if the end hosts in daisy chain are connected, then it represents Ring topology.

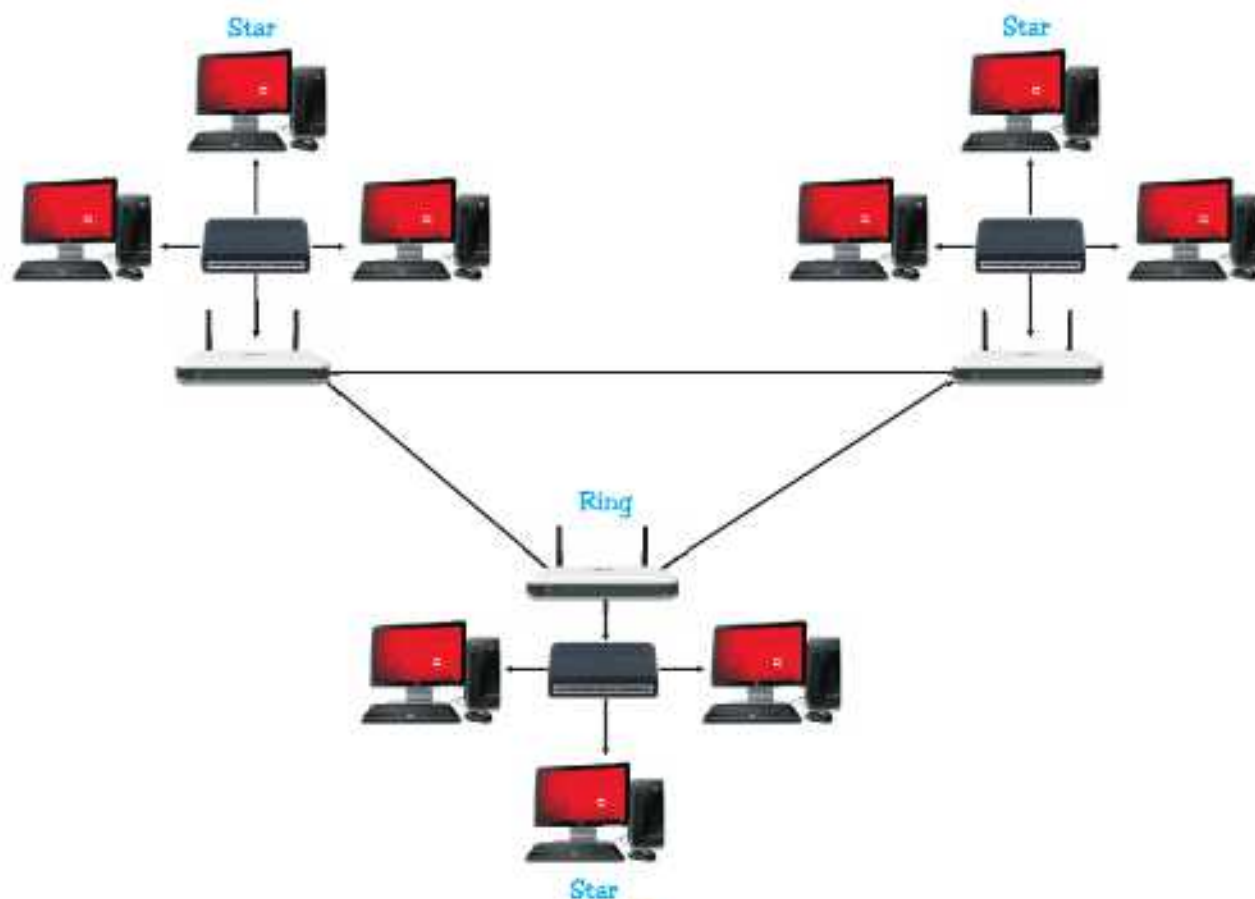
Each link in daisy chain topology represents single point of failure. Every link failure splits the network into two segments. Every intermediate host works as relay for its immediate hosts.



Hybrid Topology

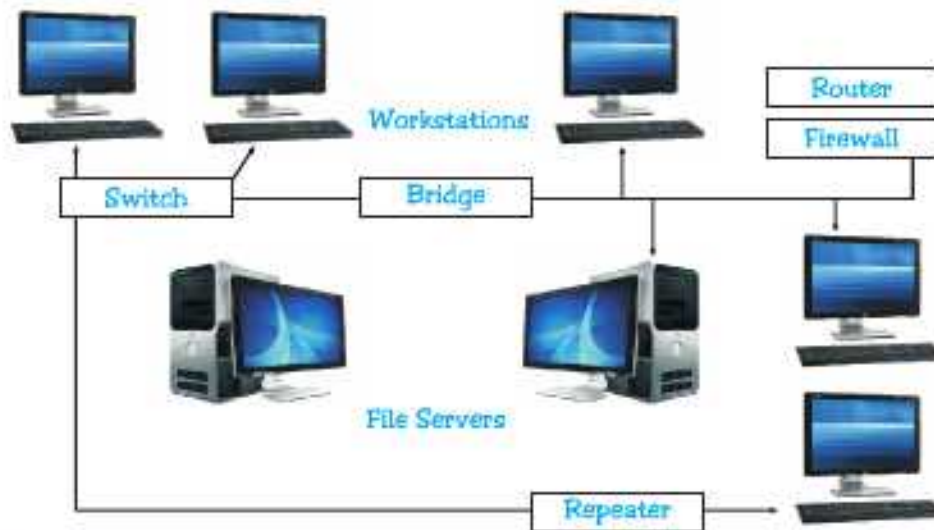
A network structure whose design contains more than one topology, is said to be hybrid topology. Hybrid topology inherits merits and demerits of all the incorporating topologies.

The below picture represents an arbitrarily hybrid topology. The combining topologies may contain attributes of Star, Ring, Bus, and Daisy-chain topologies. Most WANs are connected by means of Dual-Ring topology and networks connected to them are mostly Star topology networks. Internet is the best example of largest Hybrid topology.



Networking Hardware

Networking hardware includes all computers, peripherals, interface cards and other equipment needed to perform data-processing and communications within the network. CLICK on the terms below to learn more about those pieces of networking hardware.



File/Network Servers

The network operating system provides tools to share server resources and information with network users. For small networks, a single network server may provide access control, file and printer sharing, email, database and other services. The network server may be responding to requests from many network users simultaneously. For example, it may be asked to load a word processor program to one workstation, receive a database file from another workstation, and store an e-mail message during the same time period.

Workstations

Computers used by humans are broadly categorized as workstations. A typical workstation is a computer that is configured with a network interface card, networking software and appropriate cables. Workstations do not necessarily need large storage hard drives, because files can be saved on the file server. Almost any computer can serve as a network workstation.

Laptops/Mobile Devices

Laptops and other mobile devices are becoming more common. These devices typically have modest internal storage, but enough power to serve as a workstation for users. These machines nearly always have a wireless adapter to allow quick network connections without cumbersome cabling. In a school environment with good wireless coverage, a mobile device user can move in the campus freely, and remain continuously connected to the network.

Network Interface Cards

The Network Interface Card (NIC) provides the physical connection between the network and the computer workstation. Most NICs are internal, and they are included in the purchase of most computers. Network interface cards are a major factor in determining the speed and performance of a network.

The most common network interface connections are Ethernet cards and wireless adapters.

Ethernet Cards

Ethernet cards are usually included with a computer, although additional ethernet cards can be purchased and installed on most computers. Ethernet cards can contain connections to attach coaxial, twisted pair or fiber optics cable. When this method is used, there is always an external transceiver attached to the workstation.



Wireless Adapters

Wireless adapters are found in most portable devices, such as laptops, smart phones and tablet devices. External wireless adapters can be purchased and installed on most computers having an open USB (Universal Serial Bus) port or unused expansion slot.

Switches

An ethernet switch is a device that provides a central connection point for cables from workstations, servers and peripherals. Most switches are active, that is they electrically amplify the signal as it moves from one device to another. The predecessor of the switch is the hub, which broadcasted all inbound packets on all ports of the device, creating unnecessary network traffic.

Repeaters

The repeater electrically amplifies the signal it receives and rebroadcasts it. Repeaters can be separate devices or they can be incorporated into a concentrator. They are used when the total length of your network cable exceeds the standards set for the type of cable being used.

Bridges

A bridge is a device that allows you to segment a large network into two smaller, more efficient networks. A bridge monitors the information traffic on both sides of the network, so that it can pass packets of information to the correct location. The bridge can inspect each message and, if necessary, broadcast it on the other side of the network.

Routers

Routers are the traffic directors of the global internet. All routers maintain complex routing tables, which allow them to determine

appropriate paths for packets destined for any address. Routers communicate with each other, and forward network packets out of or into a network.



Bridges can be used to connect different types of cabling or physical topologies.

Firewalls

A firewall is a networking device that is installed at the entrance to a LAN when connecting the networks together, particularly when connecting a private network to a public network, such as the internet. The firewall uses rules to filter traffic into and out of the private network, to protect the private network users and data from malicious hackers.

Transmission Media

Twisted Pair Cable

A Twisted pair cable is made of two plastic insulated copper wires twisted together to form a single media. Out of these two wires, only one carries actual signal and another is used for ground reference. The twists between wires are helpful in reducing noise (electro-magnetic interference) and crosstalk.



There are two types of twisted pair cables:

- Shielded Twisted Pair (STP) Cable
- Unshielded Twisted Pair (UTP) Cable

STP cable comes with twisted wire pair covered in metal foil. This makes it more indifferent to noise and crosstalk.

UTP has seven categories, each suitable for a specific use. In computer networks, Cat-5, Cat-5e and Cat-6 cables are mostly used. UTP cables are connected by RJ45 connectors.

Coaxial Cable

Coaxial cable has two wires of copper. The core wire lies in the center and it is made of solid conductor. The core is enclosed in an insulating sheath. The second wire is wrapped around over



the sheath and that too in turn encased by insulator sheath. This all is covered by plastic cover.

The coaxial cable is capable of carrying high frequency signals than that of twisted pair cable.

There are three categories of coaxial cables namely, RG-59 (Cable TV), RG-58 (Thin Ethernet) and RG-11 (Thick Ethernet). RG stands for Radio Government.

Fiber Optics

Fiber Optic works on the properties of light. When light ray hits at critical angle, it tends to refract at 90 degree. The core of fiber optic cable is made of high quality glass or plastic. From one end of it, light is emitted, it travels through it and at the other end, light detector detects light stream and converts it to electric data.

Fiber Optic provides the highest mode of speed.



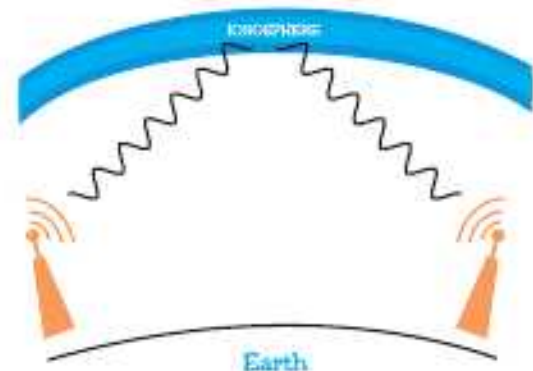
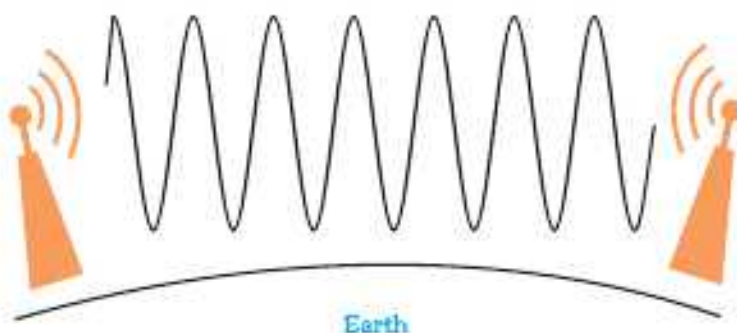
Wireless Transmission

Wireless transmission is a form of unguided media. Wireless communication involves no physical link established between two or more devices, communicating wirelessly. Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.

When an antenna is attached to electrical circuit of a computer or wireless device, it converts the digital data into wireless signals and is spread all over within its frequency range. The receptor on the other end receives these signals and converts them back to digital data.

Radio Transmission

Radio frequency is easier to generate and because of its large wavelength it can go through walls. Radio frequencies are sub-divided into six bands.



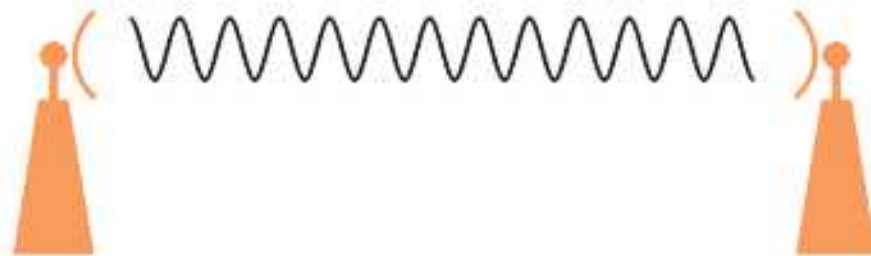
Radio waves at lower frequencies can travel through walls, whereas higher RF can travel in straight line and bounce back. The power of low frequency waves decreases sharply as they cover long distance. High frequency radio waves have more power.



Radio waves of high frequencies are flat to be absorbed by rain and other obstacles.

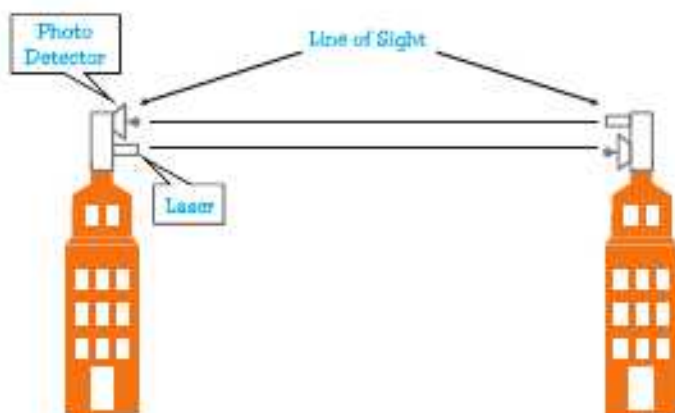
Microwave Transmission

Microwaves travel in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight. Microwave antennas concentrate the waves making a beam of them. As shown in the picture below, multiple antennas can be aligned to reach farther. Microwaves have higher frequencies and do not go through wall like obstacles.



Infrared Transmission

Infrared wave lies in between visible light spectrum and microwaves. It is used for very short range communication purposes, such as television and its remote. It travels in a straight line hence it is directional by nature. Because of high frequency range, it cannot cross wall-like obstacles.



Light Transmission

Laser tends to travel strictly in a straight line. Hence the sender and the receiver must be in the line-of-sight. Lasers cannot go through obstacles, such as walls, rain and thick fog. Additionally, laser beam is distorted by wind, atmosphere temperature or variation in temperature, in the path.



COMPUTER ETIQUETTES

Do not press the keys of the keyboard hardly.



Let's Implement.

A

Tick (✓) the correct answer.

1. It includes wireless computer keyboard and mouse, Bluetooth-enabled headphones, wireless printers and TV remotes.
a. Personal Area Network ☐ b. Local Area Network ☐
c. Wide Area Network ☐ d. Metropolitan Area Network ☐
2. It uses advanced technologies, such as Asynchronous Transfer Mode (ATM), Frame Relay and Synchronous Optical Network (SONET).
a. Personal Area Network ☐ b. Local Area Network ☐
c. Wide Area Network ☐ d. Metropolitan Area Network ☐
3. In this topology, the receiving end of one host is connected to sending end of the other and vice-versa.
a. Point-to-Point ☐ b. Bus ☐
c. Star ☐ d. Ring ☐
4. In this topology, when one host tries to communicate to a host which is not adjacent to it, the data travels through all intermediate hosts.
a. Point-to-Point ☐ b. Bus ☐
c. Star ☐ d. Ring ☐
5. This may be responding to requests from many network users simultaneously.
a. File Server ☐ b. Data Server ☐
c. Bus Server ☐ d. Network Server ☐
6. This cable is made of two plastic insulated copper wires twisted together to form a single media.
a. Twisted Pair Cable ☐ b. Coaxial Cable ☐
c. Fibre Cable ☐ d. Shielded Cable ☐
7. This cable is capable of carrying high frequency signals than that of twisted pair cable.
a. Twisted Pair Cable ☐ b. Coaxial Cable ☐
c. Fibre Cable ☐ d. Shielded Cable ☐
8. This transmission travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.
a. Microwave Transmission ☐ b. Wireless Transmission ☐
c. Satellite Transmission ☐ d. Optic Transmission ☐

B**Fill in the blanks.**

1. A network spanned inside a building and operated under single administrative system, is termed as _____.
2. In _____, all devices share single communication line or cable.
3. In _____, a host is connected to one or multiple hosts.
4. The most common network interface connections are _____ and _____.
5. _____ can contain connections to attach coaxial, twisted pair or fiber optics cable
6. An _____ is a device that provides a central connection point for cables from workstations, servers and peripherals.
7. The _____ electrically amplifies the signal it receives and rebroadcasts it.
8. The _____ uses rules to filter traffic into and out of the private network, to protect the private network users and data from malicious hackers.
9. There are two types of twisted pair cables _____ and _____.
10. _____ is easier to generate and because of its large wavelength it can go through walls.

C**Write True or False.**

1. The LAN generally expands throughout a city such as cable TV network.
2. Topologies may define both physical and logical aspects of the network.
3. All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.
4. Tree Topology imitates as extended Star topology and inherits properties of bus topology.
5. A network structure whose design contains more than one topology, is said to be hybrid topology.
6. A router is a device that allows you to segment a large network into two smaller, more efficient networks.
7. Switch communicate with each other, and forward network packets out of or into a network.
8. Infrared wave lies in between visible light spectrum and microwaves.
9. Lasers cannot go through obstacles such as walls, rain and thick fog.

D**Answer the following questions.**

1. What is a network?

2. Explain the advantages of network.
3. Define LAN, MAN and WAN.
4. Write the use of Bus, Star and Ring Topology.
5. What are the hardware components required for networking?
6. Differentiate between Switches and Repeaters.
7. Differentiate between Twisted Pair Cable and Coaxial Cable.
8. Define the following:
 - Routers
 - Firewalls
 - Microwave Transmission
 - Infrared Transmission

Refreshment Corner

- A** With the help of Internet, collect the images and information about networking and make a presentation on 'NETWORKING' in MS-Powerpoint.
- B** Identify the following computer network types and write their names.

1.



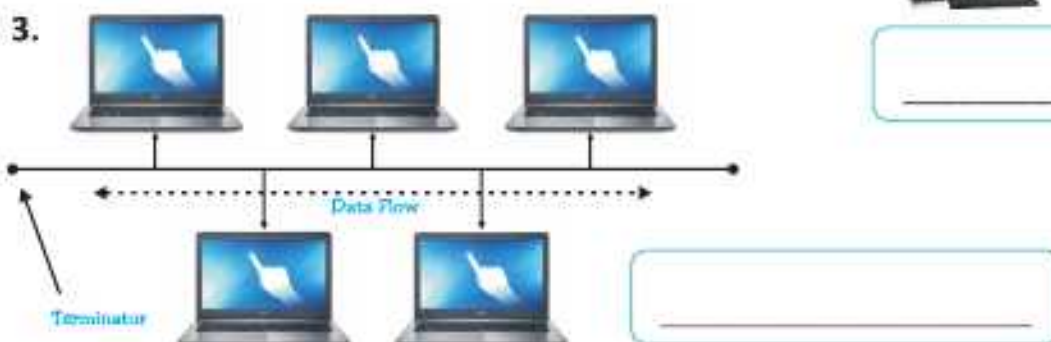
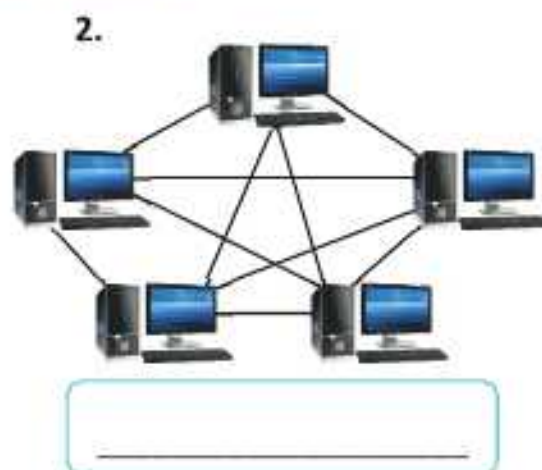
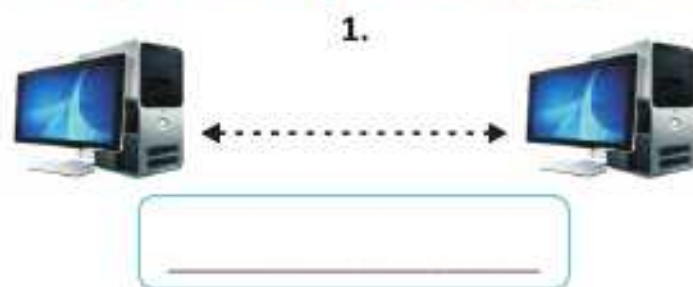
2.



3.

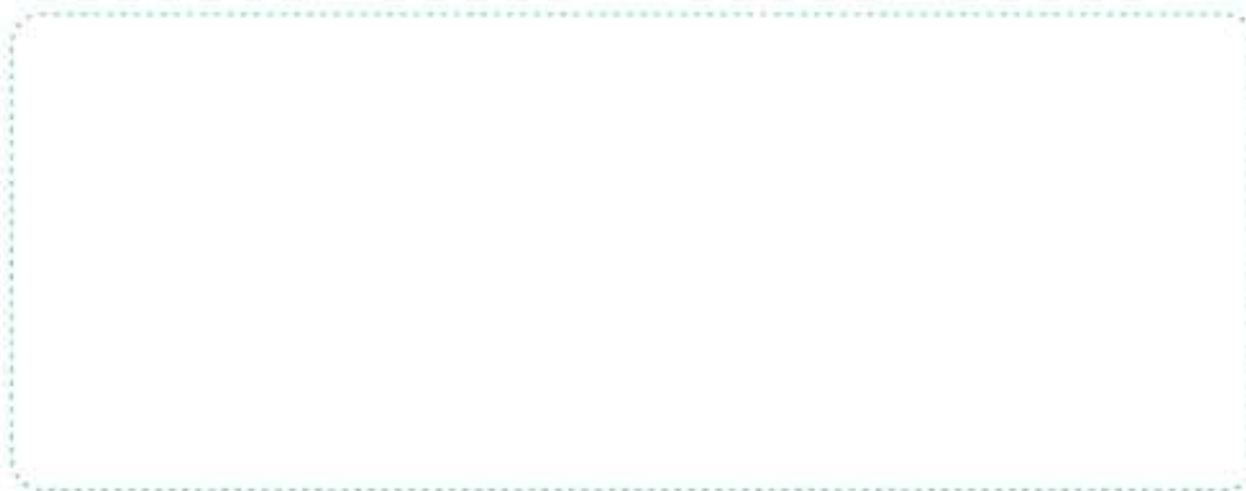


C Identify the following network topologies and write their names.



D Draw and colour the picture of fiber optics in the space given below.

Art Integration



RACKING YOUR BRAIN

Life Skills and Values

How can we use personal Area Network at our home? Justify your answer.

Just Have a Discussion

Communication

Discuss the advantages of network in the class in detail.



TEACHER'S NOTE

The students should be shown some network components used to connect the computer in a network in the lab.



WORKING WITH TABLES IN MS ACCESS 2016



Outlines

- What is a Table?
- Views of a Table
- Entering Data in a Table
- Editing Records in a Table
- Sorting Records in a Table
- Modifying the Table Design
- Components of a Table
- Creating a Table
- Adding Records in a Table
- Deleting Records in a Table
- Searching Records in a Table
- Deleting a Field

What is a Table?

A table is a database object, which is used to store the information in an organised manner. It consists of rows and columns. A database can have one or more tables.

Fields or Attributes

Roll No	Name	Address
1.	Rajeev	Sagarpur
2.	Nina	Model Town
3.	Payal	Lajpat Nagar
4.	Aarav	Sadar

Components of a Table

The important components of a table are:

1. Field or Attribute

The columns in a table are known as fields or attributes. A field is a part of record, which

consists of a single piece of data. A field stores one type of information about all the objects or items. Every field has a data type that determines the type of values that can be stored under it.

Data types are used to declare the fields of a table. A data type determines the type of data in it. Once the data type is defined, we cannot enter a different type of data in it. If we try to do so, an error message will be displayed.

Field Data Types : The various data types available in Access are described in the table given below:

Data Types	Description
Short Text	It is used to store text or a combination of text and numbers that does not require calculations, such as addresses, phone numbers, etc. The field with this data type can have a maximum of 255 characters.
Long Text	It is used to store lengthy text, i.e., up to 65,536 characters. It can be used for storing detailed information such as the synopsis of a book or a patient's medical history.
Number	It is used for numeric data that can be used in calculations.
Date/Time	It stores date and time values. You can display the date and time in various formats.
Currency	It stores currency values and then displays them in different formats.
AutoNumber	It stores an integer that increments automatically as you add records.
Yes/No	This can have only one of two possible values, such as Yes/No, True/False or On/Off.
OLE	This stores graphics or objects such as Microsoft Excel worksheet or Microsoft Word document.
Hyperlink	The fields with this data type can store links to websites or email addresses.
Attachmen	This data type is used to attach images, spreadsheet files, documents, charts and other types of supported files to the records in your database, much similar to the manner you attach files to the emails.

Calculated	Fields with this data type store results of a calculation. The calculation must refer to other fields in the same table.
Lookup Wizard...	This creates a lookup field that displays a set of values. It helps you to create a field whose values are chosen from the value in another table, query or list of values.

After setting the data type for a field, you can set its properties.

Let us learn more about field properties.

Field Properties : Every field in a table has its properties to define its characteristics and behaviour. The field properties let you have more control on data that can be entered in a particular field. Some of the field properties are as follows:

2. Record

The rows in a table are known as records. A record stores complete information about an object or an item. Each record contains the complete information about a specific entity.

Field Property	Description
Field Size	This property can be set for fields with text or number data type. It sets the maximum size for data stored in a field to accommodate text (0-255), numbers or AutoNumber data type.
Format	It controls the manner in which data gets displayed.
Input Mask	It makes data entry easier and controls the values which a user can enter into it.
Caption	This property is used to give an alternate, more descriptive name to a field.
Default Value	This property is used to specify a value that gets displayed automatically when a new record is created.
Validation Rule	It is used to put conditions on the data that can be entered in a field.
Validation Text	It is used to display an error message when the validation rule gets violated.
Required	It forces you to enter a value if required property is set for Yes.
Decimal Places	It specifies the number of decimal places for numeric data.

Allow Zero Length	It accepts a blank entry if required property is set for Yes.
--------------------------	---

3. Primary Key

A primary key is a field or a combination of fields that uniquely identifies the records in a table. A primary key field cannot have repetitive values and cannot be left blank.

Views of a Table

You can work on a table in two views – [Design View](#) and [Datasheet View](#).

DESIGN VIEW

In Design View of the table, you can enter the field names, their data types and description.

The Design View window is divided into two parts:

- **Field Grid Pane:** It is used for entering field names and their data types. You can also give an optional description about each field in this pane.
- **Field Properties Pane:** This section is used to set properties for the fields in the table.

DATASHEET VIEW

Datasheet View is used to enter data in a table. When you double-click the table name in the database window, it opens in the Datasheet View.

Switching Between Views

Microsoft Access allows you to switch from one view to another by following either of the two methods:

Method 1: Click on the [View](#) option in the [Views](#) group on the [Home](#) tab to change the desired view.

Method 2: Click on the [Design View](#) or [Datasheet View](#) buttons at the right corner of the status bar.



Creating a Table

There are various ways of creating a table in MS Access. They are:

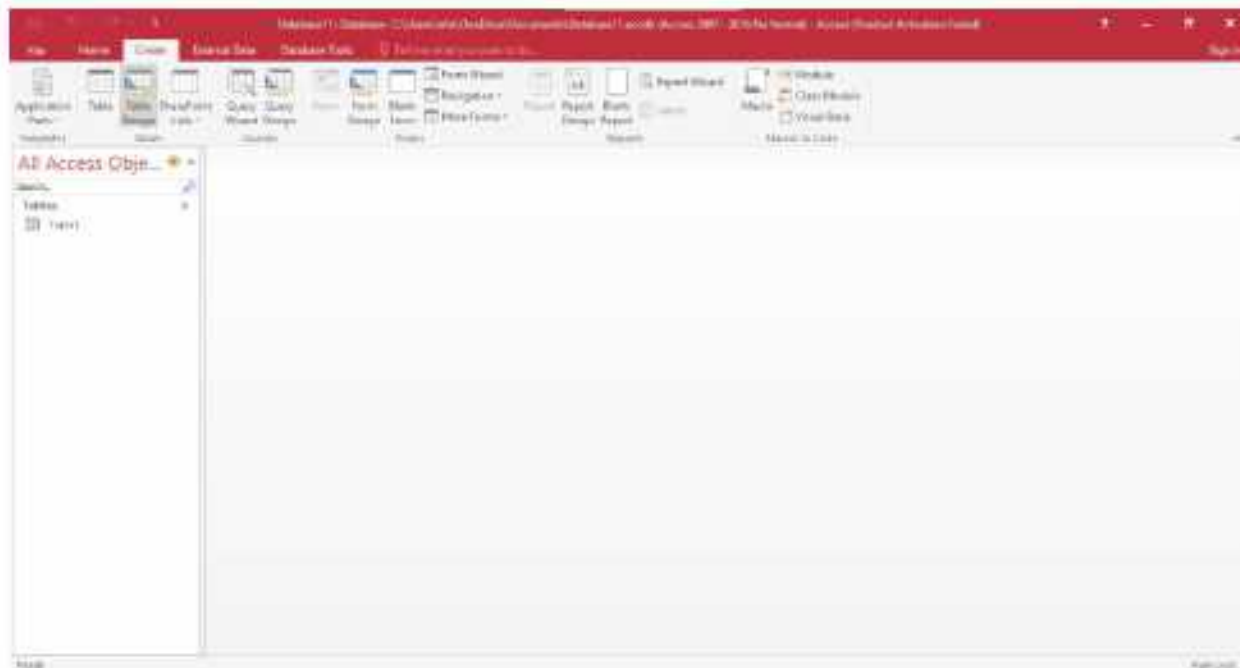
- Creating tables in Design View
- Creating tables by using Templates

- Creating tables by entering data in Datasheet View

In this chapter, you will learn how to create a table in the Design View. This lets you design the structure of a table by specifying the field names, data types and their properties.

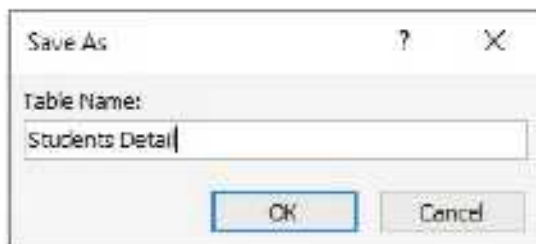
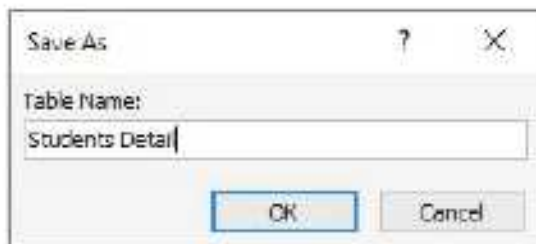
To create a table in Design View, follow these steps:

1. Click on the **Create** tab.
2. Select the **Table Design** option in **Tables** group.



3. Enter the name for the field in the **Field Name** column.
4. Select a data type for a field from the **Data Type** list.
5. Use the **Field Properties** pane to set the properties for the fields.
6. Place the cursor on the field to be set as the primary key.
7. Click on the **Primary Key** option in the **Tools** group on the **Design** tab under **Table Tools**. You will notice that a key symbol appears on the field.
8. Click on the **Save** button on the **Quick Access toolbar** or click on the **File** tab and select the **Save** option. The **Save As** dialog box appears.
9. Type the name of the table in the **Table Name:** text box.
10. Click on the **OK** button.

The table is now added to the **Navigation Pane**.





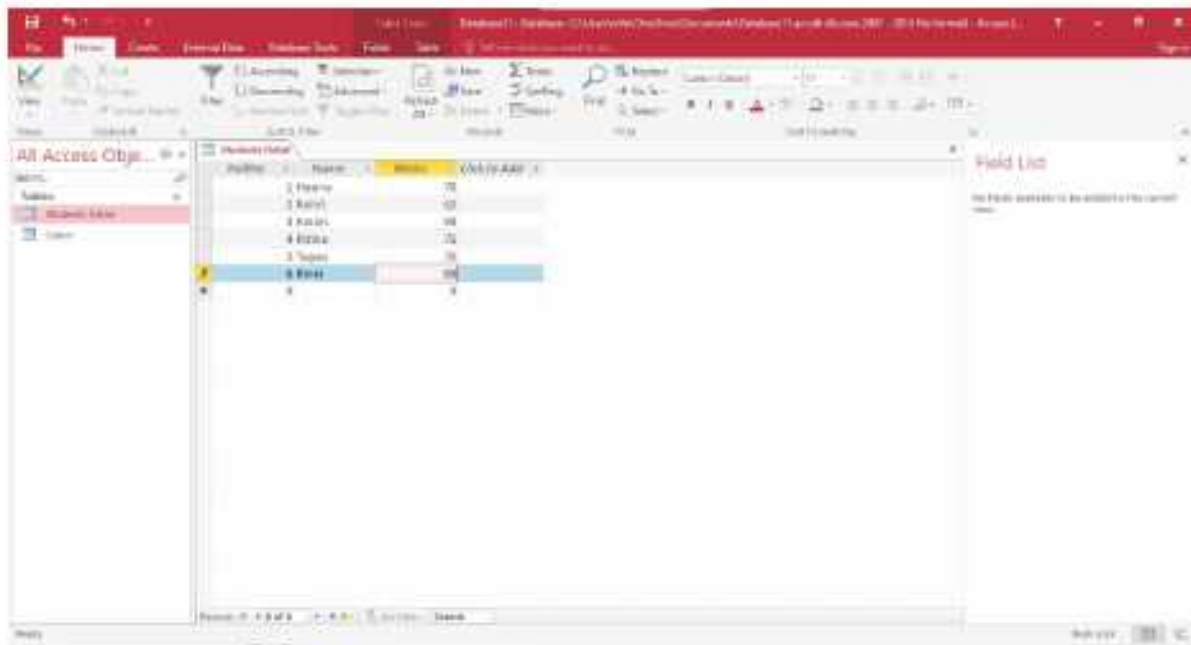
Entering Data in a Table

To enter data in a table, follow these steps:

-
- The screenshot shows the Microsoft Access application window. The title bar indicates the database is 'Database1 - Students - Access'. The ribbon is set to 'Table Design'. The main window displays the 'Students' table in Datasheet view. The 'All Access Objects' task pane on the left shows the 'Students' table selected. The 'Field List' task pane on the right shows the fields of the 'Students' table: ID, Name, Address, and Date of Birth.

2. Apart from fields, you will also find a blank row with an asterisk (*) symbol. Type the data in the appropriate fields and press either the Enter or the Tab key to move to the next field.

When the data is entered in a row, a new blank row will appear automatically. You can insert any number of records in a table.



Press the **Shift+Tab** keys to move to the previous field. If you press the **Tab** key on the last field, the cursor automatically jumps to the next row.

You can view the record number at the bottom of the screen in the Record Navigation Bar.

Adding Records in a Table

To Add a New Record

You can add a new record to a table in three ways:

Method 1: Select the **New** button in the **Records** group on the **Home** tab.



Method 2: Click on the **New (blank)** record button on the **Record Navigation Bar**.



Method 3: Simply start typing in the row below the last added record.

To Save a Record

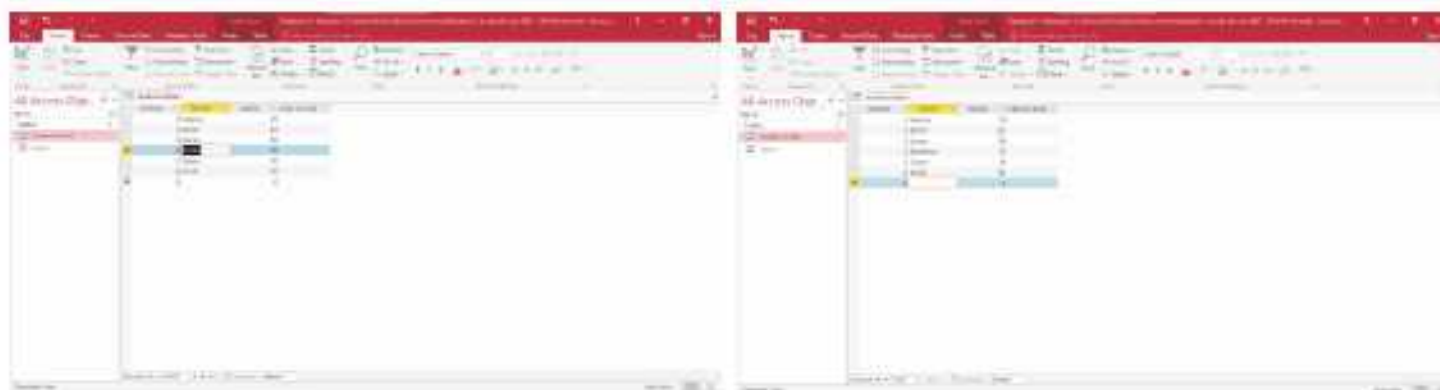
Click on the **Save** button in the **Records** group on the **Home** tab.



Editing Records in a Table

After entering data in a table, sometimes it is required to make changes in the database. To edit records in a table, follow these steps:

1. Use the **Record Navigation** buttons to find the record to edit or double-click the cell where you want to make changes.
2. Type the information into the field. A **pencil icon** appears to indicate the **Edit mode**. Click outside the record to apply the changes.

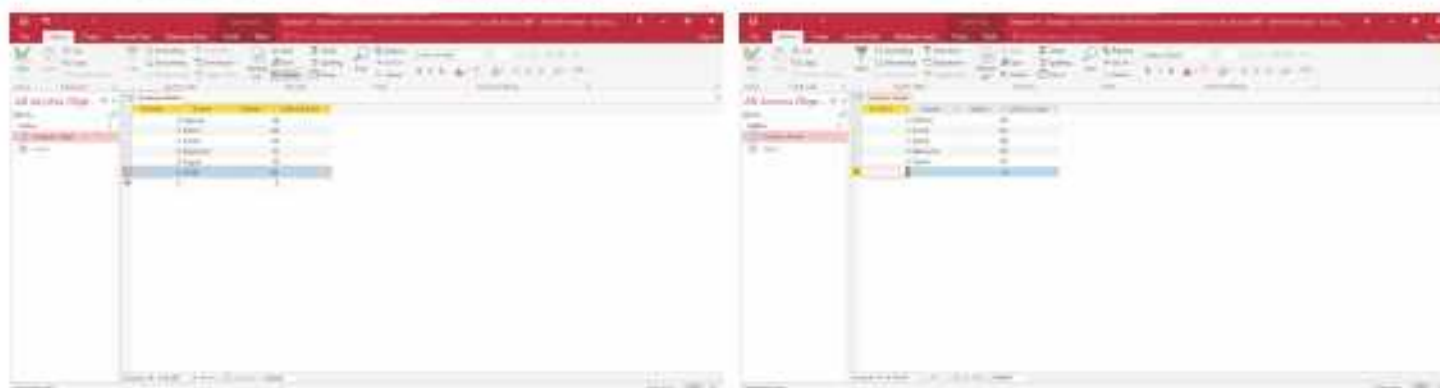


Deleting Records in a Table

To delete a record, open the table in the Datasheet View and follow these steps:

1. Click on the record that needs to be deleted.
2. Click on the **Delete** option in the **Records** group on the **Home** tab.

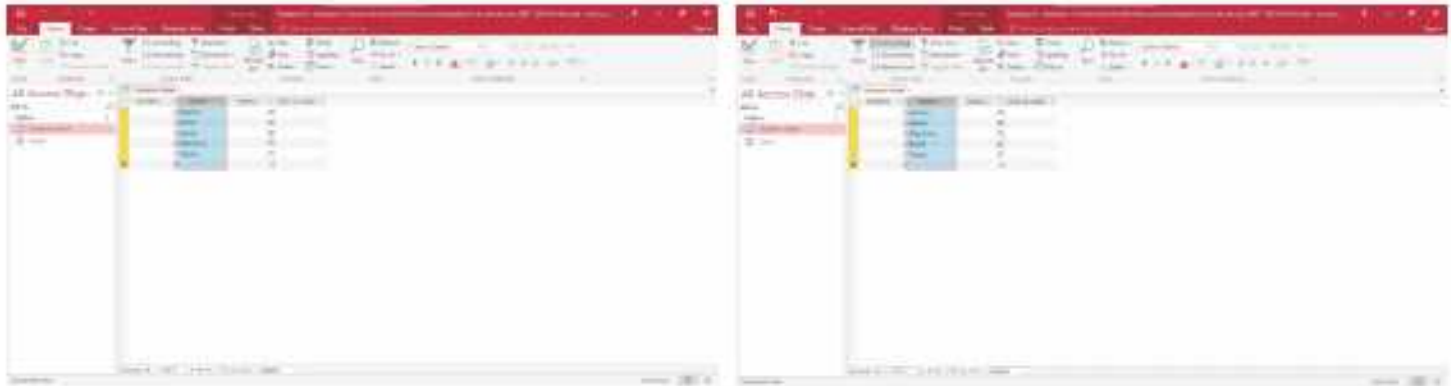
The selected row will be deleted.



Sorting Records in a Table

To sort records, open the table in Datasheet View and follow these steps:

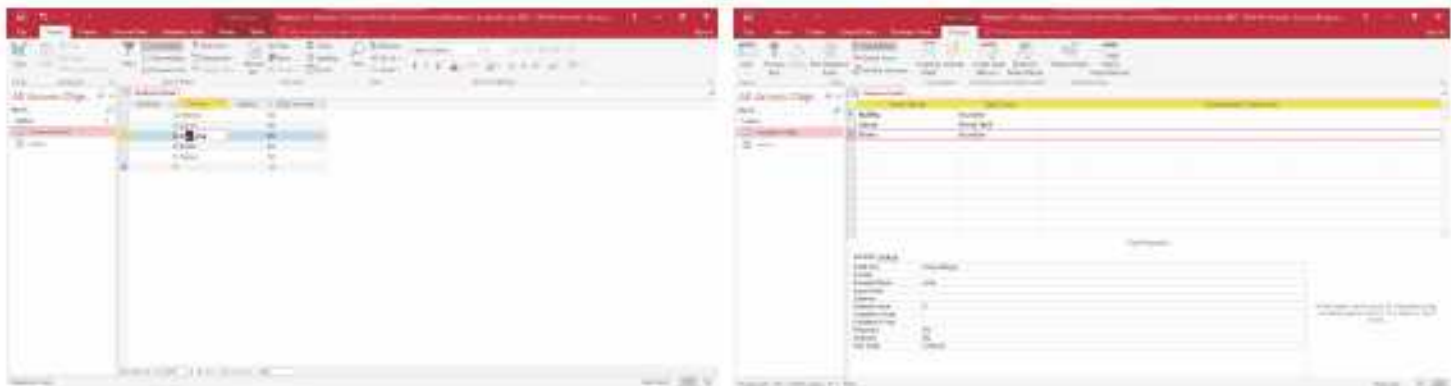
1. Select the field on the basis of which you want to sort the records.
2. Click on the  Ascending or  Descending option in the **Sort & Filter** group on the **Home** tab. The records get sorted.



Searching Records in a Table

To search for data quickly, follow these steps:

1. Enter a part or all of a word, phrase, date or number in a search box at the bottom of the Access screen.
2. Access highlights the corresponding characters in the first record that match your search. To find the next matching record, press the **Enter** key again.



Access finds all records that match your search data, anywhere in your datasheet.

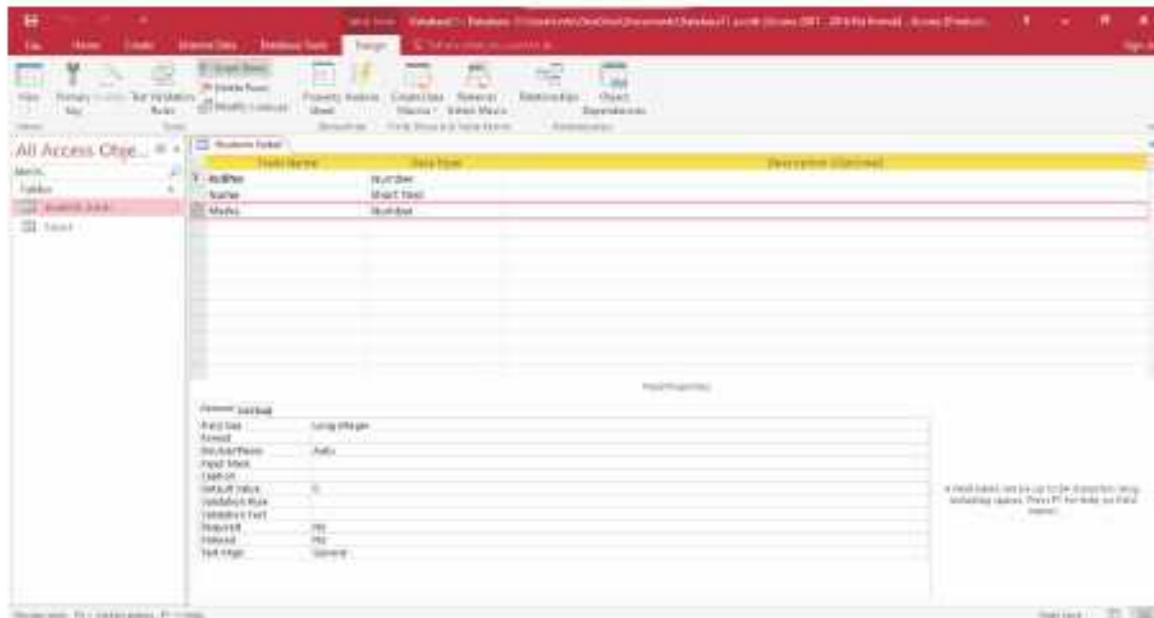
Modifying the Table Design

You can make changes to the design of a table after it has been created.

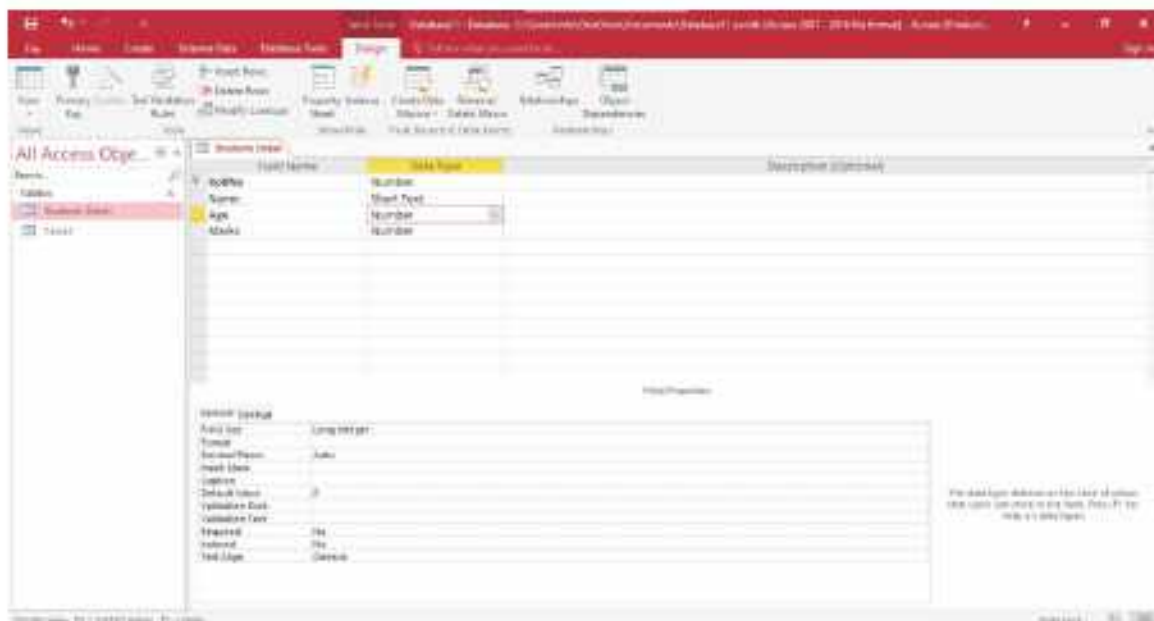
Inserting a New Field

To insert a new field, open the table in **Design View** and follow these steps:

1. Place the cursor on the field before which you want to insert a new field.
2. Click on the **Insert Rows** option in the **Tools** group on the **Design** tab under **Table Tools**.



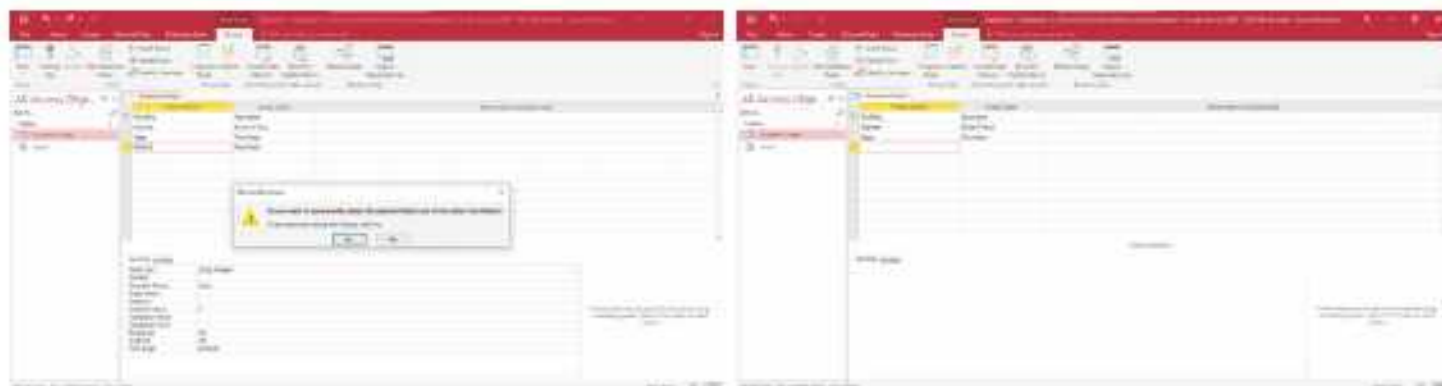
3. A new row is inserted. Enter the field name, data type and properties for the new field.



Deleting a Field

To delete a field, open the table in Design View and follow these steps:

1. Place the cursor on the field before which you want to insert a new field.
2. Click on the **Delete Rows** option in the **Tools** group on the **Design** tab under **Table Tools**.
3. Access displays a warning box confirming whether you want to delete the field permanently. Click on the **Yes** button.



The field is deleted from the table.



COMPUTER ETIQUETTES

Always clean the computer with clean and dry cloth.



Let's Implement.

A

Tick (✓) the correct answer.

- Which of the following data types lets you store an integer that is incremented automatically as you add records?
 a. Currency ☐ b. Yes/No ☐ c. AutoNumber ☐ d. Long Text ☐
- Find the odd term out.
 a. Date/Time ☐ b. Field Size ☐
 c. Required ☐ d. Allow Zero Length ☐
- A field that has a unique value for every record and cannot be left blank, is called the:
 a. Special key ☐ b. Primary key ☐ c. Unique key ☐ d. DBMS key ☐
-  is the symbol for:
 a. sorting ☐ b. Design View ☐ c. Datasheet View ☐ d. primary key ☐
- Which of the following field properties is used to put conditions on the data that can be entered in a table?
 a. Field Size ☐ b. Format ☐ c. Validation Rule ☐ d. Required ☐

B

Fill in the blanks.

- _____ view is used to enter data in a table.
- In _____ view of the table, you can enter the field names, their data types and description.

3. A table consists of _____ and _____.
4. The _____ data type is used to store lengthy text that is up to 65,536 characters.
5. The _____ data type is used to store text or a combination of text and numbers.

C Write True or False.

1. A database can have only one table.
2. Number data type is used for numeric data that will be used in calculations.
3. Caption property is used to give an alternate name to a field.
4. Field Properties Pane is used to set properties for the fields in the table.
5. Primary key option is available on the Home tab.

D Answer the following questions.

1. What do you understand by the term 'table'? Name some important components of a table.
2. What is a primary key? How do we set primary key in Access?
3. Explain the two parts of the Design View Window?
4. How will you sort the data within a table?
5. What do you mean by data types?



Label the pictures given below.





Experiential Learning

Create a table with the given fields. Save the database with the name 'Teacher's Details'.

Reg_No	F_Name	Subject	School	Address	Mobile_No.
--------	--------	---------	--------	---------	------------

- Create a table and enter 10 records in it.
- Make changes in a table. Rename the field name 'Mobile_No' to 'Contact_No'.
- Set 'Reg_No' as the Primary key.

- Insert a new field, 'Date_of_Joining' at the end.
- Move the field 'Subject' after the 'School' field.
- Edit records from the rows 4 to 7 by replacing their information with the new values.
- Delete the first two records.
- Sort the records in an ascending order.
- Save the table and exit MS Access 2016.

Competency Based Questions.

1. Consider the table given alongside and answer the following questions.
 - a. Can Item_No be taken as a primary key? If not, why?
 - b. What is the data type of the field Item_No?

Item_No	Name	Quantity	Purchase Date
101	Pen	120	14-03-2024
102	Pencil	200	24-03-2024
103	Eraser	120	13-04-2024
104	Sharpener	150	17-04-2024

2. Observe the table given below and answer the following questions.
 - a. What is the name of the table?
 - b. Name the view in which the table is displayed.
 - c. How many fields and records does the table have?

Student				
Roll_No	Name	Address	Marks	Click to Add
1	Akshay	Meerut	80	
2	Chandani	Modinagar	75	
3	Deepak	Meerut	86	
4	Farah	Gzb	67	
5	Govind	Gzb	70	
6	Himanshu	Modinagar	76	
7	Jiten	Meerut	98	

RACKING YOUR BRAIN

Life Skills and Values

Should you delete or edit data of a database created by your friend or teacher? Justify your answer.

Just Have a Discussion

Communication

Discuss the various ways of creating a table in MS Access in the class.



TEACHER'S NOTE

- The importance of having a primary key in a table should be explained to the students.
- The differences between Excel and Access should also be explained to the students.



WORKING WITH QUERIES, FORMS AND REPORTS



Outlines

- Query
- Forms
- Reports
- Printing Reports
- Creating a Query
- Creating a Form
- Creating a Report

Query

A Query is a method of getting answers to questions about the data. The query can involve a single table or multiple tables to answer a question about the data. A query is used to retrieve selective data based on specific criteria. Some of its examples are:

- The list of students who got grade A1 in all subjects.
- The list of employees who get salaries above ₹ 50,000 a month.

Creating a Query

Just like tables, there is a Query Design view in which queries are created, and a Query Datasheet view in which results of the queries are displayed.

You can create a query by these two methods:

- (a) Query Wizard (b) Query Design

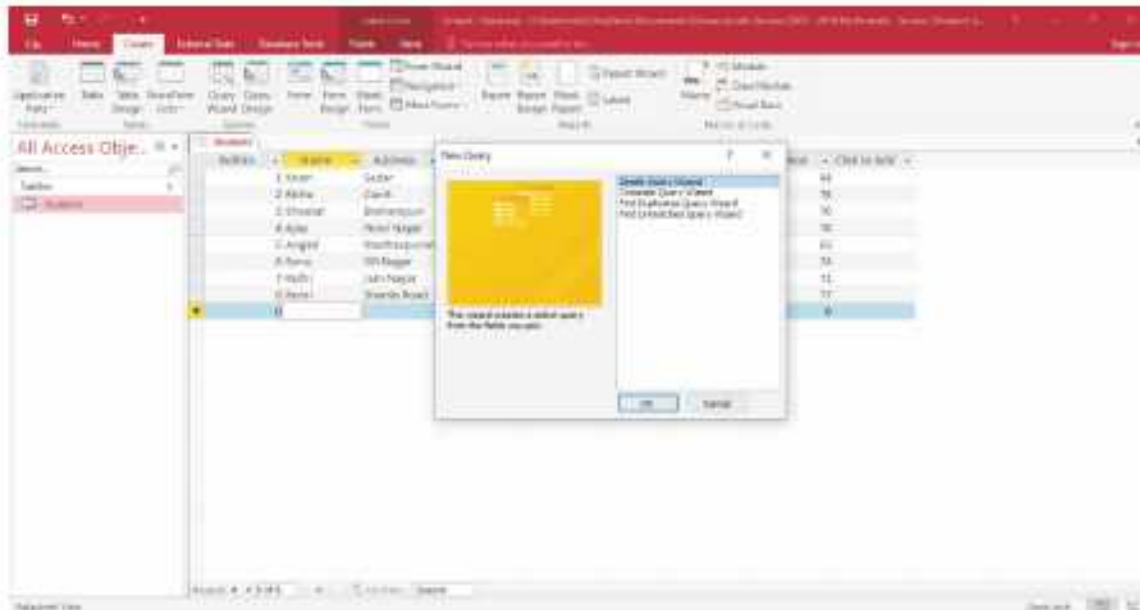
Let us create a query based on a table named 'Students'.

RollNo	Name	Address	Hindi	English	Maths	Computer	Science	Click to Add
1	Kiran	Sadar	89	87	85	86	84	
2	Richa	Centt	76	75	71	72	76	
3	Sheetal	Brahmapuri	75	74	72	71	70	
4	Ajay	Noor Nagar	74	71	72	71	70	
5	Angad	Madhavpuram	68	62	61	64	63	
6	Sonu	GG Nagar	79	77	78	76	74	
7	Ridhi	Jain Nagar	77	75	76	71	72	
8	Roni	Sharda Road	78	75	74	76	77	

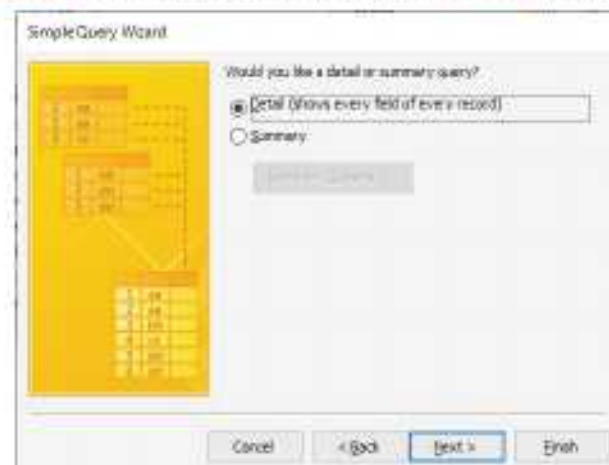
Using Query Wizard

To create a query using Query Wizard, follow these steps:

1. Click on the **Create** tab.
2. Click on the **Query Wizard** option in the **Queries** group.
3. A **New Query** dialog box appears. Select **Simple Query Wizard**.
4. Click on the **OK** button.

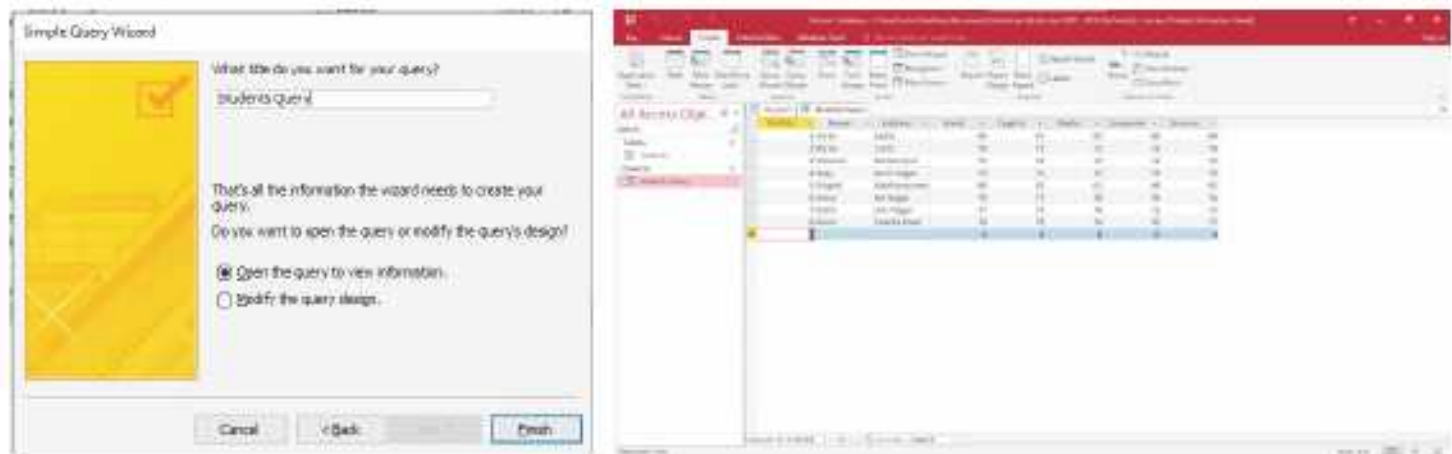


5. The **Simple Query Wizard** screen appears. Select the table that contains your data in the **Tables/Queries** box.
6. Now, add the fields you want to see in the query results. To add a field, select it in the **Available Fields** list and then click **>**, or just double-click it. To remove fields, select them in the **Selected Fields** list and click **<**.
7. Click on the **Next** button.
8. Select **Detail (shows every field of every record)** option and click on the **Next** button.



9. The **Simple Query Wizard** screen appears. Type a query name in the text box.
10. Select the **Open the query to view information** option.
11. Click on the **Finish** button.

The query result will appear in the Datasheet View.

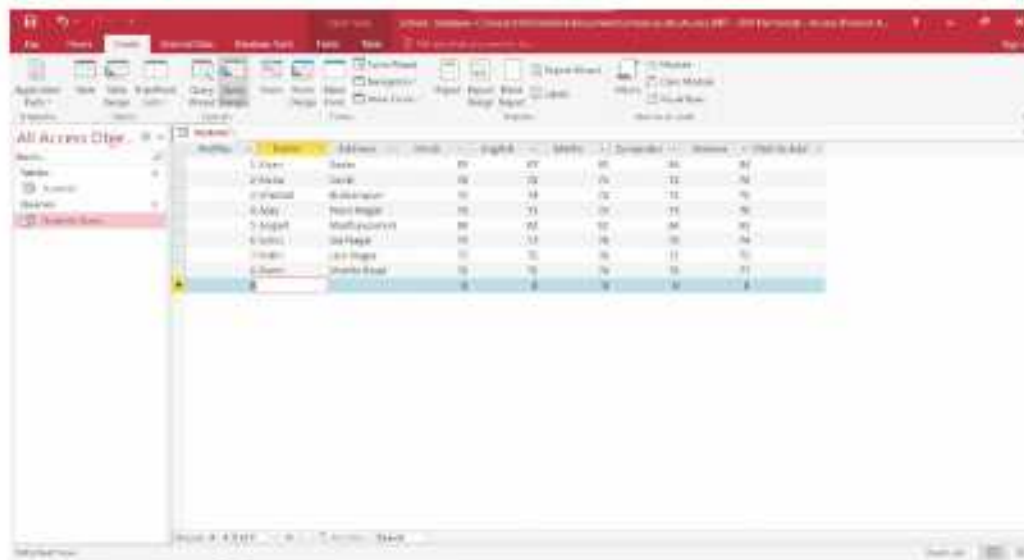


Using Query Design

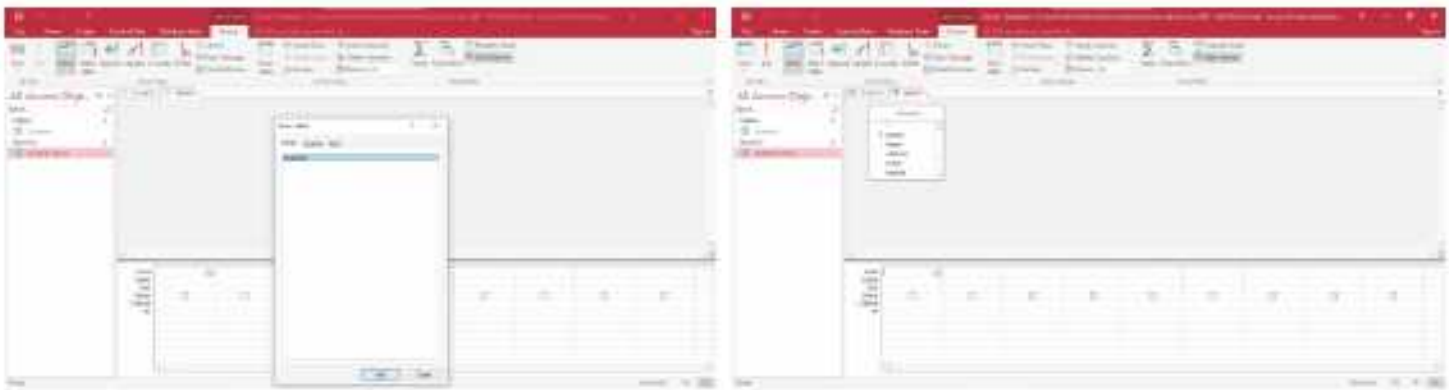
Let us create a query based on a table named 'Student' to filter the details of students whose marks of Maths are more than 70.

To create a query using Query Design, follow these steps:

1. Click on the **Create** tab.
2. Click on the **Query Design** option in the **Queries** group.

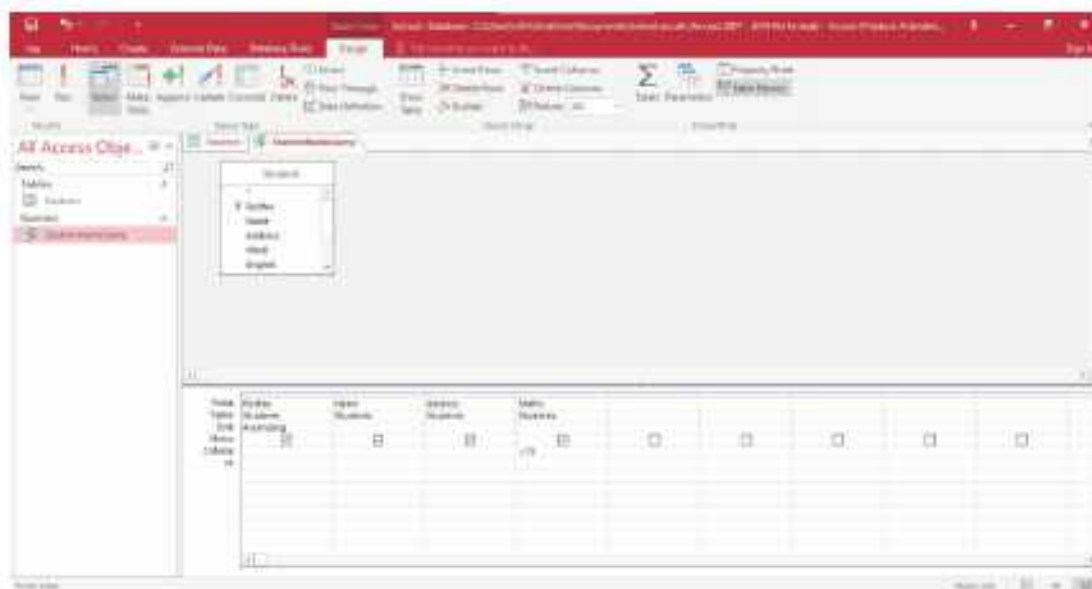


3. The **Show Table** dialog box appears. Select the table on which you want to base your query.
4. Click on the **Add** button.
5. After adding the table, click on the **Close** button to close the **Show Table** dialog box.

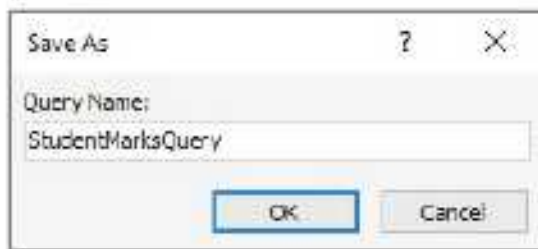


The selected table appears in a small window in the upper section of the **Query Design view**. The lower section of the Query Design view displays the **Design Grid**.

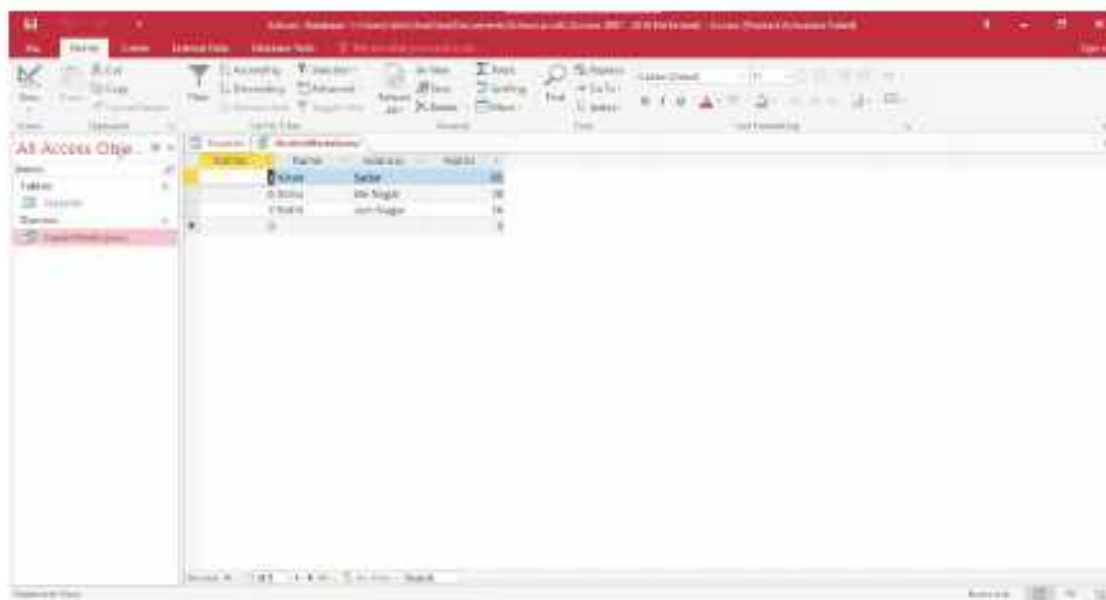
6. Add the fields to be used in the query in the Design Grid, by double-click on the field name in the small table window in the upper section of the Query Design view or drag and drop fields from the small table window in the upper section, to the field column in the Design Grid.
7. If you want the Query results to be sorted, then select **Ascending** or **Descending** order from the **Sort** box, under that field.
8. Clear the **Show** checkbox for fields that should not appear in the Query Result.
9. Type the search condition in the Criteria row of the field, on the basis of which you want to choose or filter the records.



10. Click on the **Save** option on the **Quick Access Toolbar**.
11. The **Save As** dialog box appears. Type the name of the query.
12. Click on the OK button.
13. Click on the Run option in the **Results** group on the **Design** tab, under **Query Tools** to execute the query.



The query results appear in the **Datasheet View**, which looks like a table.



Forms

Forms provide an easy way to enter, edit, delete and view data in a table. Forms are made up of elements such as text boxes and labels. These elements are called controls. They allow you to add and update data in a table, one record at a time.

You can work on a form in three views:

- **Form View**  : You can enter, edit and view data in Form view.
- **Layout View**  : You can view the data and make changes in the form, such as changing the size of the various controls.
- **Design View**  : Design view displays only the structure of your form. In this view, you cannot see the underlying data, but you can perform some tasks that cannot be performed in layout view, such as viewing header and footer.

Creating a Form

Microsoft Access provides many ways to create a form. Two are discussed here:

- (a) Form Wizard (b) Form option

Let us create a form based on a table named 'Student'.

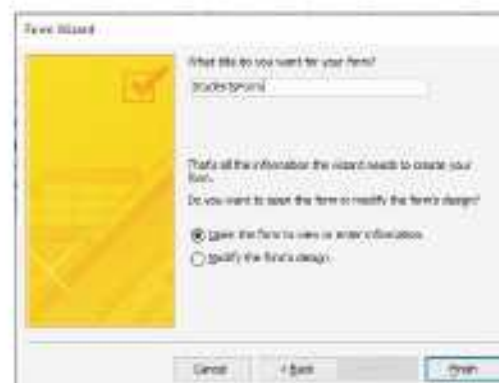
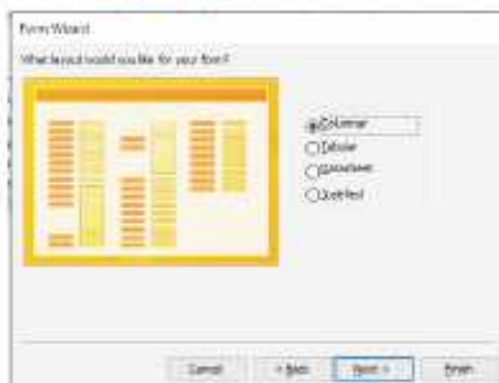
Using Form Wizard

To create a form using Form Wizard, follow these step:

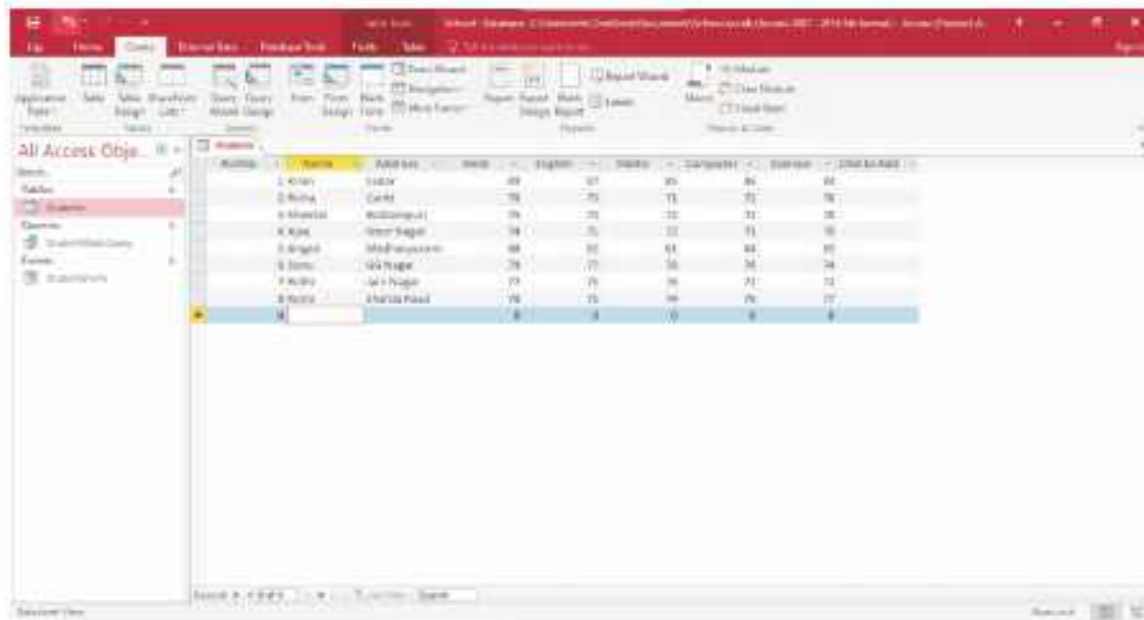
1. Click on the **Form Wizard** option in the **Forms** group on the **Create** tab.
2. The **Form Wizard** screen (1 of 3) appears. Select the table from the **Tables/Queries** drop-down menu.
3. Select the fields you want in the form.
4. Click on the **Next** button.



5. The **Form Wizard** screen (2 of 3) appears. Select a form layout. The default selection is **Columnar**, which you can change if required.
6. Click on the **Next** button.
7. The **Form Wizard** screen (3 of 3) appears. Give a name to the form.
8. Select the option **Open the form to view or enter information**.
9. Click on the **Finish** button.



The formatted form will appear on the screen.

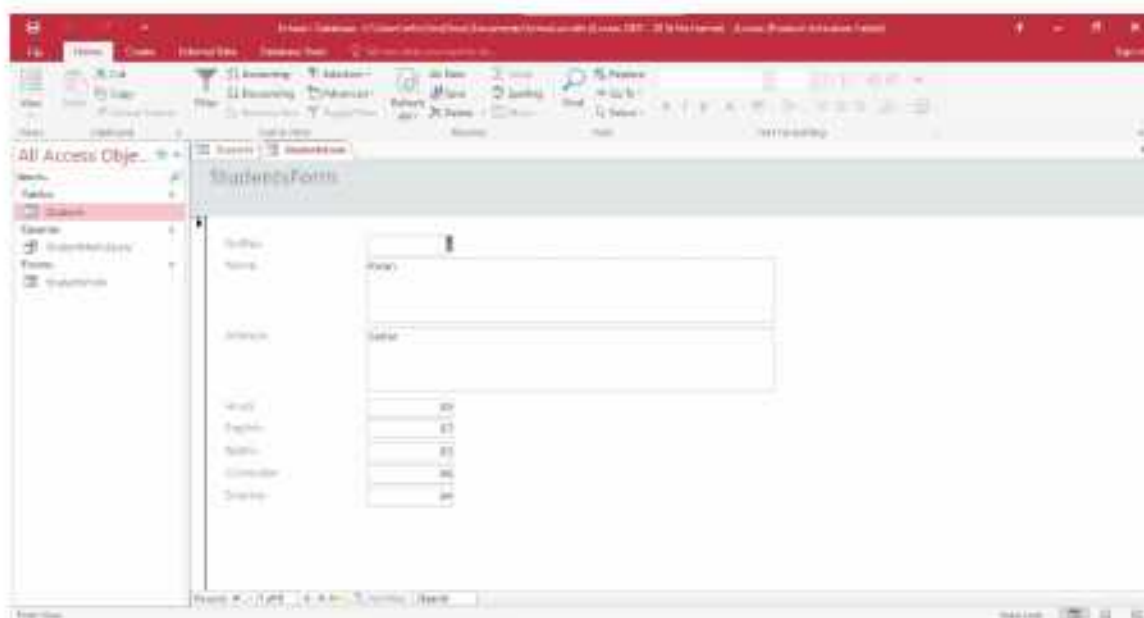


Using Form Option

To create a form using Form option, follow these steps:

1. Click on the table in the **Navigation Pane** on which the form should be based.
2. Click on the **Form** option in the **Forms** group on the **Create** tab.

A form is created and it opens in the **Layout View**.

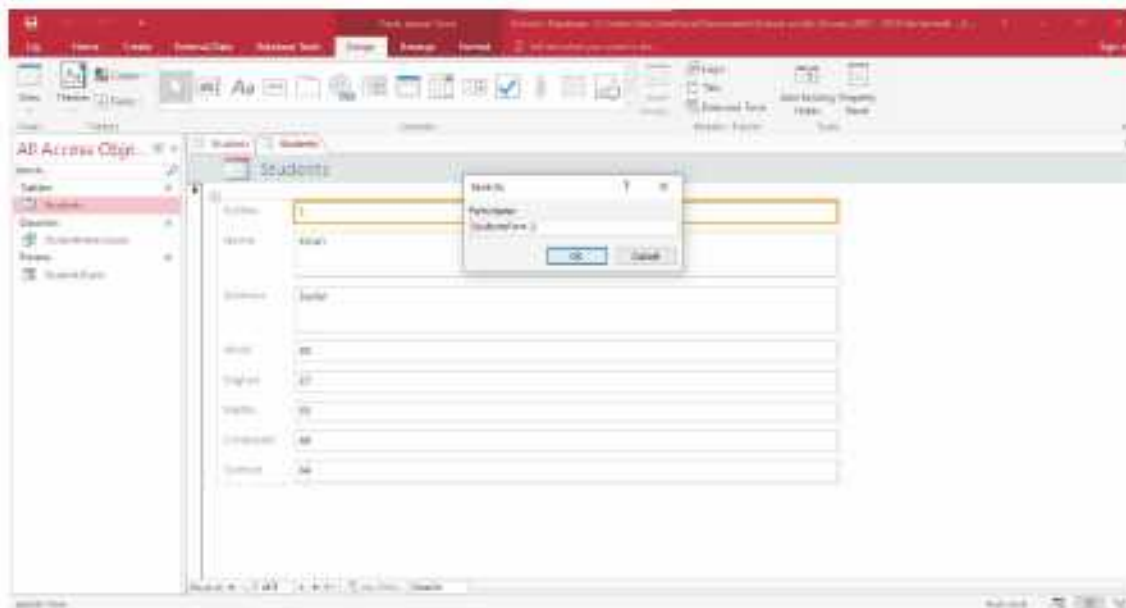


You can change the view to the **Form View** and enter or display data in the table. Three new tabs — **Design**, **Arrange** and **Format** — appear in the Ribbon.

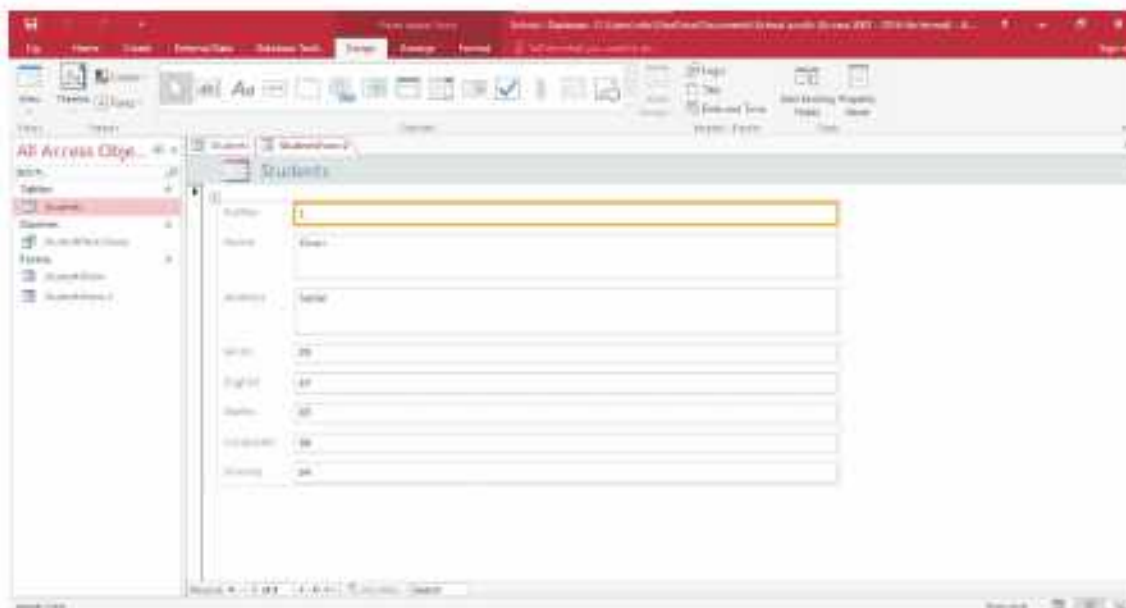
3. Click on the **Save** option in the **Quick Access Toolbar**.
4. The **Save As** dialog box appears. Type the name of the Form.



5. Click on the **OK** button.



You will notice that the Form object gets added to the Navigation Pane.



You can add a logo and a title to your form by making use of the pic-25 and pic-26 options in the **Header/Footer** group on the **Design** tab under **Form Layout Tools**.

Reports

Reports are an effective way to present your data in printed format. You can customise the appearance of tabular data on a report. Thus, you can display the information the way you want to see it. But, you cannot make changes to the table data in a report.

You can work on a report in four views:

- **Report View**  : You can view data in this view but cannot make any design-related changes.
- **Print Preview**  : This view displays the report as it will appear after printing.
- **Layout View**  : In this view, you can view the data and make changes in the report such as changing the size of various controls.
- **Design View**  : This view displays only the structure of your report. In this view, you cannot see the underlying data, but you can perform some tasks that you cannot perform in the Layout View, such as viewing the header and footer sections, etc.

To switch between the different views of a report, click on the down arrow under the **View** option in the **Views** group on the **Home** tab.

Creating a Report

Microsoft Access provides many ways to create a report. Two are discussed here:

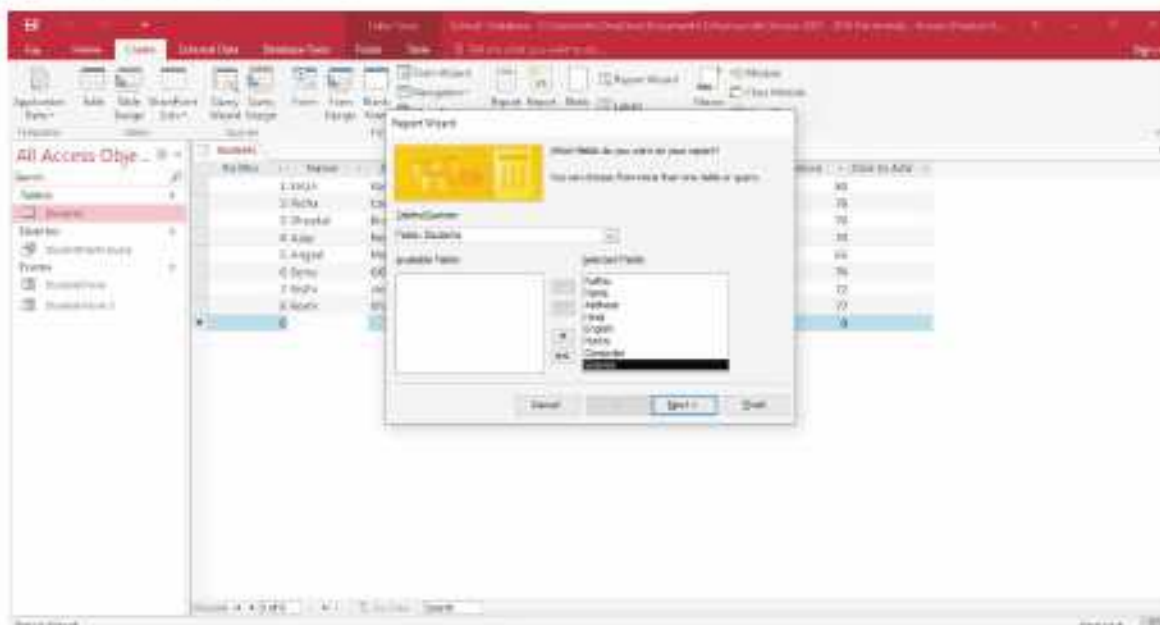
(a) Using the Report Wizard

(b) Using the Report option

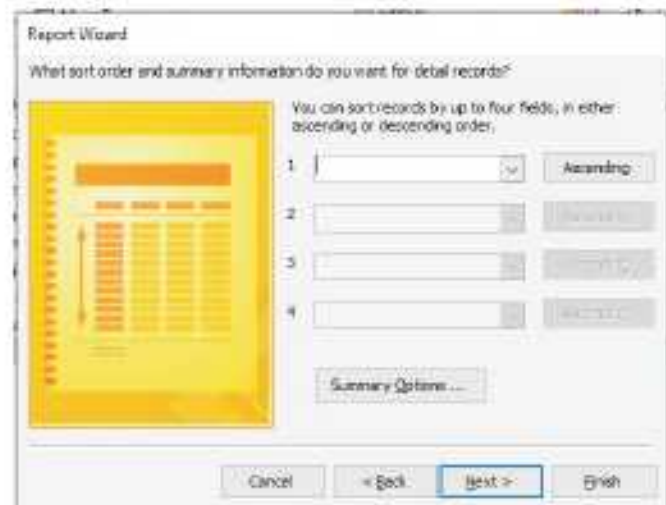
Using the Report Wizard

To create a report using the **Report Wizard**, follow these steps:

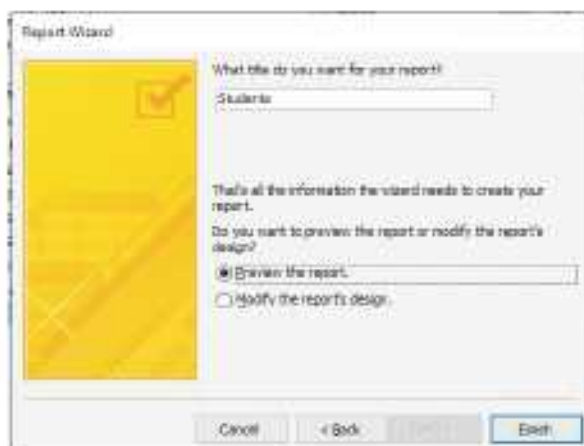
1. Click on the **Report Wizard** option in the **Reports** group on the **Create** tab.
2. The **Report Wizard** screen (1 of 5) appears. Select the table or query.
3. Add the fields that you want to see in the report results.
4. Click on the **Next** button.



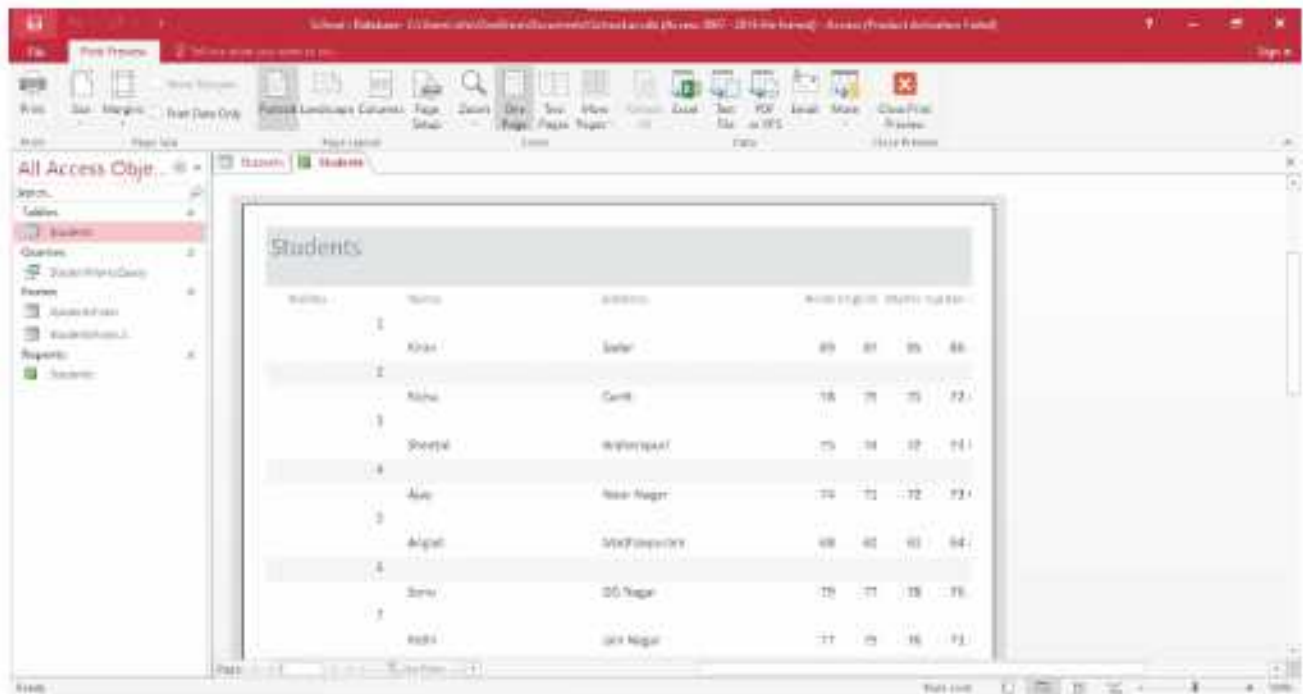
5. The **Report Wizard** screen (2 of 5) appears. Select the grouping level within the report and click on the **Next** button. (In this example, the grouping level is by Roll No.)
6. The **Report Wizard** screen (3 of 5) appears. In this screen, you can set the sort order and the summary options. You can sort data by up to four fields. By default, sorting is in ascending order. Click on the **Ascending** button to change it to **Descending**.
7. Click on the **Next** button.



8. The **Report Wizard** screen (4 of 5) appears. Select the type of **Layout**.
9. In the **Orientation** section, select either **Portrait** or **Landscape**.
10. Click on the **Next** button.
11. The **Report Wizard** screen (5 of 5) appears. Type a title for the report, which will also be the name assigned to the report.
12. Select **Preview the report** option.
13. Click on the **Finish** button.



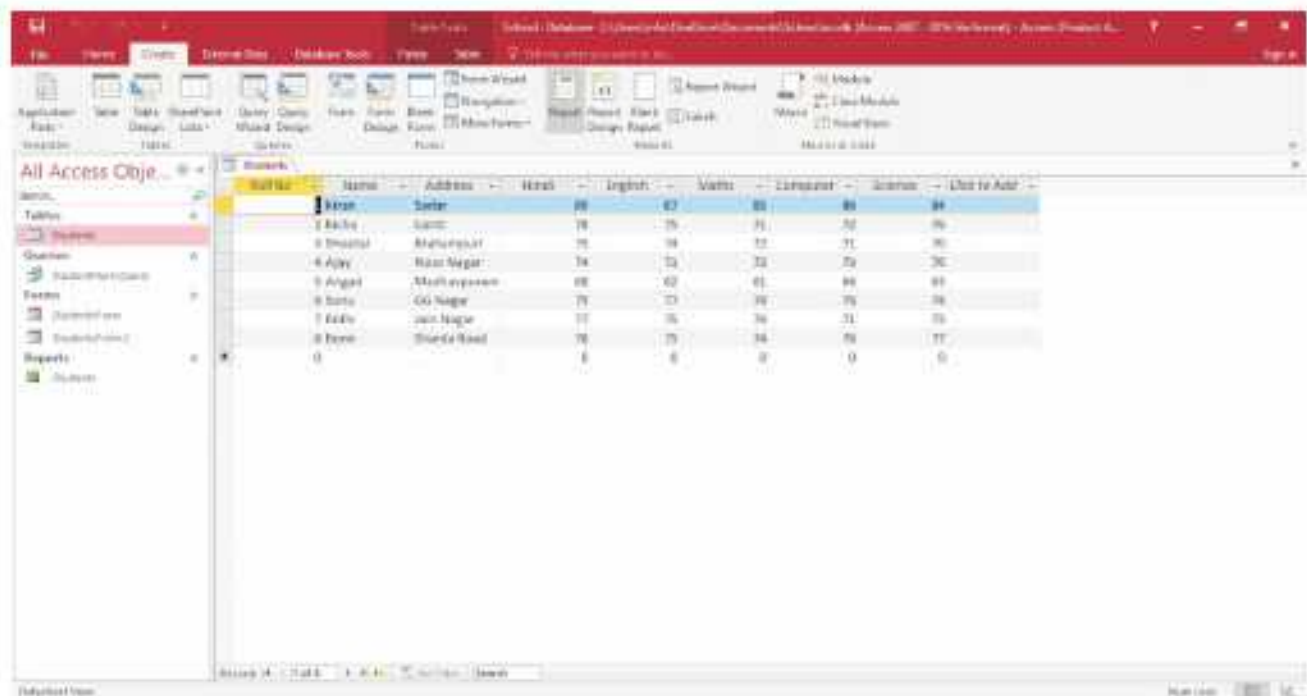
The report will appear on the screen.



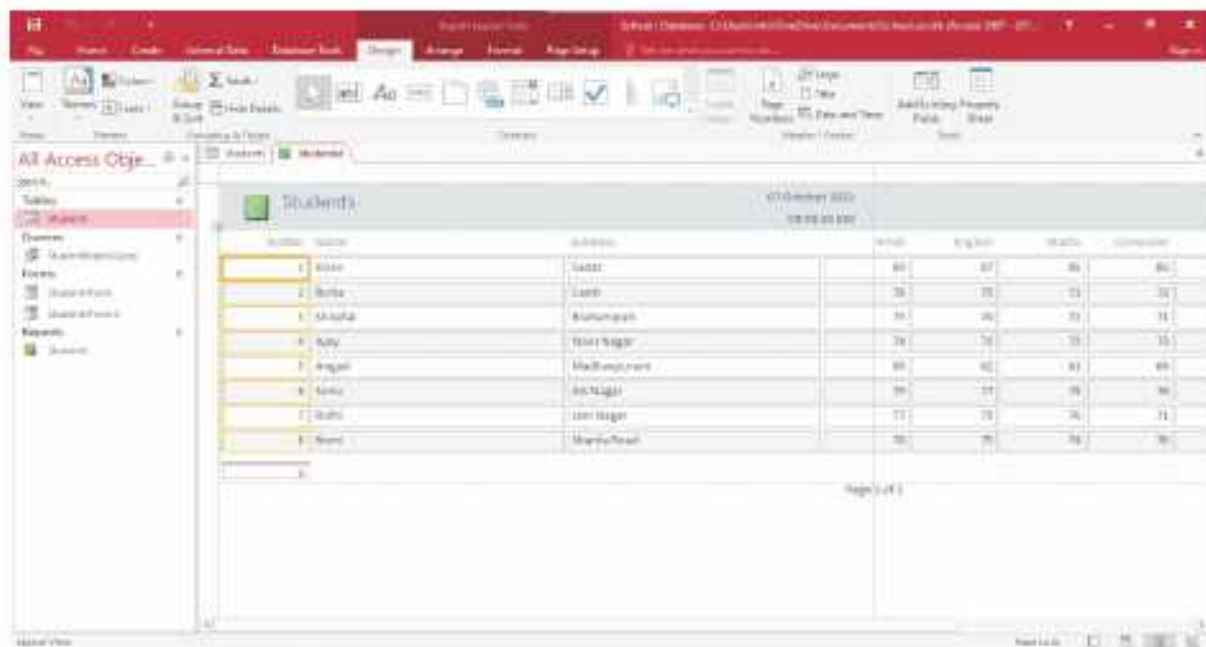
Using the Report Option

To create a report using the Report option, follow these steps:

1. Click on the table or query for which you want to create a report.
2. Click on the **Report** option in the **Reports** group on the **Create** tab.



The report is created and opens in the **Layout view**. Four new tabs: **Design**, **Arrange**, **Format** and **Page Setup** appear in the Ribbon.



3. Click on the **Save** option in the Quick Access Toolbar.
4. The **Save As** dialog box appears. Type the name of the Report.
5. Click on the **OK** button.



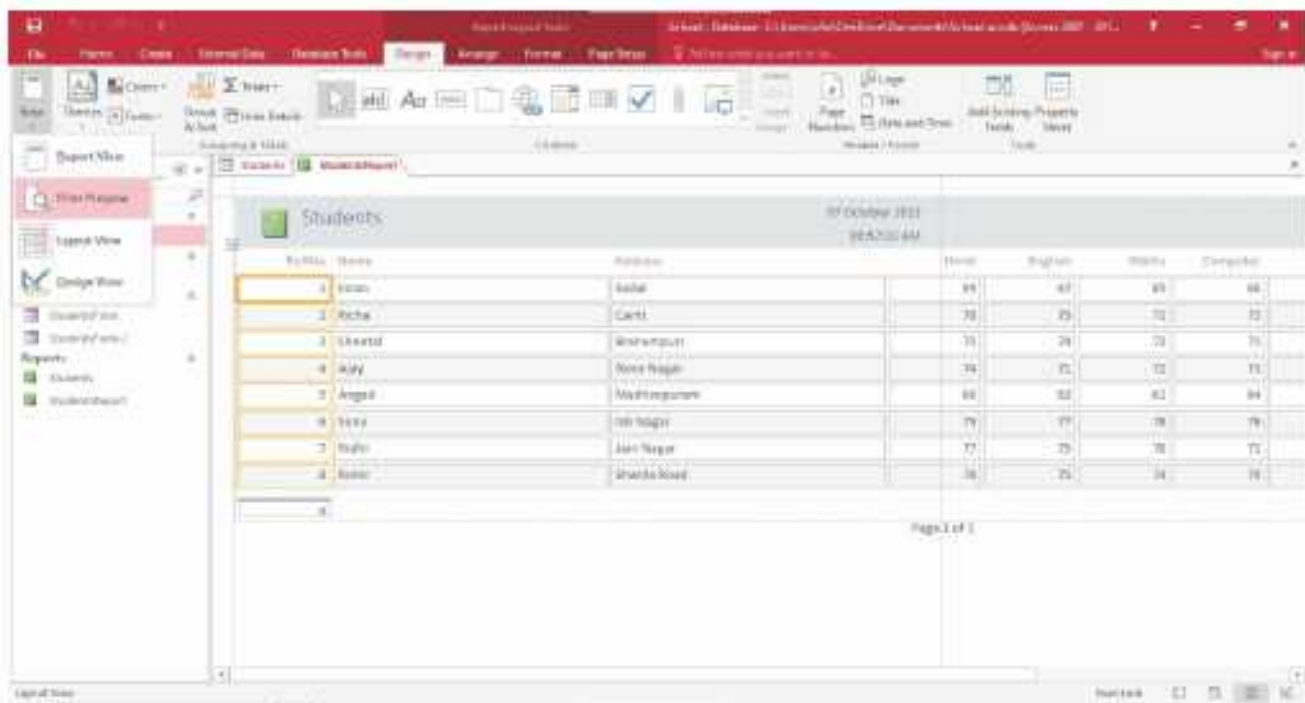
- The generated report may contain fields that you may not want. Click on the unwanted cell containing the field and press the **Delete** key to remove unnecessary fields.
- You can drag and change the width and the size of the controls.

Printing a Report

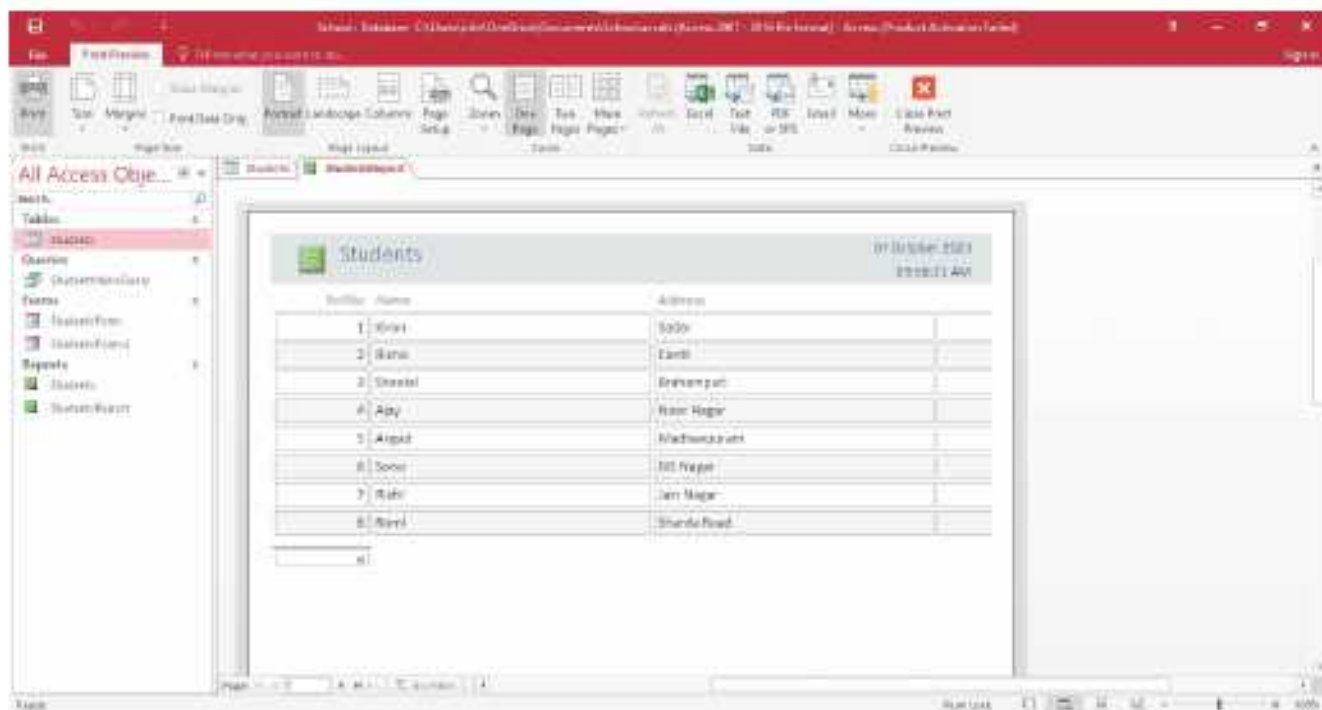
You can print reports using the **Print** option on the **File** tab or by using the **Print Preview View** of a report.

This view shows how the report will appear on a page. It also lets you choose the page orientation, page setup, etc.

1. Click on the down arrow under the **View** option in the **Views** group on the **Home** tab.
2. Click on the **Print Preview** option. The report is displayed in the manner in which it would appear on a page.



3. Select the appropriate values for Page Size, Margins and Orientation from the Ribbon.
4. Click on the Print option to print the report.



Click on the **Close Print Preview** button in the **Close Preview** group on the **Print Preview** tab to return the View to the **Report** or the **Layout View**.



Let's Implement.

A Tick (✓) the correct answer.

- _____ view is not a view for displaying a report.
a. Datasheet ☐ b. Design ☐ c. Both a and b ☐ d. None of these ☐
- There are _____ views to display a Form.
a. two ☐ b. three ☐ c. four ☐ d. five ☐
- A report has _____ views.
a. two ☐ b. three ☐ c. four ☐ d. five ☐
- _____ is not a view for working with a Form object.
a. Print ☐ b. Layout ☐ c. Both a and b ☐ d. None of these ☐
- _____ option is not present in a Query Design Grid.
a. Criteria ☐ b. Filter ☐ c. Sort ☐ d. None of these ☐
- The Query Design option is present on the _____ tab.
a. House ☐ b. File ☐ c. Format ☐ d. Create ☐

B Fill in the blanks.

- A _____ is a method of getting answers to questions.
- The _____ view is used to enter, edit and view data of the Form.
- Queries, forms and reports are _____ of Access.
- _____ view displays only the structure of your form.
- A _____ is an effective way to present your data in printed form.
- The Run option in the _____ group on the Design tab is used to execute the query.

C Write True or False.

- The Design View of the form displays only the structure of the form.
- You can create a query by using the Query Wizard only.
- The Query Wizard option is present in the Queries group.
- The Form option is present in the Forms group on the Create tab.
- You cannot customise the appearance of tabular data on a report.
- You can print reports using the Print option.

D Answer the following questions.

- What is a Query?
- Differentiate between Form and Report.
- How can you specify multiple conditions while designing a query?

4. What is the use of the Print Preview and Design View of a report?
5. What is the use of the Layout View and Design View of a form?
6. Write the steps to print a report.

Refreshment Corner

A Identify the following options of MS Access 2016 and write their names.

1.  _____
2.  _____
3.  _____
4.  _____
5.  _____

B Identify the following Views of a Report and write their names. Also, write their uses.

1.  _____
2.  _____
3.  _____
4.  _____

Practical Session

Experiential Learning

- Create a database named 'School' based on the following information:

Table Name : Student	
Fields	Data Type
Admission_No	Number
Name	Text
Father's Name	Text
Date_of_Birth	Date/Time
Age	Number
Marks	Number
Percentage	Number

Query:

Query Name: Query_Percentage

To display the Name, Date of Birth, Marks and Percentage of students with more than 75 per cent

Form:**Form Name: Form_Entry**

To view and enter records into the table 'Student'

Report:**Report Name: Report_Student**

To display the details of all students

Report:**Name: Report_Experience**

Report based on Query (Query_percentage) that displays Name, Date_of_Birth, Marks and Percentage of students with more than 75 per cent

Competency Based Questions

1. Observe the figure given below and answer the following questions.

Field:	Roll_No	Name	Marks		
Table:	Studentsdetail	Studentsdetail	Studentsdetail		
Sort:		Descending			
Show:	☐	☒	☒	☐	☐
Criteria:			>75		
on:					

- a. On which table is the query based?
 - b. Which field will be visible in the query result?
 - c. What is the area shown in the figure known as?
2. JP company has a huge database of employees. The company wants to maintain the privacy of each employee, and has given instructions to computer operators to update all records, one at a time. Which database object should they use to perform this task?

RACKING YOUR BRAIN

Life Skills and Values

Your friend created a database of his/her classmates and entered 30 records in it. Now, he/she wants to show the data of those students who have scored less than 50% marks. But, he/she don't know the process. Then, what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the usefulness of the query in the class in detail.

**TEACHER'S NOTE**

The teacher should hold discussions with the students about the usefulness of the query, form, and report objects in MS Access 2016.



COMPUTER VIRUS



Outlines

- Virus
- How Virus Spreads?
- Malware
- Trojan Horse
- Steps to Prevent Virus
- Importance of Regular Updates
- Virus Symptoms
- Sweeper
- Computer Worms
- Spyware
- Antivirus
- Using an Antivirus Software

Computers and data have become the lifeline of every organisation. If something goes wrong with them, business almost comes to a standstill. This could cause a loss of thousands or even millions of hours of hard work research. Hence, we must take preventive measures to protect our system and data from virus attacks, physic damage, data loss and hacking. Many times, while working on a computer, it gets hanged, restarts frequently and sometimes data gets lost. Generally, we call it a virus attack. However, in reality, it is not only the virus that does but Worms, Trojans, Spyware, etc. are also equally responsible for improper functioning of a computer. A common mistake which people make while discussing about a computer virus is to refer a Computer Worm or Trojan Horse as a virus. Though Trojan, Worm and Virus are often used interchangeably, but they are not exactly the same. Viruses, Worms and Trojan Horses are all malicious programs that can cause damage to a computer, but these programs have certain differences. Knowing these differences can help us to protect our computers from their damaging effects.

Virus

A computer virus is a program or a set of programs that disrupts the normal functioning of a computer. A virus infects or destroys the data. It enters a computer without the permission or knowledge of the user. The full form of VIRUS is Vital Information Resources Under Seize. The essential resources of a computer, like memory, processing speed, data and programs

are hampered when a virus spreads into the system. Almost all viruses are attached to an executable file, which means that if a virus exists on a computer, it will not infect the computer unless you run or open the malicious program.

Types of Viruses

There are different types of viruses but broadly they can be classified into the following categories:

Boot Sector Virus : This type of virus infects the boot sector of a hard drive. A disk or hard drive comprises of smaller sections known as sectors. The first sector is called the boot sector. It carries the master boot record, which is responsible for reading and loading the operating system. Boot Sector Virus becomes active at the time of booting the machine.

Examples: Disk killer, Stone virus, etc.

Program File Virus : This virus infects the executable files or programs such as those with extension, like .com, .exe, .ovl, .drv, .sys, etc. These programs get loaded in the memory during execution. The virus becomes active in the memory by replicating itself and infecting files on the disk.

Examples: Sunday, Cascade, etc.

Multipartite Virus : A multipartite virus is a hybrid of a Boot Sector and Program File virus. The virus gets entry into the computer system through some infected media and then resides inside the computer memory. After that, this virus moves into the hard drive, infects the boot sector and starts spreading itself in the system by infecting the executable files.

Examples: Invader, Flip, Tequila, etc.



The first PC virus was a boot sector virus named as Brain, released in 1987 by the Farooq Alvi Brothers in Lahore, Pakistan.

Macro Virus : This virus infects the files which are created using such application softwares that support macros such as Word, Excel and PowerPoint. A macro is a series of programming steps that allows automation of many actions with a single keystroke. When the infected document is opened, the macro virus gets loaded into the main memory. Once a macro virus gets active on to your computer, every document you produce will become infected.

Examples: Melissa. A, Relax, Bablas, etc.

Popular Viruses

- Jerusalem or Israeli Virus
- Brain or Pakistani
- Brain
- April First
- Ping Pong or Bouncing Ball
- Stoned (Marijuana)
- Dark Avenger

Virus Symptoms

Given below are some of the symptoms that indicate that the system is virus affected:

- Computer starts doing things on its own like moving the cursor all by itself, randomly opening and closing the windows, etc.
- Computer may shut down and restart frequently.
- Malfunctioning may be noticed in several applications.
- Certain drives on the computer become inaccessible.
- Unexpected error messages pop up.
- Menus and dialog boxes may appear distorted.
- Speed of computer drops.



The Creeper Virus was first detected on ARPANET in the early 1970s.

How Virus Spreads?

Virus programs are self-replicating programs, which propagate by infecting more and more computer files, thus resulting in the loss of speed and destruction of data. Computer viruses usually spread in one of the three ways:

MyDoom is considered to be the most expensive virus in the world and in cyber security history, having caused an estimated financial damage of \$38.5 billion.

Via Removable Storage Device

Sharing and transferring files via flash drives is a very common practice today. If this drive contains malicious files, it will infect the computer with which it is attached. Likewise, if a flash drive is inserted in an infected computer, it will also be infected with the virus. This infected drive then becomes the vector and capable of spreading the virus onto other computers if the operating system on those machines has an enabled **AutoRun** feature. The **AutoRun** function launches installers and other programs automatically when a flash drive or CD is inserted.

A virus can also be embedded in a normal file on a flash drive even if the **AutoRun** feature is disabled. The computer will become infected when that file is opened.

From Internet

Computer viruses spread very quickly through Internet, because they can be transmitted to

many site simultaneously. There are many websites which while visiting, try to install adware bugs that will cause a flood of pop ups to appear on your screen. This adware often allows for other rogue programs to be installed and before you know it, your computer gets filled with filthy material.

Through E-mail Attachments

Viruses may be sent as an executable file with an e-mail attachment and mislead the recipient into clicking on it.

Sweeper

Sweeper is another type of virus which looks like an Antivirus program and the user unintentionally download it. This virus makes use of Google services to install the malicious code on the computer. Once installed, it creates a variety of harmless, fake malware files in the computer, which get automatically started when the user logs Windows. It controls the browsing activity of the user, changes system files, etc. Without the user's knowledge, it performs a scan and prompts security alerts by displaying the list of these fake malware files. These alerts are shown to scare the user into thinking that his computer is infected. One should ignore these alerts and make use of proper steps to remove the malicious code.

Malware

The term Malware is made up of two words, malicious and software. It is a generic term used for such programs that enter into a computer and damage it without the user's knowledge. Malware not only interrupts the normal functioning of a computer, but also steals important information. Some examples of data stealing malware are:

Bancos

It is an information stealer that waits for a user to access banking websites and while surfing, it steals the sensitive information.

Gator

It secretly monitors web-surfing habits and sends this information to its server, which then throws related pop-up advertisements.

Computer Worms

A worm is similar to a virus in design and it is considered to be a sub-class of a virus. Unlike Virus, it is a self-replicating program, *i.e.*, having the capability to spread without any human intervention. Due to this self-replicating nature, the end result in most cases is heavy consumption of system memory (or network bandwidth), causing web servers, network servers and individual computers to stop responding. In recent worm attacks such

as the much-talked-about Blaster Worm, the worm was designed to tunnel into a system and allowed malicious users to control the computers remotely.

Trojan Horse

The name Trojan Horse is derived from the wooden horse used by the Greek Army to conquer the city of Troy. The Trojan horse, at first glance will appear to be a useful software, but will actually damage the computer once it is installed. Users on the receiving end of a Trojan horse, are usually tricked into opening them. Trojans have different behaviour once they gain access into the system. For example, some Trojans change the desktop by adding silly desktop icons, some prevent applications from running, especially antivirus applications. Trojans are also known to create a backdoor on a computer that gives an easy and unauthorised access to malicious users, thus allowing your confidential or personal information to be compromised. Unlike viruses and worms, Trojans do not reproduce by infecting other files, nor do they self-replicate. They require human action to run.

In 1981, Richard Skrenta wrote, Elk Cloner, the first computer virus to appear outside a single computer or lab where it was created.

Spyware

Spyware is a type of malicious software that gains entry into a system and monitors the user activity without his knowledge and then reveals the collected information to another interested party.

Spyware programs can collect various types of personal information, such as Internet surfing habits, visited web pages, passwords and even credit card numbers. They have the ability to scan files on the hard drive, snoop other applications, such as chat programs or word processors. While doing so, they consistently transmit this information in the background to someone else who will either sell the information to another party or use it for advertising/marketing purposes.

Sometimes, they also change the computer's settings, which results in slow connection speed and wrong functionality of other programs.

Stealth Viruses are difficult to detect by antivirus software as they use various methods to avoid detection.

Steps to Prevent Virus

In today's time, data holds the same value as money or gold, so we should take every possible step to prevent it from getting lost or damaged. Generally, computer virus infects

and spreads from one computer to another by transferring infected data, using e-mail, flash drives or other sources. To prevent the virus attack, follow the given tips:

- Equip the PC with some antivirus program and keep it updated.
- Check whether your antivirus software has the potential to detect spyware and malware, if not, you should install some anti-spyware program also on your system.
- Install a firewall on your system to control the incoming and outgoing network traffic.
- Always scan the pen drive before copying files.
- Avoid downloading and installing pirated software from the Internet.
- Scan the downloaded files before opening them.
- Scan the hard disk twice a month.
- Take the backup of imported files every day.
- Use Internet and e-mail attachments very carefully. Never click on an attachment in an e-mail, unless it is from some genuine and trusted source.

Antivirus

Antivirus is a computer program, which is designed to identify, prevent, and remove viruses from a computer. It performs the following tasks in a computer:

- Scans computer files to look for known viruses from virus dictionary and tries to remove them.
- Identifies suspicious behaviour of any computer program which might indicate infection.
- Scans incoming e-mails, which might contain virus in the attachment.

Some popular antivirus software in use nowadays, are as follows:

- McAfee
- Quick Heal
- Symantec Norton Antivirus
- AVG Antivirus
- Microsoft Security Essentials

Free and Paid Tools

Free Antivirus Softwares: Avast, AVG, Avira, Bitdefender, Panda, Sophos, and Comodo.

Paid Antivirus Softwares: Kaspersky, McAfee, Webroot Secure Anywhere, Trend Micro, and Quick Heal.

How Antivirus Works?

All antivirus applications operate on similar principles. There are two main components in an antivirus:

- A virus database that contains a list of file patterns which are used to identify a virus.
- The second component of an antivirus is a scan engine which scans files and compares them with the pattern files to verify if there is any infection.

The scan engine can be run when the user explicitly activates it or it can also be set to run automatically anytime a file is opened or a program is executed. This is called real-time scan.

Real-time scan runs continuously in the background. It provides maximum protection by scanning the opened or saved files and displays a notification message when it detects a security risk.

Importance of Regular Updates

We have discussed in the above section that every antivirus has a virus database that contains a list of file patterns, which are used to identify a virus. Therefore, to catch different and new types of viruses, the virus database of the antivirus has to be updated frequently. Simply installing antivirus software in a computer does not guarantee 100% protection, as the installed antivirus program may not be able to detect any new virus programs. Frequently updating the antivirus software, will ensure that the system is always armed and well prepared for the war against all types of virus threats. Latest virus protection is crucial for protecting your valuable documents. Protecting your computer and your personal files from virus threat is a continuous on-going activity that you need to be vigilant about. Most antivirus software companies provide the facility to download the latest updates from their websites.

Using an Antivirus Software

We are using the AVG antivirus software here. Likewise, you can use any other antivirus, like Norton Antivirus, Quickheal, McAfee, etc., whichever is loaded on your system.

The steps to use an Antivirus software are as follows:

- To open the AVG antivirus software, click on **Start > AVC AVG Protection**.
- The '**AVG Internet Security**' window appears.
- Click on the **Scan now** option.
- The scanning process starts.
- If viruses are found, the antivirus will inform the same to the user and will either delete the virus or quarantine it (isolate it so that it cannot affect the system).
- When the scan process gets over, the message, **Whole Computer Scan Finished**, is displayed.
- You can also customize the scanning process by clicking on the gear icon adjacent to the **Scan Now** button and choosing any of the displayed options enlisted below:
 - a. The **Scan Whole Computer** to scan the whole computer.

- b. Scan Specific Files or Folders to scan the particular files or folders.
- c. The Anti-Rootkit scan to scan for dangerous rootkits inside applications' drivers or libraries.



COMPUTER ETIQUETTES

Always keep the mouse on the mouse pad.



Let's Implement.

A

Tick (✓) the correct answer.

1. Which virus infects the boot record or master boot record?
 a. Worm ☐ b. Boot Virus ☐ c. Spyware ☐
2. Which virus is named after the wooden horse used by the Greek army to conquer the city of Troy?
 a. Dark Avenger ☐ b. Stoned Virus ☐ c. Trojan Horse ☐
3. When was the first PC virus released?
 a. 1897 ☐ b. 1985 ☐ c. 1987 ☐
4. Which one of the following is the name of the virus?
 a. Jan First ☐ b. May First ☐ c. April First ☐
5. Who released the first PC virus?
 a. Duke Brothers ☐ b. Wright Brothers ☐ c. Farooq Alvi Brothers ☐

B

Fill in the blanks.

1. A Computer virus _____ the normal operation of the computer.
2. Computer virus results in loss of _____ and _____ of data.
3. Stone virus is a type of _____ virus.
4. _____ virus was first detected on ARPANET.
5. _____ is the most expensive virus in the world.
6. _____ is designed to identify, prevent, and remove virus from a computer.

C

Write True or False.

1. A Computer virus is similar to biological virus.
2. All virus programs cause harm to data or programs.
3. Trojan Horses are self replicating.
4. A worm is similar to a virus in design.
5. Spyware creates a backdoor on a computer.
6. We should equip our computer with antivirus programs.

D Answer the following questions.

1. What is a computer virus? Name its types.
2. Write any three possible symptoms of virus in a computer?
3. What do you understand by the term spyware?
4. Explain the working of Trojan Horse?
5. Write any three steps required to protect your computer from Virus attack.



► Draw and colour the icon of any one antivirus in the space given below.

Art Integration



Competency Based Questions

1. Varsha's computer has got infected with a virus, resulting in the frequent hanging of the system. Which software will you suggest her to install in order to solve this problem?
2. Sumit's computer is not working properly. The executable files and programs with extensions, like .com, .exe, .ovl, .drv, .sys on his PC have got infected. What type of virus has infested his computer?

RACKING YOUR BRAIN

Life Skills and Values

Your friend forgets to remove his/her pen drive from your PC after finishing his/her work. What will you do:

- a. Check the content of his/her pen drive
- b. Keep it with yourself
- c. Return to him immediately

☐
☐
☐

Just Have a Discussion

Communication

Discuss the use of antivirus in the class in detail.



Teacher's Note

- A demonstration of how to scan a computer using antivirus software should be arranged in the class.
- The importance of using antivirus software should be explained to the students.



QUICK REVIEW 1

Tick (✓) the correct option.

1. It includes wireless computer keyboard and mouse, Bluetooth-enabled headphones, wireless printers and TV remotes.
a. Personal Area Network ☐ b. Local Area Network ☐
c. Wide Area Network ☐ d. Metropolitan Area Network ☐
2. Find the odd term out.
a. Date/Time ☐ b. Field Size ☐
c. Required ☐ d. Allow Zero Length ☐
3. A report has _____ views.
a. two ☐ b. three ☐ c. four ☐ d. five ☐
4. Which one of the following is the name of the virus?
a. Jan First ☐ b. May First ☐ c. April First ☐ d. June First ☐
5. This may be responding to requests from many network users simultaneously.
a. File Server ☐ b. Data Server ☐
c. Bus Server ☐ d. Network Server ☐
6. It uses advanced technologies, such as Asynchronous Transfer Mode (ATM), Frame Relay and Synchronous Optical Network (SONET).
a. Personal Area Network ☐ b. Local Area Network ☐
c. Wide Area Network ☐ d. Metropolitan Area Network ☐
7. A field that has a unique value for every record and cannot be left blank, is called the:
a. Special key ☐ b. Primary key ☐ c. Unique key ☐ d. DBMS key ☐
8. _____ is not a view for working with a Form object.
a. Print ☐ b. Layout ☐ c. Both a and b ☐ d. None of these ☐
9. Who released the first PC virus?
a. Duke Brothers ☐ b. Wright Brothers ☐
c. Farooq Alvi Brothers ☐ d. None of these ☐
10. The Query Design option is present on the _____ tab.
a. House ☐ b. File ☐
c. Format ☐ d. Create ☐



HTML IMAGE AND LIST



Outlines

- HTML Images
- HTML Lists

HTML Images

Images are very important to beautify as well as to depict many complex concepts in simple way on your web page. This tutorial will take you through simple steps to use images in your web pages.

Insert Image

You can insert any image in your web page by using `<img>` tag. Following is the simple syntax to use this tag.

```

```

The `<img>` tag is an empty tag, which means that it can contain only list of attributes and it has no closing tag.

Example : To try the following example, let's keep our HTML file test.htm and image file test.png in the same directory:



You can use PNG, JPEG or GIF image file based on your comfort, but make sure you specify correct image file name in src attribute. Image name is always case sensitive.

The alt attribute is a mandatory attribute which specifies an alternate text for an image, if the image cannot be displayed.

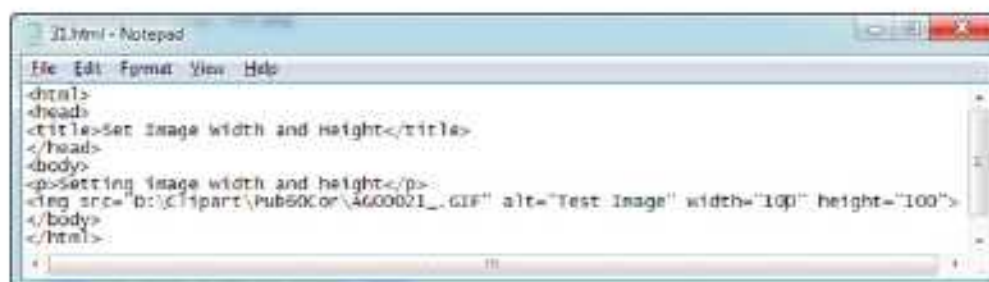
Set Image Location

Usually we keep the our all images in a separate directory. So let's keep HTML file test.htm in our home directory and create a subdirectory image inside the home directory where we will keep our image test.png.

Set Image Width/Height

You can set image width and height based on your requirement using width and height attributes. You can specify width and height of the image in terms of either pixels or percentage of its actual size.

Example :



```
31.html - Notepad
File Edit Format View Help
<html>
<head>
<title>Set Image width and height</title>
</head>
<body>
<p>Setting image width and height</p>

</body>
</html>
```

This will produce the following result:

Set Image Alignment

By default, image will align at the left side of the page, but you can use align attribute to set it in the center or right.

Example :



```
31.html - Notepad
File Edit Format View Help
<html>
<head>
<title>Set Image Alignment</title>
</head>
<body>
<p>Setting image Alignment</p>

</body>
</html>
```

This will produce the following result:



HTML Lists

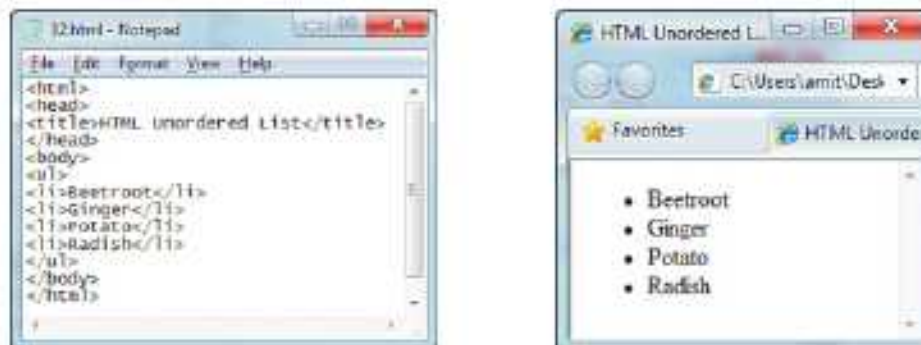
HTML offers web authors three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain:

- **** - An unordered list. This will list items using plain bullets.
- **** - An ordered list. This will use different schemes of numbers to list your items.
- **<dl>** - A definition list. This arranges your items in the same way as they are arranged in a dictionary.

HTML Unordered Lists

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML **** tag. Each item in the list is marked with a bullet.

Example :



The Type Attribute

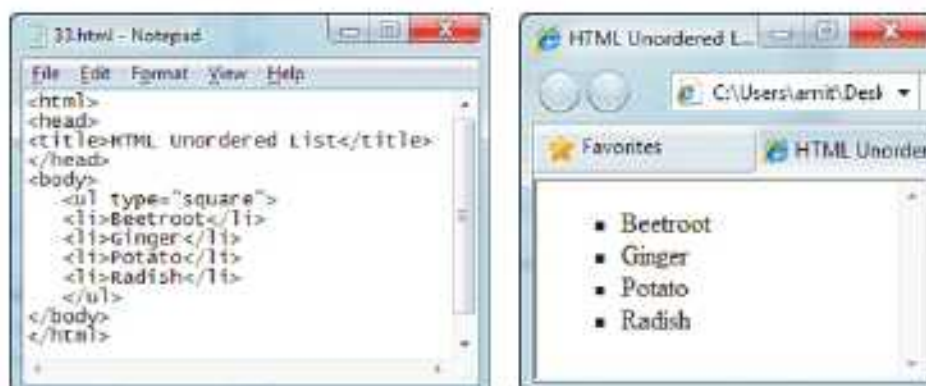
You can use type attribute for **** tag to specify the type of bullet you like. By default, it is a disc. Following are the possible options:

<ul type="square">

<ul type="disc">

<ul type="circle">

Example : Following is an example where we used **<ul type="square">**



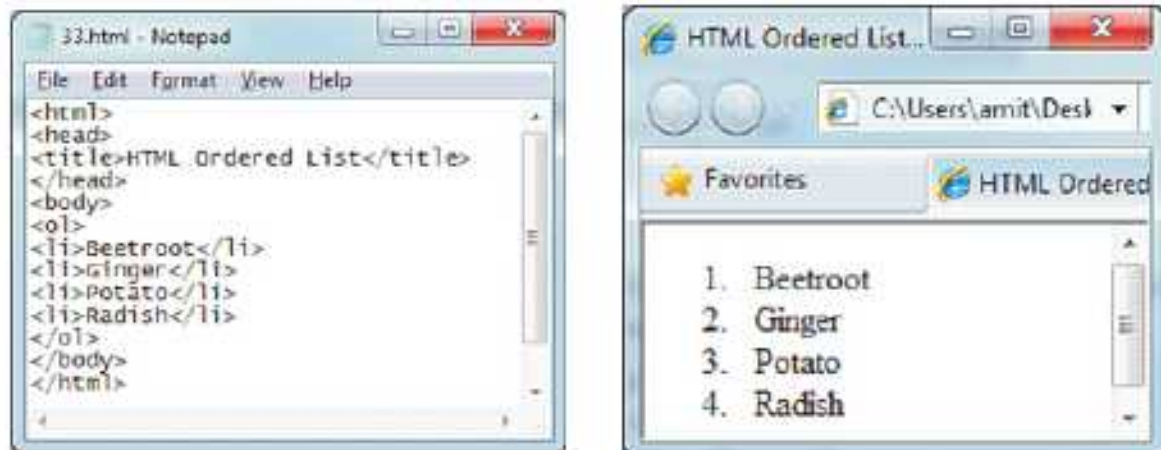
Same process will work for attribute 'disc' and 'circle'.

- For **<ul type="disc">** and
- For **<ul type="circle">** bullets will be displayed.

HTML Ordered Lists

If you are required to put your items in a numbered list instead of bulleted, then HTML ordered list will be used. This list is created by using `<ol>` tag. The numbering starts at one and is incremented by one for each successive ordered list element tagged with `<li>`.

Example :



The Type Attribute

You can use type attribute for `<ol>` tag to specify the type of numbering you like. By default, it is a number. Following are the possible options:

`<ol type="1">` - Default-Case Numerals.

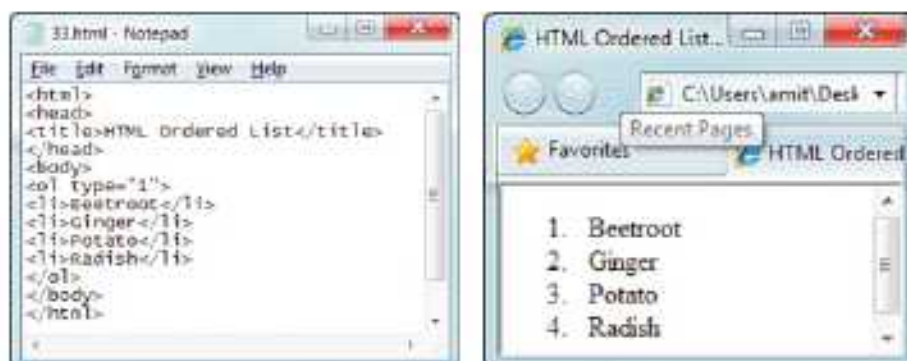
`<ol type="I">` - Upper-Case Numerals.

`<ol type="i">` - Lower-Case Numerals.

`<ol type="a">` - Lower-Case Letters.

`<ol type="A">` - Upper-Case Letters.

Example : Following is an example where we used `<ol type="1">`



Similarly, For `<ol type = "I">`, 'I, II, III...'

For `<ol type = "i">`, 'i, ii, iii...'

For `<ol type = "a">`, 'a, b, c...'

For `<ol type = "A">`, 'A, B, C...'

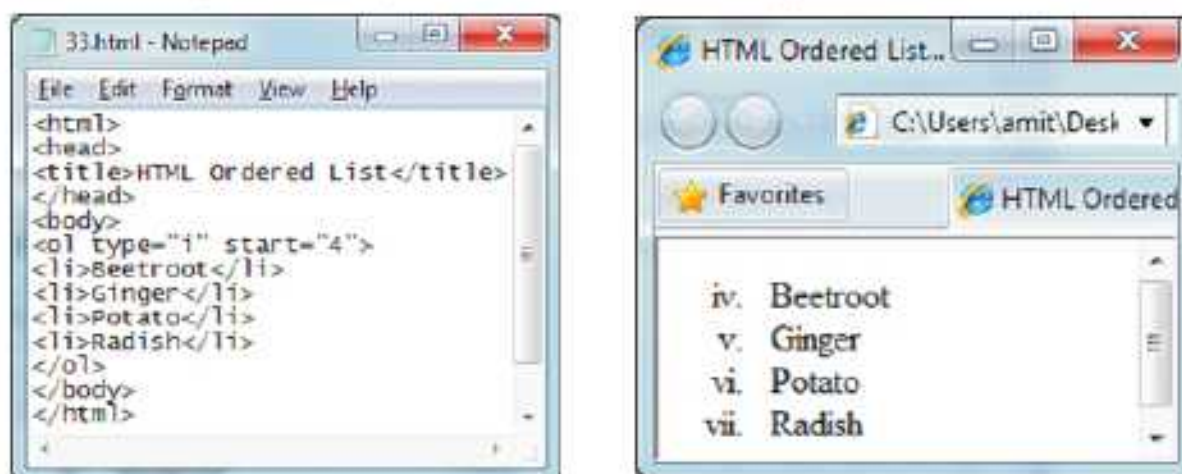
The list will be displayed.

The Start Attribute

You can use Start attribute for tag to specify the starting point of numbering you need. Following are the possible options:

- <ol type="1" start="4"> - Numerals start with 4.
- <ol type="I" start="4"> - Numerals start with IV.
- <ol type="i" start="4"> - Numerals start with iv.
- <ol type="a" start="4"> - Letters start with d.
- <ol type="A" start="4"> - Letters start with D.

Example : Following is an example where we used <ol type="i" start="4" >



HTML Definition Lists

HTML and XHTML support a list style which is called definition list where entries are listed, like in a dictionary or encyclopedia. The definition list is the ideal way to present a glossary, list of terms or other name/value list.

Definition List makes use of the following three tags.

- <dl> - Defines the start of the list
- <dt> - A term
- <dd> - Term definition
- </dl> - Defines the end of the list

Example :





COMPUTER ETIQUETTES

Never eat or drink near the computer.



Let's Implement.

A

Tick (✓) the correct answer.

- You can insert any image in your web page by using this tag.
a. <amg> ☐ b. <alt> ☐
c. <src> ☐ d. ☐
- This attribute specifies an alternate text for an image, if the image cannot be displayed.
a. olt ☐ b. clt ☐
c. alt ☐ d. clt ☐
- You can use PNG, JPEG or GIF image to specify correct image file name in this attribute.
a. crc ☐ b. arc ☐
c. src ☐ d. mrc ☐
- This tag will list items using plain bullets.
a. ☐ b. <dl> ☐
c. ☐ d. <dd> ☐
- This tag arranges your items in the same way as they are arranged in a dictionary.
a. <dl> ☐ b. ☐
c. ☐ d. <da> ☐
- You can use this attribute for tag to specify the type of numbering you like.
a. Type ☐ b. Start ☐
c. Alt ☐ d. Src ☐
- You can use this attribute for tag to specify the starting point of numbering you need.
a. Start ☐ b. Type ☐
c. Src ☐ d. Alt ☐
- This is the ideal way to present a glossary, list of terms or other name/value list.
a. Definition list ☐ b. Ordered list ☐
c. Unordered list ☐ d. Declare list ☐

B

Fill in the blanks.

- The _____ tag is an empty tag, which means that it can contain only list of attributes and it has no closing tag.

2. You can set image width and height based on your requirement using width and height _____.
3. You can specify border thickness in terms of _____ using border attribute.
4. _____ list will use different schemes of numbers to list your items.
5. An _____ list is a collection of related items that have no special order or sequence.
6. You can use type attribute for _____ tag to specify the type of bullet you like.
7. HTML and XHTML support a list style which is called _____.
8. Definition List makes use of following three tags: _____, _____ and _____.

C Write True or False.

1. You can specify width and height of the image in terms of either pixels or percentage of its actual size.
2. By default, image will have a border around it.
3. By default, image will align at the right side of the page.
4. You can use align attribute to set it in the center or right.
5. HTML offers web authors two ways for specifying lists of information.
6. All lists must contain one or more list elements.
7. If you are required to put your items in a numbered list instead of bulleted, then HTML unordered list will be used.
8. The numbering starts at one and is incremented by one for each successive ordered list element tagged with .

D Answer the following questions.

1. Define tag with example.
2. What is the use of src and alt attribute?
3. How can you set image alignment?
4. What do you mean by HTML list?
5. Define Unordered Lists with example.
6. What is the use of type attribute for tag?
7. Define Ordered Lists with example.
8. Define type attribute for tag.
9. What is the use of start attribute for tag?
10. Define the use of following tags.

• <dl>

• <dt>

• <dd>



Write HTML code to create the following web pages.

Windows Primer - *Learn how to do the following :*

1. [How do I Create a New Folder?](#)
2. [How do I Copy and Move Files and Folders?](#)
3. [How to Setup Desktop background?](#)
4. [How to change System Date and Time?](#)
5. [How to Format The Disk?](#)
6. [How to Search Files?](#)

How do I Create a New Folder?

- Start > Programs > Accessories > Windows Explore
- In the left pane, click on the drive or folder in which you wish to create new folder.
- File menu > New option > Folder option
- Type the required name of the folder and press Enter.

[Back to Top](#)



This is Jack. He's an odd little boy that just got an empty box for his birthday. He may look happy, but he's a little disappointed. Now he'll have to put a few things in the box.

Jack would really like.....

- Roller blades
- Magic tricks
- Anything Pokemon
- A motorized Lego set

You can get all this (and more) at an online toy store. Two very popular ones are [toyusa.com](#) and [eToys](#).

RACKING YOUR BRAIN

Life Skills and Values

Your friend wants to create a web page for his/her father's restaurant where he can display menu of the restaurant. How can you help him/her in doing so?

Just Have a Discussion

Communication

Divide the class into two groups and discuss on 'Different types of lists in HTML'.



TEACHER'S NOTE

The teacher should demonstrate to the students in detail how to create lists.



TABLES AND LINKS



Outlines

- HTML Tables
- HTML Image Links
- HTML Text Links
- HTML Email Links

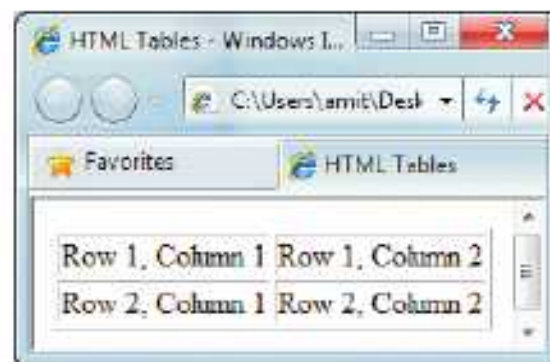
HTML Tables

The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.

The HTML tables are created using the `<table>` tag in which the `<tr>` tag is used to create table rows and `<td>` tag is used to create data columns or cells.

Example :

```
38.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML tables</title>
</head>
<body>
<table border="1">
<tr>
<td>Row 1, column 1</td>
<td>Row 1, column 2</td>
</tr>
<tr>
<td>Row 2, column 1</td>
<td>Row 2, column 2</td>
</tr>
</table>
</body>
</html>
```



Here **border** is an attribute of `<table>` tag and it is used to put a border across all the cells. If you do not need a border then you can use `border="0"`.

Table Heading

Table heading can be defined using `<th>` tag. This tag will be put to replace `<td>` tag, which is used to represent actual data cell. Normally, you will put your top row as table heading as shown below, otherwise you can use `<th>` element in any row.

Example :



```
<html>
<head>
<title>HTML Table Header</title>
</head>
<body>
<table border="1">
<tr>
<th>Name</th>
<th>Salary</th>
</tr>
<tr>
<td>Sunil Kumar</td>
<td>7000</td>
</tr>
<tr>
<td>Anil Gupta</td>
<td>9000</td>
</tr>
</table>
</body>
</html>
```

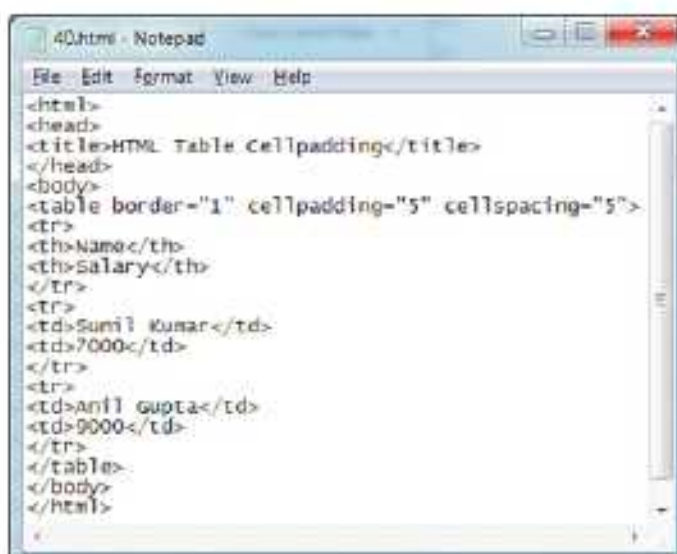


Name	Salary
Sunil Kumar	7000
Anil Gupta	9000

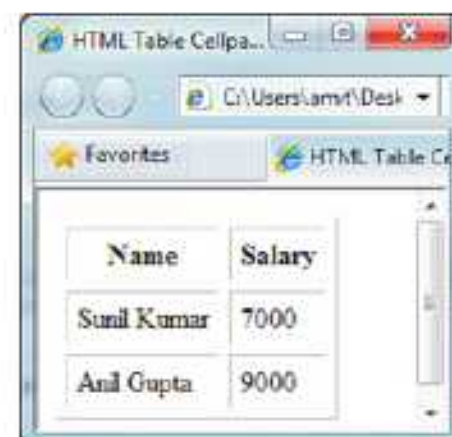
Cellpadding and Cellspacing Attributes

There are two attributes called *cellpadding* and *cellspacing* which you will use to adjust the white space in your table cells. The *cellspacing* attribute defines the width of the border, while *cellpadding* represents the distance between cell borders and the content within a cell.

Example :



```
<html>
<head>
<title>HTML Table Cellpadding</title>
</head>
<body>
<table border="1" cellpadding="5" cellspacing="5">
<tr>
<th>Name</th>
<th>Salary</th>
</tr>
<tr>
<td>Sunil Kumar</td>
<td>7000</td>
</tr>
<tr>
<td>Anil Gupta</td>
<td>9000</td>
</tr>
</table>
</body>
</html>
```



Name	Salary
Sunil Kumar	7000
Anil Gupta	9000

Colspan and Rowspan Attributes

You will use **colspan** attribute if you want to merge two or more columns into a single column. In a similar way, you will use **rowspan** if you want to merge two or more rows.

Example :

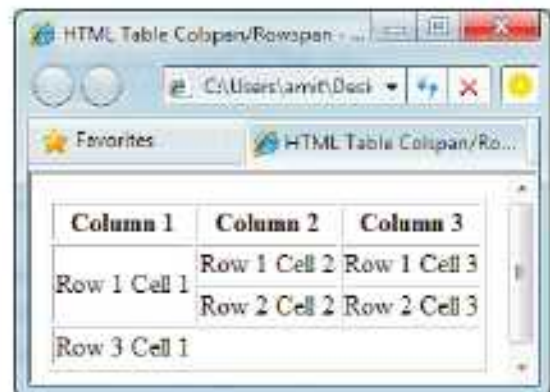
```
41.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table Colspan/Rowspan</title>
</head>
<body>
<table border="1">
<tr>
<th>Column 1</th>
<th>Column 2</th>
<th>Column 3</th>
</tr>
<tr>
<td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr>
<td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr>
<td colspan="3">Row 3 Cell 1</td></tr>
</table>
</body>
</html>
```

This will produce the following result:

Tables Backgrounds

You can set table background using one of the following two ways:

- **bgcolor** attribute— You can set background color for whole table or just for one cell.
- **background** attribute— You can set background image for whole table or just for one cell.



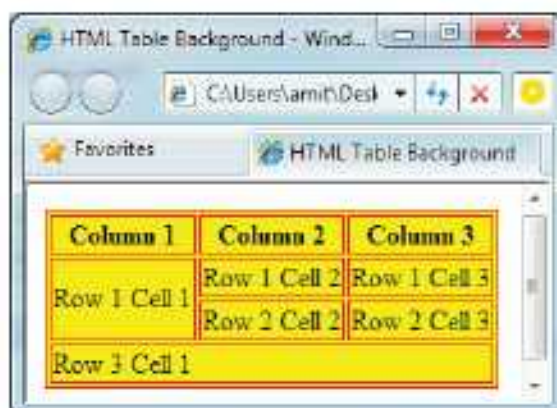
A screenshot of a web browser window titled "HTML Table Colspan/Rowspan". The address bar shows "CAUsers\amith\Desktop". The table displayed is as follows:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

You can also set border color also using **bordercolor** attribute.

Example :

```
42.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table Background</title>
</head>
<body>
<table border="1" bordercolor="red" bgcolor="yellow">
<tr>
<th>Column 1</th>
<th>Column 2</th>
<th>Column 3</th>
</tr>
<tr>
<td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr>
<td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr>
<td colspan="3">Row 3 Cell 1</td></tr>
</table>
</body>
</html>
```



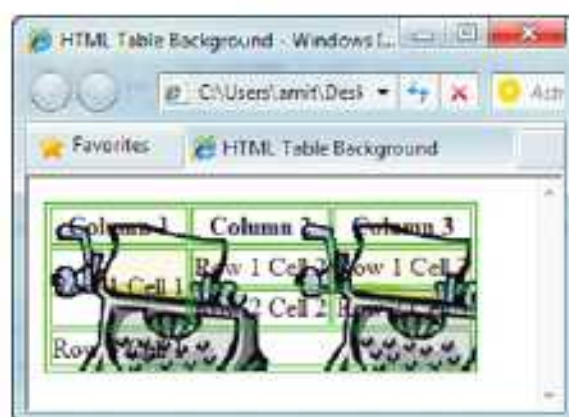
A screenshot of a web browser window titled "HTML Table Background - Wind...". The address bar shows "CAUsers\amith\Desktop". The table displayed has a yellow background and a red border. The table structure is identical to the previous one:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

This will produce the following result:

Here is an example of using **background** attribute. Here we will use an image available in images directory.

```
43.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML table background</title>
</head>
<body>
<table border="1" bordercolor="green" background="D:\Clipart\hoseanim\AG00003_.GIF">
<tr>
<th>Column 1</th>
<th>Column 2</th>
<th>Column 3</th>
</tr>
<tr>
<td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr>
<td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
<tr>
<td colspan="3">Row 1 Cell 1</td></tr>
</table>
</body>
</html>
```



This will produce the following result. Here background image did not apply to table's header.

Table Height and Width

You can set a table **width** and **height** using width and height attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

Example :

```
44.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table width/height</title>
</head>
<body>
<table border="1" width="400" height="150">
<tr>
<td>Row 1, column 1</td>
<td>Row 1, column 2</td>
</tr>
<tr>
<td>Row 2, column 1</td>
<td>Row 2, column 2</td>
</tr>
</table>
</body>
</html>
```

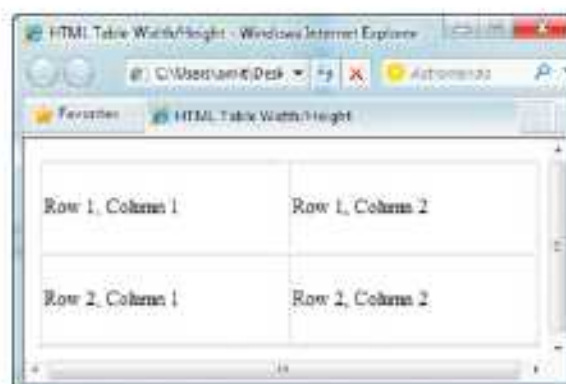


Table Caption

The **caption** tag will serve as a title or explanation for the table and it is shown at the top of the table. This tag is deprecated in newer version of HTML/XHTML.

Example :

```
45.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table Caption</title>
</head>
<body>
<table border="1" width="100%">
<caption>This is the caption</caption>
<tr>
<td>row 1, column 1</td><td>row 1, column 2</td>
</tr>
<tr>
<td>row 2, column 1</td><td>row 2, column 2</td>
</tr>
</table>
</body>
</html>
```

This will produce the following result:

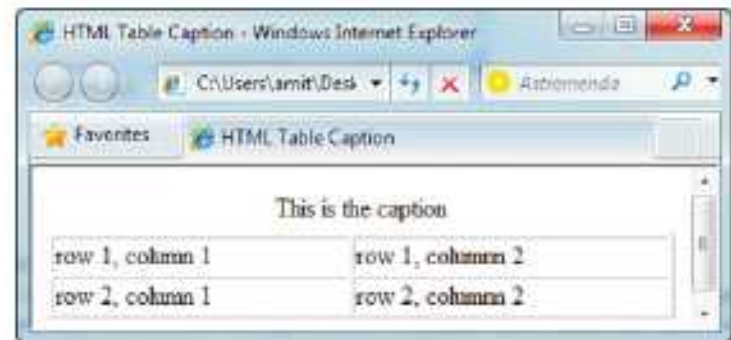


Table Header, Body and Footer

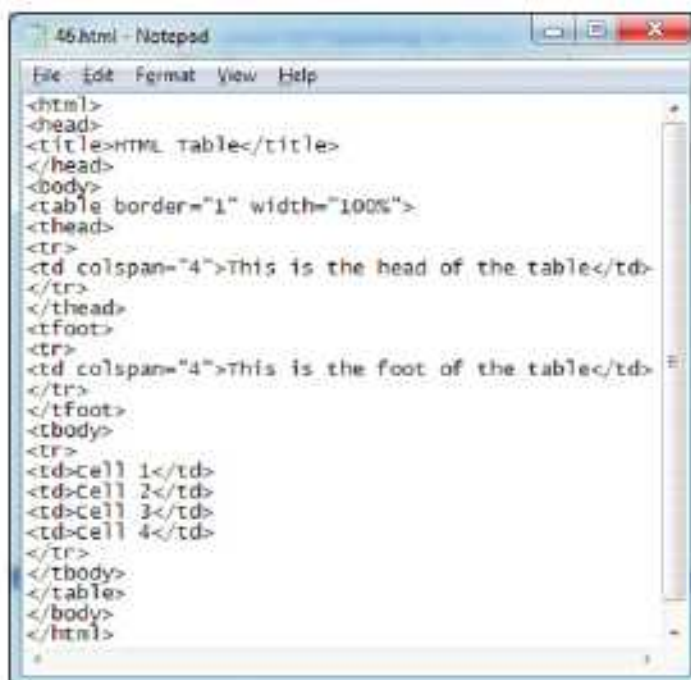
Tables can be divided into three portions: a header, a body and a footer. The head and foot are rather similar to headers and footers in a word-processed document that remain the same for every page, while the body is the main content holder of the table.

The three elements for separating the head, body and foot of a table are:

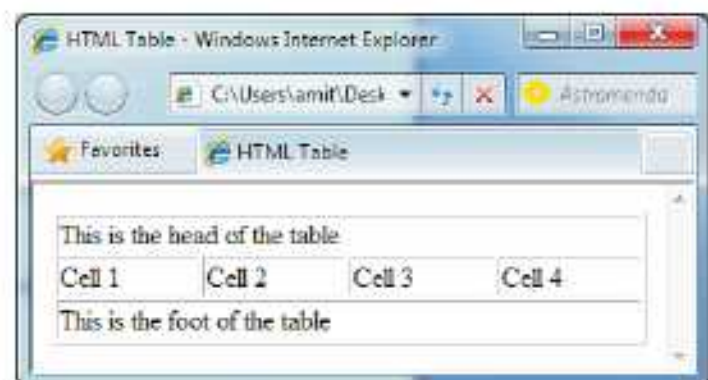
- `<thead>` - to create a separate table header.
- `<tbody>` - to indicate the main body of the table.
- `<tfoot>` - to create a separate table footer.

A table may contain several `<tbody>` elements to indicate different pages or groups of data. But it is notable that `<thead>` and `<tfoot>` tags should appear before `<tbody>`

Example :



This will produce the following result:



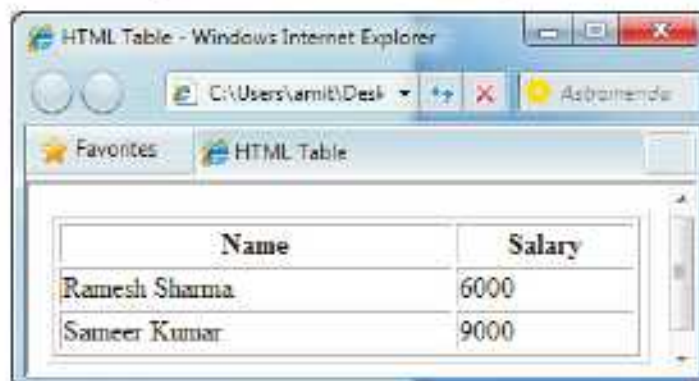
Nested Tables

You can use one table inside another table. Not only tables, you can use almost all the tags inside table data tag `<td>`.

Example : Following is the example of using another table and other tags inside a table cell.

```
47.html - Notepad
File Edit Format View Help
<html>
<head>
<title>HTML Table</title></head>
<body>
<table border="1" width="100%">
<tr><td><table border="1" width="100%">
<tr><th>Name</th> <th>salary</th> </tr>
<tr><td>Ramesh Sharma</td> <td>6000</td> </tr>
<tr><td>Sameer Kumar</td> <td>9000</td> </tr>
</table></td></tr>
</table>
</body>
</html>
```

This will produce the following result:



The screenshot shows a web browser window titled "HTML Table - Windows Internet Explorer". The address bar shows "C:\Users\amit\Desktop". The page content displays a table with two columns: "Name" and "Salary". The table contains two rows of data: "Ramesh Sharma" with a salary of "6000", and "Sameer Kumar" with a salary of "9000".

Name	Salary
Ramesh Sharma	6000
Sameer Kumar	9000

HTML Text Links

A webpage can contain various links that take you directly to other pages and even specific parts of a given page. These links are known as hyperlinks.

Hyperlinks allow the visitors to navigate between Websites by clicking on words, phrases, and images. Thus, you can create hyperlinks using text or images available on a webpage.

Linking Documents

A link is specified using HTML tag `<a>`. This tag is called **anchor tag** and anything between the opening `<a>` tag and the closing `</a>` tag becomes a part of the link and a user can click that part to reach to the linked document. Following is the simple syntax to use `<a>` tag.

```
<a href="Document URL" ... attributes-list>Link Text</a>
```

Example : Let's try the following example which links <http://www.tutorialspoint.com> at your page:

```
35.html - Notepad
File Edit Format View Help
<html>
<head>
<title>Hyperlink Example</title>
</head>
<body>
<p>Click the following link</p>
<a href="http://www.tutorialspoint.com" target="_self">Tutorials Point</a>
</body>
</html>
```

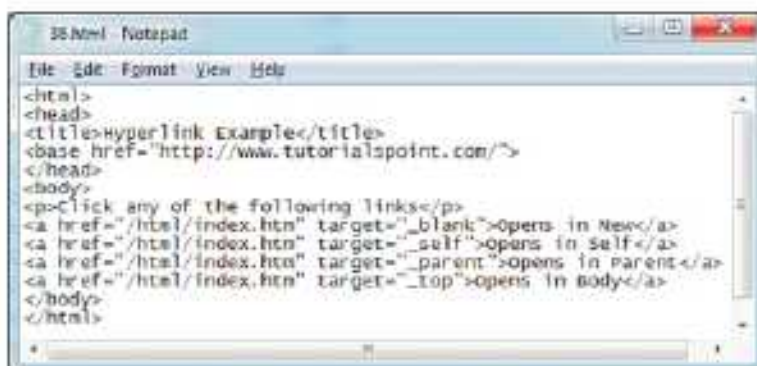
This will produce the following result, where you can click on the link generated **Tutorials Point** to reach to the home page of Tutorials Point.



The Target Attribute

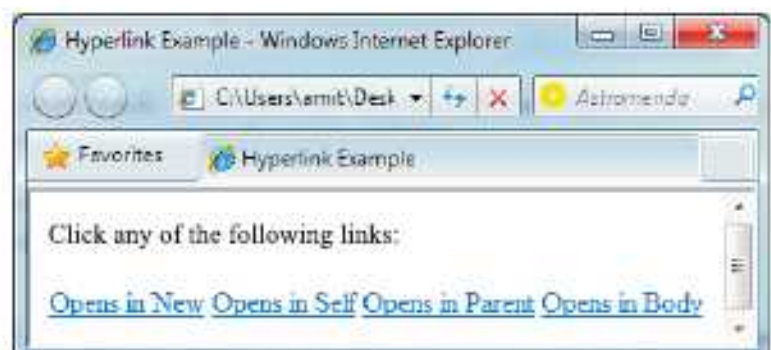
We have used Target attribute in our previous example. This attribute is used to specify the location where linked document is opened. Following are the possible options:

Option	Description
_blank	Opens the linked document in a new window or tab.
_self	Opens the linked document in the same frame.
_parent	Opens the linked document in the parent frame.
_top	Opens the linked document in the full body of the window.
_targetframe	Opens the linked document in a named <i>target frame</i> .



Example : Try the following example to understand the basic difference in few options given for target attribute.

This will produce the following result, where you can click on different links to understand the difference between various options given for target attribute.

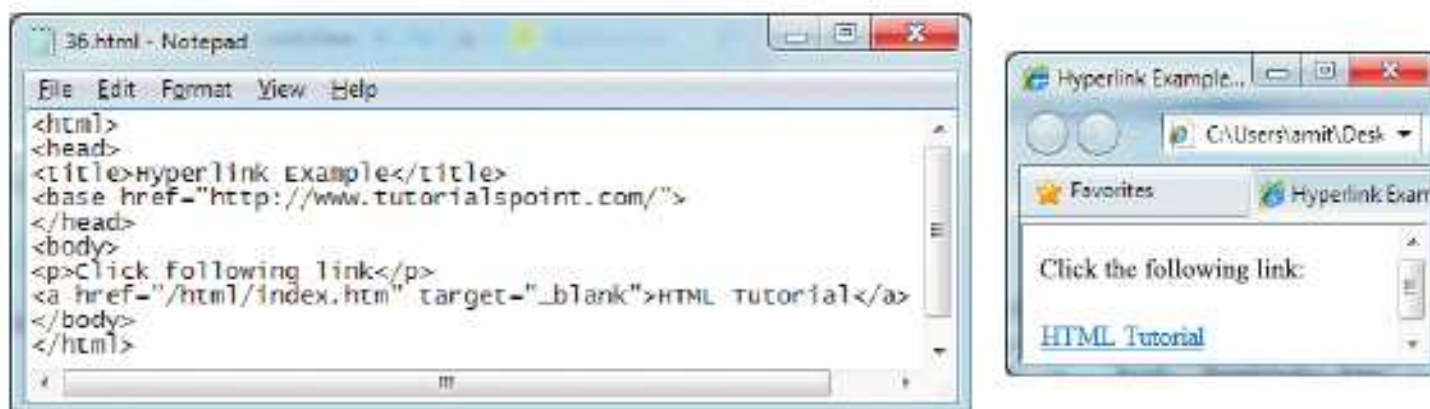


Use of Base Path

When you link HTML documents related to the same website, it is not required to give a complete URL for every link. You can get rid of it if you use `<base>` tag in your HTML document header. This tag is used to give a base path for all the links. So your browser will concatenate given relative path to this base path and will make a complete URL.

Example : Following example makes use of `<base>` tag to specify base URL and later we can use relative path to all the links instead of giving complete URL for every link.

This will produce the following result, where you can click on the link generated **HTML Tutorial** to reach to the HTML tutorial site. Now given URL `<a href="/html/index.htm">` is being considered as `<a href="http://www.tutorialspoint.com/html/index.htm">`.



Linking to a Page Section

You can create a link to a particular section of a given webpage by using the **name** attribute. This is a two step process.

First create a link to the place where you want to reach within a webpage and name it using `<a...>` tag as follows:

```
<h1>HTML Text Links <a name="top"></a></h1>
```

Second step is to create a hyperlink to link the document and place where you want to reach:

```
<a href="/html/html_text_links.htm#top">Go to the Top</a>
```

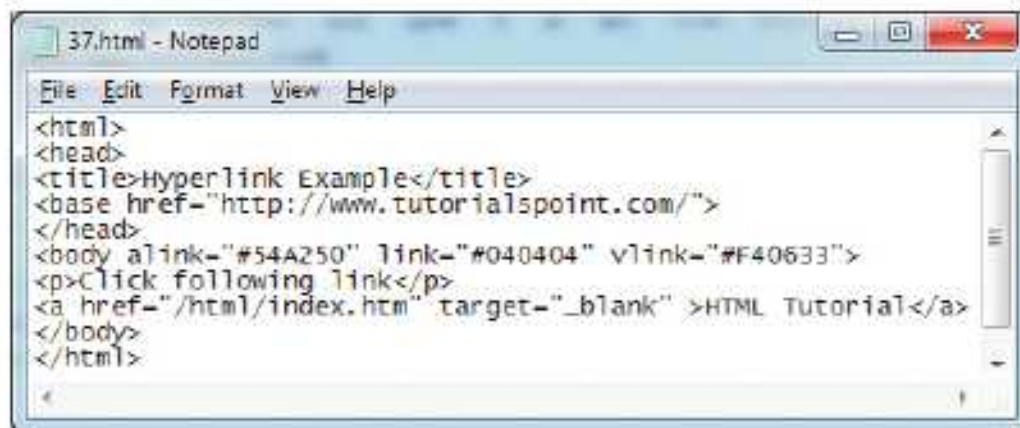
This will produce the following link, where you can click on the link generated 'Go to the Top' to reach the top of the HTML Text Link tutorial.

[Go to the Top](#)

Setting Link Colours

You can set colours of your links, active links and visited links using **link**, **alink** and **vlink** attributes of <body> tag.

Example: Save the following in test.htm and open it in any web browser to see how **link**, **alink** and **vlink** attributes work.

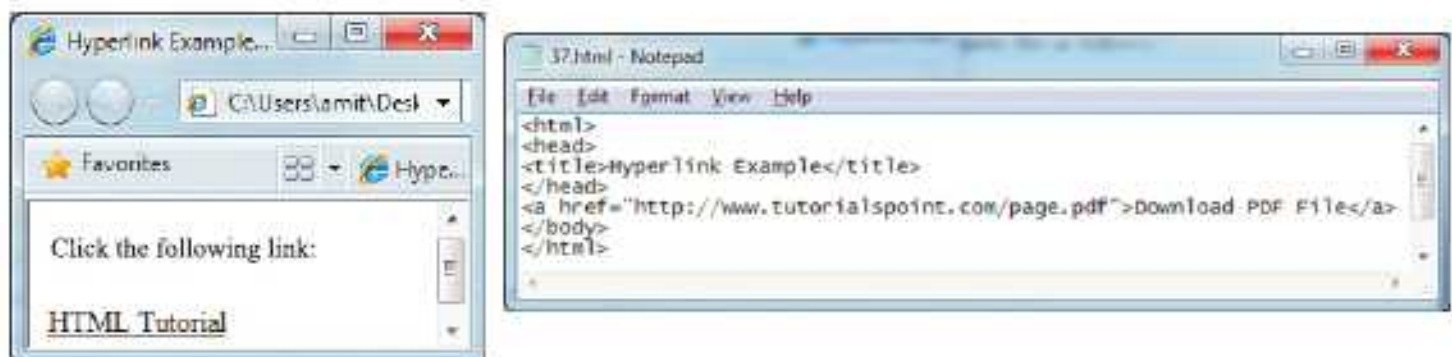


```
37.html - Notepad
File Edit Format View Help
<html>
<head>
<title>Hyperlink Example</title>
<base href="http://www.tutorialspoint.com/">
</head>
<body alink="#54A250" link="#040404" vlink="#F40633">
<p>Click following link</p>
<a href="/html/index.htm" target="_blank" >HTML Tutorial</a>
</body>
</html>
```

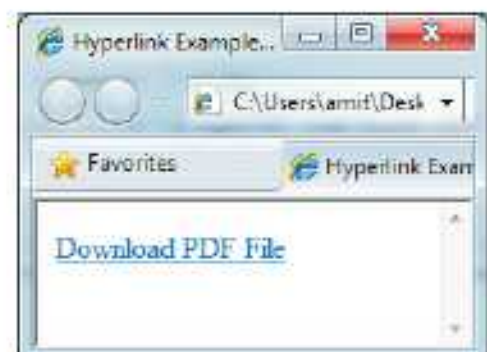
This will produce the following result. Just check colour of the link before clicking on it, next check its colour when you activate it and when the link has been visited.

Download Links

You can create text link to make your PDF or DOC or ZIP files downloadable. This is very simple, you just need to give complete URL of the downloadable file as follows:



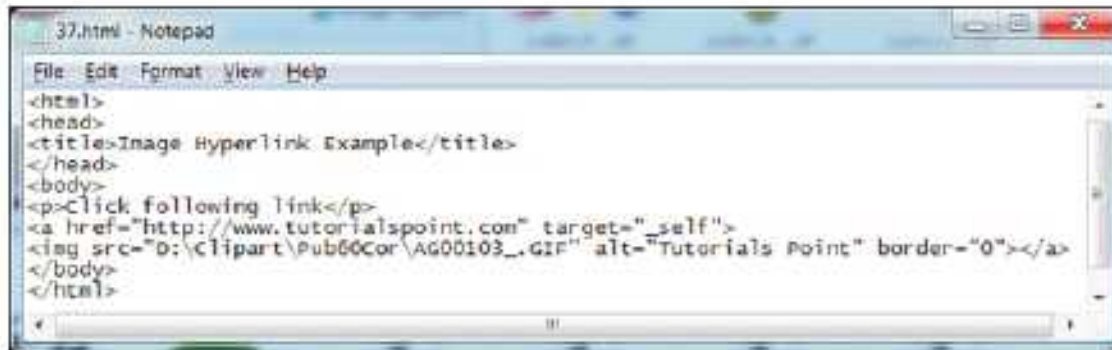
This will produce the following link and will be used to download a file.



HTML Image Links

We have seen how to create hypertext link using text and we have also learnt how to use images in our webpages. Now we will learn how to use images to create hyperlinks.

Example : It's simple to use an image as hyperlink. We just need to use an image inside hyperlink at the place of text as shown below :



This will produce the following result, where you can click on the images to reach to the home page of Tutorials Point.

This was the simplest way of creating hyperlinks using images.



HTML Email Links

It's not difficult to put an HTML email link on your webpage, but it can cause unnecessary spamming problem for your email account. There are people who can run programs to harvest these types of emails and later use them for spamming in various ways.

HTML Email Tag

HTML `<a>` tag provides you option to specify an email address to send an email. While using `<a>` tag as an email tag, you will use `mailto:email address` along with href attribute. Following is the syntax of using `mailto` instead of using `http`.

```
<a href="mailto:abc@example.com">Send Email</a>
```

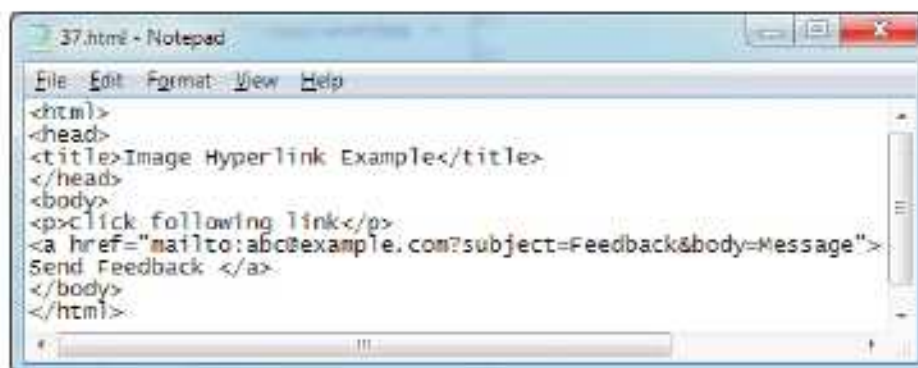
This code will generate the following link which you can use to send an email.

Send Email

Now if a user clicks this link, it launches one Email Client (like Outlook Express, etc.) installed on your user's computer. There is another risk to use this option to send an email because if the user do not have an email client installed on their computer, then it would not be possible to send email.

Default Settings

You can specify a default email subject and email body along with your email address. Following is the example to use default subject and body.



```
37.html - Notepad
File Edit Format View Help
<html>
<head>
<title>Image Hyperlink Example</title>
</head>
<body>
<p>click following link</p>
<a href="mailto:abc@example.com?subject=Feedback&body=Message">
Send Feedback </a>
</body>
</html>
```



This code will generate the following link which you can use to send an email.



COMPUTER ETIQUETTES

1. Hold both arms straight in front of the body at shoulder level.
2. Open your hands, palms down, and stretch your fingers as far apart from each other as possible.



Let's Implement.

A

Tick (✓) the correct answer.

1. They allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.
a. HTML links ☐ b. HTML lists ☐
c. HTML tabs ☐ d. HTML tables ☐
2. This represents the distance between cell borders and the content within a cell.
a. Cellpadding ☐ b. Cellspacing ☐
c. Cellcolspan ☐ d. Cellrowspan ☐

3. This attribute is used when you want to merge two or more columns into a single column.

a. Caption	☐	b. Colspan	☐
c. Rowspan	☐	d. Hyperlink	☐
4. This attribute is used to set background color for the whole table or just for one cell.

a. Bgcolor	☐	b. Caption	☐
c. Rowspan	☐	d. Background	☐
5. This tag will serve as a title or explanation for the table and it shows up at the top of the table.

a. Heading	☐	b. Header	☐
c. Footer	☐	d. Caption	☐
6. It allows the visitors to navigate between Websites by clicking on words, phrases and images.

a. Hyperpage	☐	b. Hypertext	☐
c. Hyperweb	☐	d. Hyperlinks	☐
7. You can create a link to a particular section of a given webpage by using this attribute.

a. Header	☐	b. Name	☐
c. Footer	☐	d. Body	☐
8. This tag is used to indicate the main body of the table.

a. <ttext>	☐	b. <thead>	☐
c. <tbody>	☐	d. <tfoot>	☐

B Fill in the blanks.

1. The _____ tag is used to create table rows and _____ tag is used to create data cells.
2. There are two attributes called _____ and _____ which you will use to adjust the white space in your table cells.
3. The _____ attribute defines the width of the border.
4. You will use _____ if you want to merge two or more rows.
5. _____ attribute is used to set background image for the whole table or just for one cell.
6. Tables can be divided into three portions: _____, _____ and _____.
7. The _____ attribute is used to specify the location where linked document is opened.

8. A link is specified using HTML tag _____.

C

Write True or False.

1. The HTML tables are created using the <tt> tag.
2. Border attribute of <table> tag is used to put a border across all the cells.
3. You can also set border colour using bgcolor attribute.
4. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.
5. <head> tag is used to create a separate table header.
6. <foot> tag is used to create a separate table footer.
7. You can set colours of your links, active links and visited links using link, alink and vlink attributes of <body> tag.
8. You can create text link to make your PDF or DOC or ZIP files downloadable.

D

Answer the following questions.

1. Differentiate between Cellpadding and Cellspacing attributes.
2. Define Colspan and Rowspan attributes with example.
3. What are the methods to set table background? Explain.
4. Define Table Caption with example.
5. What do you mean by Table Header, Body and Footer?
6. What are Hyperlinks?
7. Write the use of Target attribute.
8. What is the use of Base Path?
9. Write the difference between image and email link.



Refreshment Corner



► **Create a web page using the html code.**

S. No.	Name	Images
1.	TAJ MAHAL	

2.	WATER FALL	
3.	FLOWERS	
4.	BIRDS	

Competency Based Questions

1. Sarthak wants to merge two or more columns into a single column. Which attribute should he use for this purpose?
2. Arinjay wants to adjust the white space in his table cells. Which attributes should he use for this purpose?

RACKING YOUR BRAIN

Life Skills and Values

While creating links in your website, is it ethical to connect these links with any other website? Give reasons to for your answer.

Just Have a Discussion

Communication

Discuss the different tags of HTML in the class in detail.



TEACHER'S NOTE

Explain how to create webpages to students in the class.



ADVANCED FEATURES OF KRITA



Outlines

- Layers
- Animation in Krita
- Creating Animation

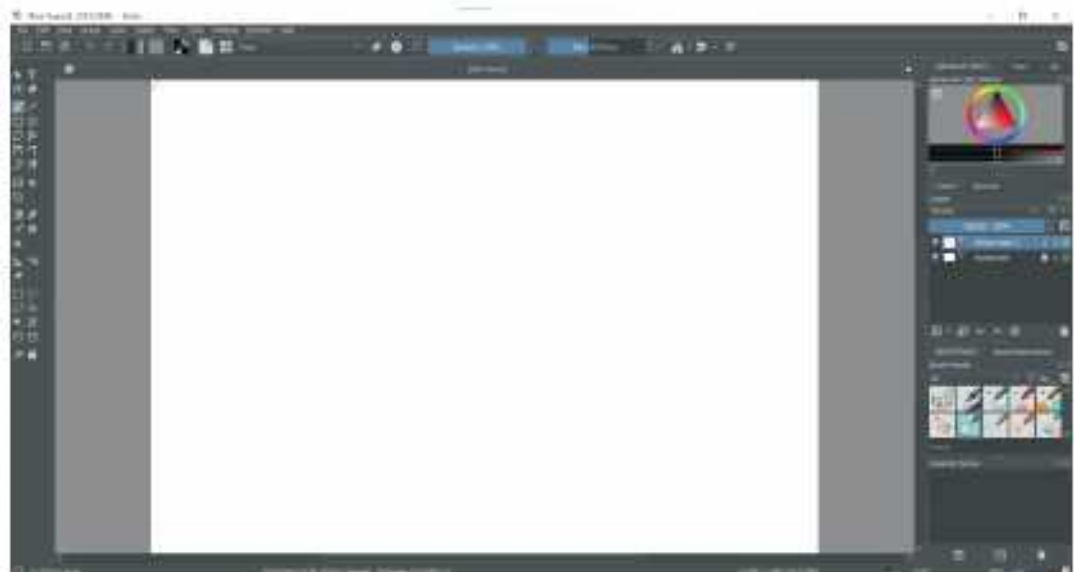
Krita is a photo-editing software. With a photo-editing software, a simple looking picture can be enhanced to an extent that it looks as if captured by an expert photographer.

The word 'Krita' comes from the Swedish language; where 'Kri' means 'crayons' and 'Rita' means 'to draw'. Krita is an open-source software designed for digital drawing, photo editing and 2D animation.

Layers

Layers are transparent sheets containing objects which are stacked on top of each other, so that the individual properties of an object are preserved. The layers help to edit an object without affecting the other objects.

To work on a particular object, it is essential to first select the layer on which the object has been added or created. To access a particular object, you need to first access its layer. All the layers of an image are shown in the Layers panel.

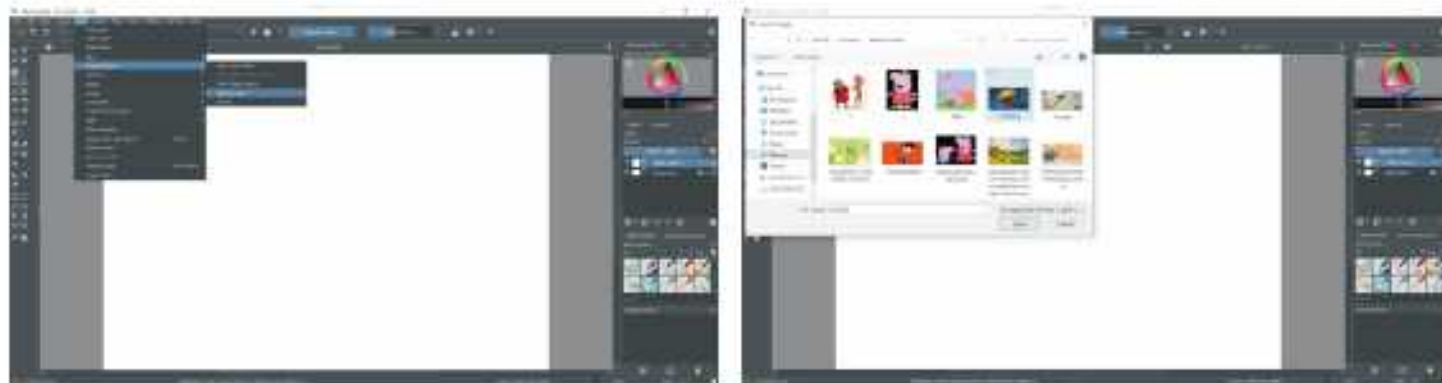


Managing Layers

Managing layers means to create, delete, duplicate, hide, rename and insert an object as a new layer.

Creating a New Layer

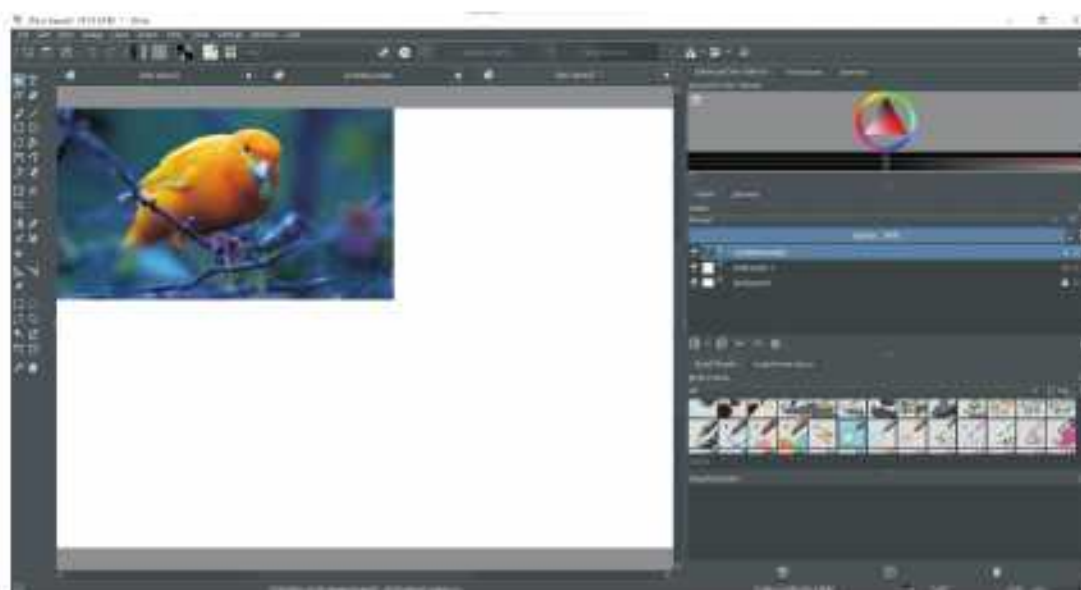
To create a new layer, click on the Add Layer button pic-2 at the bottom left corner of the Layers docker. When you click on this button, a new layer is added and shown in the Layers docker.



Duplicate an Existing Layer

You can duplicate a layer along with its contents and properties. To duplicate a layer, follow these steps:

1. First select the layer which you want to duplicate.
2. Click on the **Duplicate layer** or **mask button** at the bottom of the Layers docker.



Adding Objects As Layers

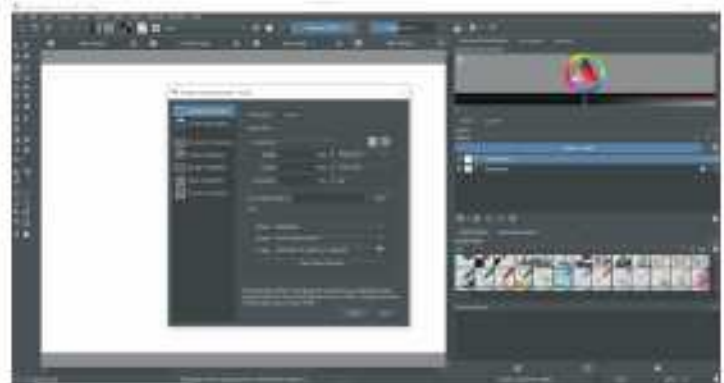
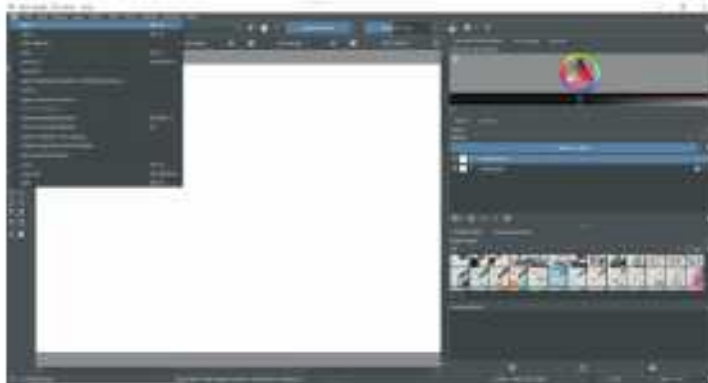
To add objects as layers, follow these steps:

1. Click on the **Layers** menu.

2. Then, click on the **Import/Export** option.
3. Select the **Import/Export** option. The **Import Image** dialog box appears.



4. Select the image or object that you want to insert in the workspace.
5. Once the image is inserted in the workspace, the Layers panel will now show an unnamed layer as shown below.



You can always change the name of the layer in the Layers panel and give it a name of your choice.

Showing and Hiding Layers

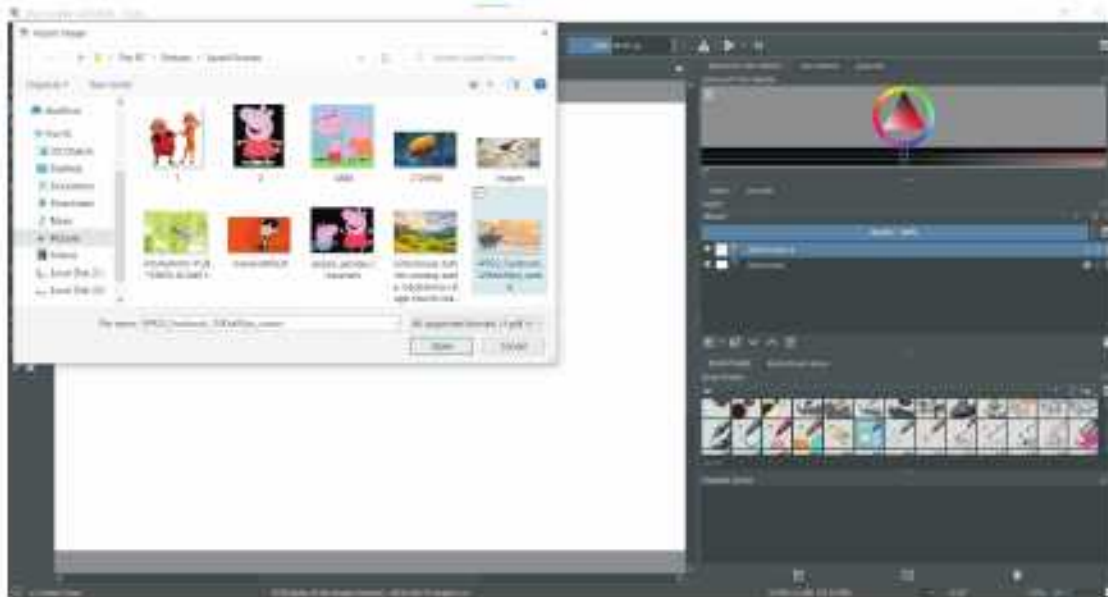
While working with an image with the different layers, you have the option to Hide/Show different layers. This helps you to focus on the elements or objects of an individual layer at a time by hiding all the other layers.

- To hide or show the layers, click on the **Eye** icon shown before the layer name in the Layers panel.

Renaming a Layer

A layer can be renamed by you as per your need. You can rename a layer using the following steps:

1. Double-click on the layer you want to rename in the Layers panel.
2. Type the new name and press the **Enter** key.



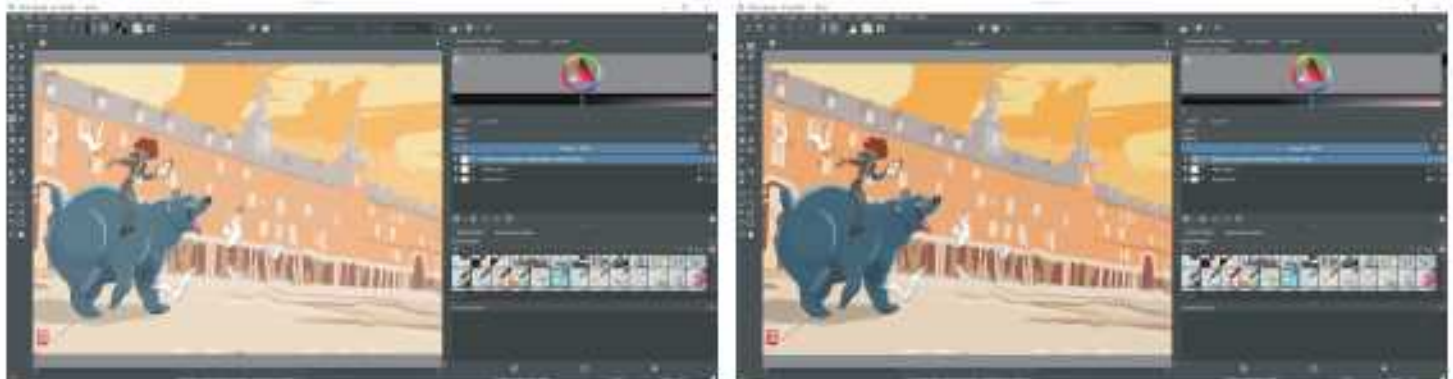
Deleting a Layer

In case, you do not wish to work on a particular layer or had added a layer that you do not want now, then you can remove that layer by simply deleting it from the Layers panel.

- You can delete a layer by selecting the layer in the Layers panel and then by pressing the **Delete** key from the keyboard.

or

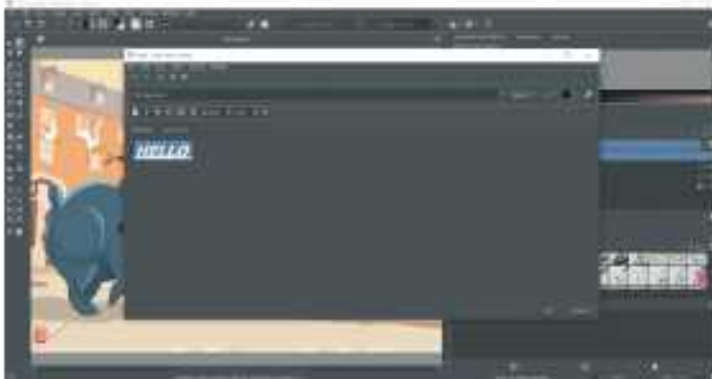
- Right-click on the layer you want to delete. From the Context menu, select the **Remove Layer** option.



Moving a Layer

Layers can be moved up and down in the Layers docker. To move a layer, follow these steps:

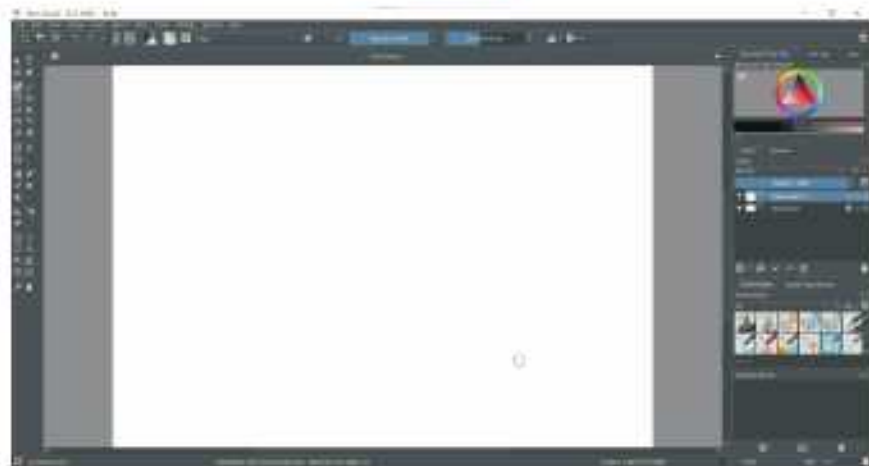
1. Select the layer in the **Layers** docker.
2. Click on the **Move layer or mask up** or **Move layer or mask down** button at the bottom of the layers docker.



Working with Multiple Layers

Now that, you have understood the concept of layers, let us create a project where you will work with the multiple layers.

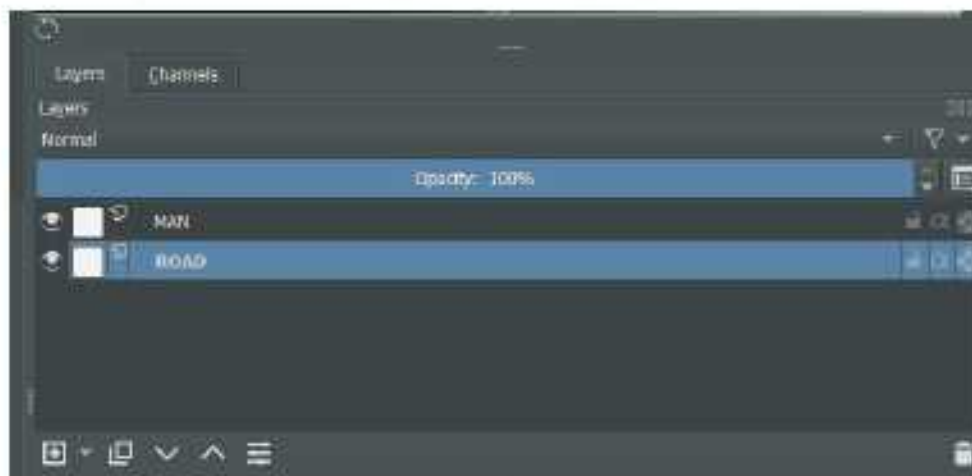
1. Create a new file by clicking on the **File New** option.
2. The Create new dialog box is displayed. Click on the **Create** button to create a new file.



3. To add an image, click on the **Layers** menu and then click on the **Import/Export** option. Select the **Import Layer** option. This will show the **Import Image** dialog box.
4. Select the image object you want to insert in the workspace.



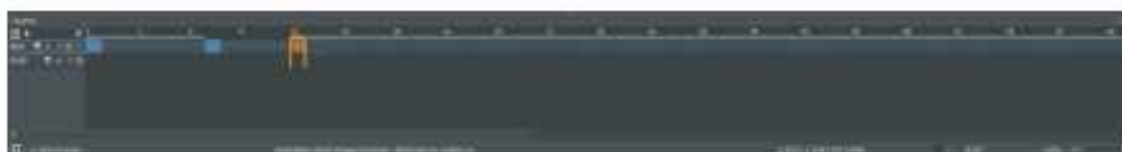
- From the Toolbox, select the [Transform a layer](#) pic-22 or [Selection tool](#). Using the mouse, drag around the border of the image to cover the workspace area. Observe that the new image is added as a layer in the [layers](#) panel.



- Now, select the [Text](#) tool and place it on the image. Drag the borders to increase the size of the text box.
- The [Edit Text](#) dialog box will be displayed. Add the text in the box and click on the [Save](#) button. You can change the font style and size, etc., using the formatting options available in the [Edit Text](#) dialog box.
- After making the changes to the text, click on the [Close](#) button to quit the [Edit Text](#) dialog box.



The text is now added to the image. Observe that a new layer is also added to the [Layers](#) panel.



Animation in Krita

Animation is a method by which images are manipulated in a manner so that they appear as moving objects. For this, the workspace needs to be changed to Animation mode and the Animations docker need to be added in Krita. To add the animation dockers, follow these steps:

- Click on the [Choose Workspace](#) button present on the top right side of the Toolbar. This will display the workspaces panel.

2. Here, select the [Animation](#) option.

You will see the animation dockers and stage on your workspace as shown in the figure.



Before creating the animation, let us first know about the components of the Animation workspace.

The Workspace

The workspace is the arrangement of the various Krita elements, such as Tools Panel (containing different dockers), Stage, Timeline and so on.

Stage

The stage is a drawing board where you can create graphics, animation and make modifications too. It is a large space present in the center of the workspace.

Tools Panels

The tools panel consists of the various tools to draw, paint and modify the objects in the workspace. It is divided into three parts— Toolbox, Layers and Animation.

- **Toolbox:** This docker has the different tools that are used for drawing, painting and selecting the objects.
- **Layers:** This docker is used to manage all the layers used in the animation.
- **Animation:** This docker consists of the controls that are used for presenting the animation.

Timeline

The timeline is a rectangular window, which is present at the bottom of the stage. It is the area where one controls the sequencing and timing of the graphics and the other elements of animation. The major components of the timeline are [Layers](#), [Frames](#) and [Playhead](#).

Frames

Animation in Krita is created with the help of frames. Each frame is defined as the little rectangular cell that appears on the timeline. Every fifth frame displays a number that you can find on the top of the timeline.

Keyframes

Keyframes are special types of frames, where you define some changes to an object's properties for an animation, like position, colour, shapes, etc.

Layers

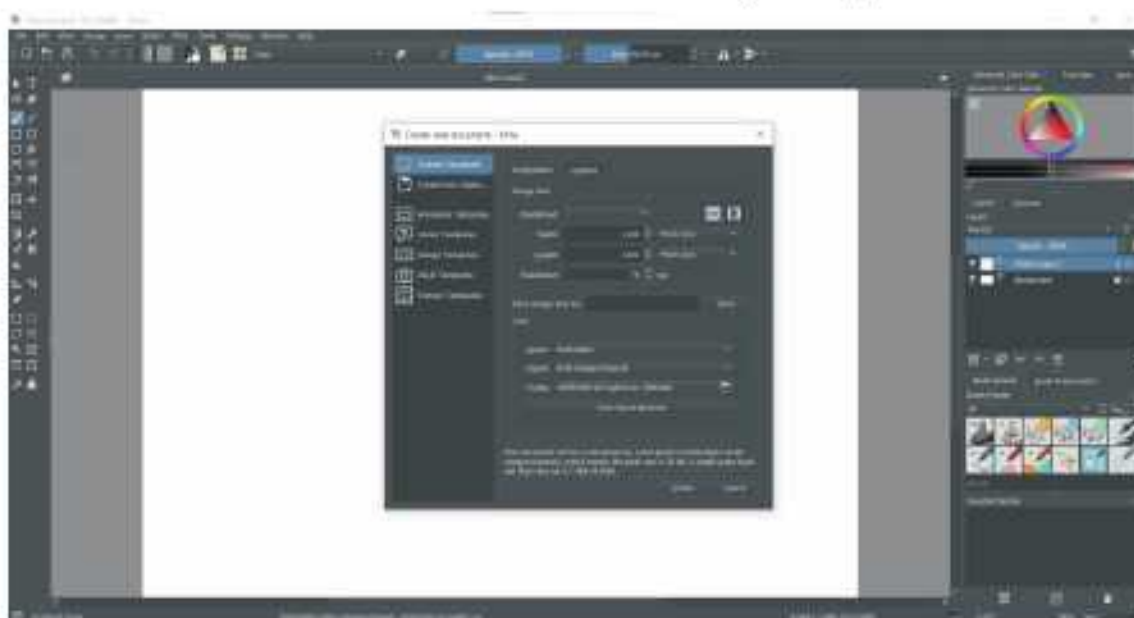
Layers are like transparent sheets stacked on top of one another; each containing a different image that appears on the stage. You can draw and edit the objects on one layer without affecting the objects on another layer.

Creating Animation

Let us create the first animation in Krita: **A jumping man on the road**. For this, follow the steps given below:

Creating a New Document: You will need two layers to make a jumping man on the road: one for the road and another for the man. So, create a new document with two layers. To do so, follow these steps:

- Click on **File> New**. The **Create new document** dialog box appears.

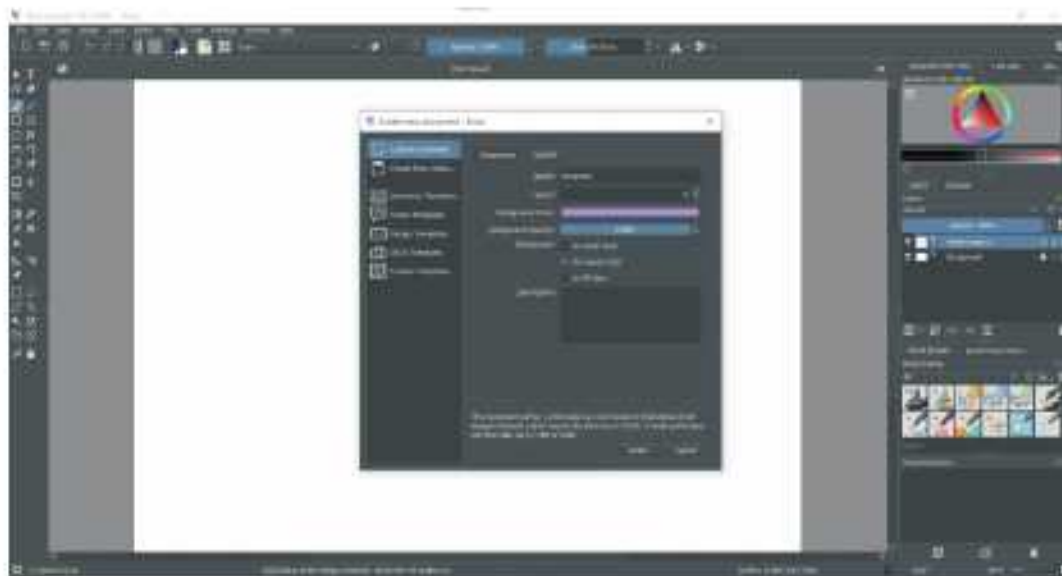


- In the **Create new document** dialog box, add the required values as given below:

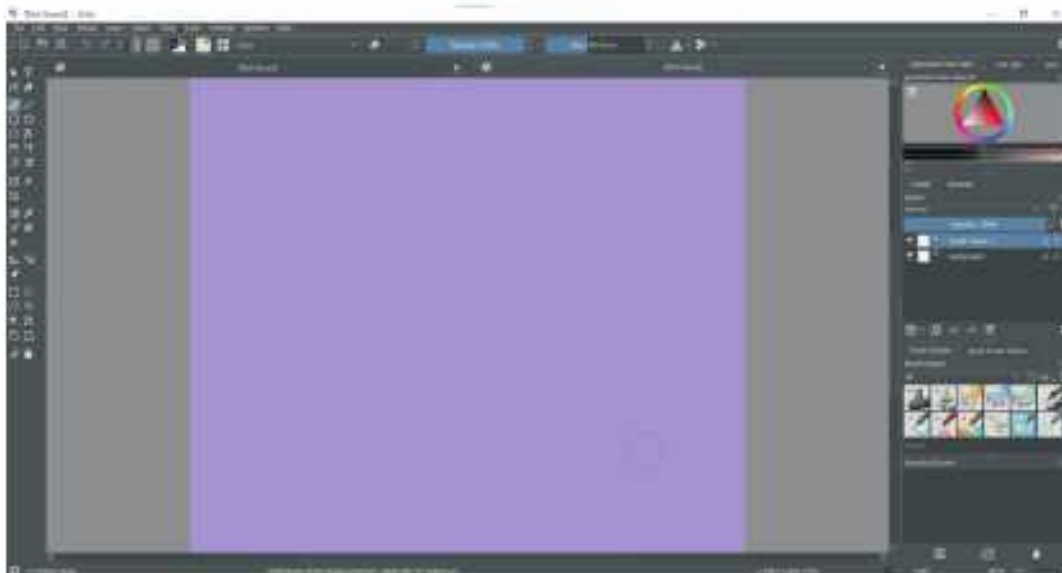
Width	1200 Pixels (px)
Height	1024 Pixels (px)
Resolution	74 ppi

- Now, click on the Content tab and update the values as given below:

Layers	2
Background Color	Select any colour of your choice
Background Opacity	100%
Background	As canvas color



- After updating all the required values, click on the Create button. This will **create** a New Krita document as shown in the figure.



Renaming Layers

For better understanding, rename the layers first. Rename Layer1 as Road and rename Layer2 as Man. You can rename a layer using the following steps:

- Double-click on the layer you want to rename in the **Layers** panel.
- Type the new name and press the **Enter** key.



Adding Animation

To add animation, follow these steps:

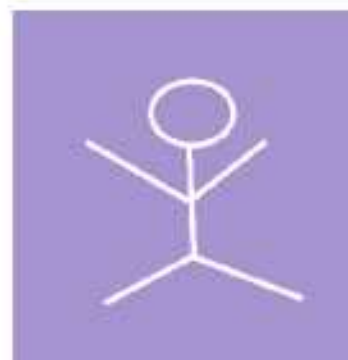
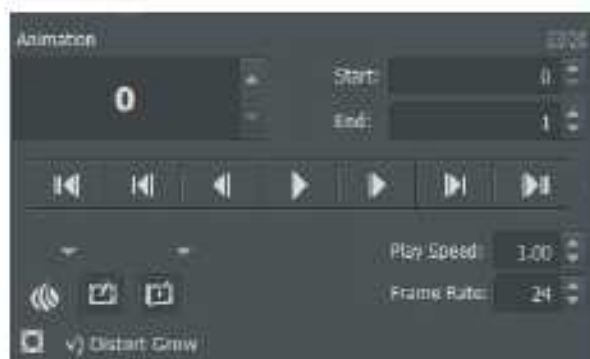
- Before you start drawing on the Road layer, make sure you hide the other layer(s). For example, to hide the Man layer simply click on the  icon.
- Now, select the layer in which you want to draw an object.
- Select any drawing tool from the Toolbox and draw an image that looks similar to the one given below in the work area.
- Click on the Road layer and draw a horizontal line using the Line tool.
- Now, the next thing to do so is add another frame in the timeline. To add a blank frame in the timeline, click on the **Add blank frame**  button.
- Repeat the previous step and add new frames at three (0, 6, 12) points, as shown in the figure.



- Click on each new frame and draw the images with the drawing tools.



- Now, make both the layers visible and add the following information in the **Animation** panel as shown.
- To view the animation, click on the **Play**  button.



Properties of Animation Panel

Start	First frame of the animation
End	Last frame of the animation
Play Speed	The speed at which the animation will be played in the preview mode
Frame Rate	The speed at which the animation will be played when embedded with the other apps



COMPUTER ETIQUETTES

Children should not use the computers for more than one hour per day. This can be in two sessions and preferably for educational purposes.



Let's Implement.

A Tick (✓) the correct answer.

- The word 'Krita' comes from the _____ language.
a. Spanish ☐ b. Swedish ☐
c. French ☐ d. English ☐
- Click on the _____ workspace from the Workplaces panel to add the animation controls.
a. Animation ☐ b. Timeline ☐
c. Keyframes ☐ d. Frames ☐
- _____ controls the sequencing and timing of the graphics and the other elements of animation.
a. Keyframe ☐ b. Frame ☐
c. Layer ☐ d. Timeline ☐
- _____ are like a stack in which the objects are placed one over the other or side-by-side.
a. Layers ☐ b. Import ☐
c. Both a and b ☐ d. None of these ☐
- To hide or show the layers, click on the _____ icon.
a. Layers ☐ b. Export ☐
c. Import ☐ d. Eye ☐

6. Create a new file by clicking on the _____ option.

a. File

☐

b. New

☐

c. Open

☐

d. None of these

☐

7. _____ on the layer you want to delete.

a. Left-click

☐

b. Double-click

☐

c. Single-click

☐

d. Right-click

☐

B Fill in the blanks.

1. Double-click on the layer you want to _____.

2. Layers can be moved up and down in the Layers _____.

3. You can add animation controls to the workspace by selecting the _____ option from the Workspaces panel.

4. A _____ is a special frame that enables you define some changes to an object's properties.

5. _____ are like a stack in which the objects are placed one over the other or side-by-side.

6. You can use the _____ tool to add text in the image.

7. When an image is inserted in the workspace, the Layers panel shows an _____ layer.

C Write True or False.

1. Krita is used for creating 3D animation.

2. Krita supports working with multiple layers.

3. You can rename a layer as per your need.

4. Once a layer is added in the Layers panel, you cannot delete it.

5. To access a particular object, you first need to access its layer.

6. The layers help to edit an object without affecting the other objects.

7. We cannot duplicate a layer along with its contents and properties.

D Answer the following questions.

1. Differentiate between a frame and a keyframe.

2. What is the significance of timeline in Krita?

3. What are layers used for?

4. Explain the concept of layers.

5. Write down the steps to hide/show layers in Krita.

6. What are the three parts of tools panel? Explain them.

7. Write the steps to rename a layers.



Refreshment Corner



- ▶ Draw and colour the logo of the Krita in the space given below.



Practical Session



Experiential Learning

- ▶ Create a beautiful scene of nature in Krita and use animation in it.

Competency Based Questions

1. Ashi has created an animation in Krita. While playing it, she realised that the speed of the animation is slow, she wants to increase the speed of the animation. Suggest the animation property that she can use to do so.
2. Sakshi is editing her pictures using Krita. She has added multiple in a single layer. After finishing, she realised that she does not want to use the layer with multiple objects. Suggest what Sakshi can do to get rid of the same layer.

RACKING YOUR BRAIN

Life Skills and Values

What will you do if you are unable to work with Animation in Krita and your friend knows it well? Will you ask your friend for help or will you prefer taking help from your teacher? Justify your answer.

Just Have a Discussion

Communication

Discuss how to create layers in Krita in the class in detail.



TEACHER'S NOTE

- The teacher should demonstrate to the students how to create layers in Krita.
- Demonstrate how to create animation in Krita to students in detail.



ITERATIVE STATEMENTS IN PYTHON



Outlines

- The For Statement
- Infinite Loop
- The While Statement
- Jumping Statements

Children! In the previous class, you reviewed the basic concepts of Python, such as operators and conditional statements. Now, you will learn about the iterative statements in detail. In a program, there are many situations where we need to repeat same set of tasks again and again. In Python, the statements that are used to repeat a set of instructions are called **iterative** or **looping** statements. Looping statements are very useful and necessary for developing applications.

Python provides two types of looping statements:

1. for
2. while

Let's discuss these constructs in brief.

The For Statement

The for statement executes a simple or compound statement for a fixed number of times. The syntax of the for statement is given below:

for <variable> in <iterator>:

Statements

Program 1: To print all the days of a week

```
Python 3.11.5 - C:\Users\hp\AppData\Local\Programs\Python\Python311\15.py (3/115)
File Edit Format Run Options Window Help
#Program to print all the days of the week
days = ('Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday')
for x in days:
    print(x)
```

You will get the following output:



```
Python 3.11.5 (tags/v3.11.5:ccc6ba5, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
----- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py -----
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
>>>
```

Program 2: To print a pattern using for loop.



```
15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to print a pattern using for loop

for a in range(1,10):
    print('*'*a)
```

You will get the following output:



```
Python 3.11.5 (tags/v3.11.5:ccc6ba5, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py
*
**
***
****
*****
*****
*****
*****
*****
>>>
```

The 'in' Operator

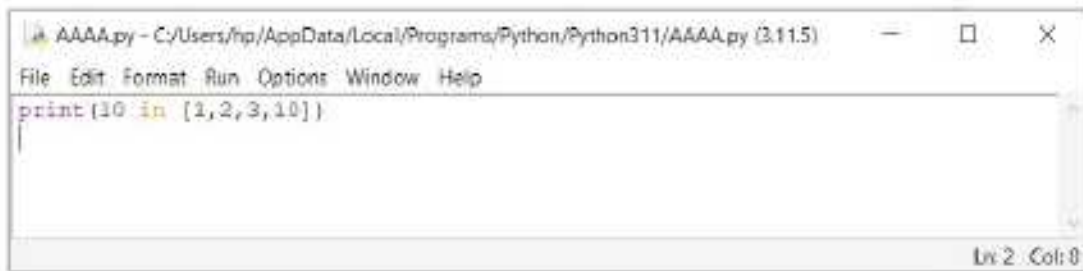
The 'in' operator is used to check if a given value exists in the sequence or not. It evaluates to be True if it finds a value in the specified sequence, else it returns False.

Syntax:

Value in [sequence]

For example, if you want to check whether the number 10 is present in a given sequence of numbers, [1, 2, 3, 10], execute the following code:

Program 3:



```
AAAA.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/AAAA.py (3.11.5)
File Edit Format Run Options Window Help
print(10 in [1, 2, 3, 10])
Ln 2 Col: 0
```

Output:



```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:ccc6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/AAAA.py =====
True
>>>
Ln 6 Col: 0
```

Program 4:



```
AAAA.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/AAAA.py (3.11.5)
File Edit Format Run Options Window Help
for i in (10, 20, 30, 40):
    print(i)
Ln 3 Col: 0
```

Output:



```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:ccc6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
- RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/AAAA.py
10
20
30
40
>>>
Ln 9 Col: 0
```

Using the Range() Function

The range() function is an in-built function of Python. This function generates a list which is a sequence type. A sequence is a succession of values bound together by a single name. The range() function is used in the for loop to iterate over the numbers.

The general syntax of using the range() function is given below.

Range(n) : Generates a set of whole numbers starting from 0 to (n-1). Example: range(6) is equivalent to [0, 1, 2, 3, 4, 5].

Range(start, stop): Generates a set of whole numbers starting from 'start' to 'stop-1'.

Example: range(3, 9) is equivalent to {4, 5, 6, 7, 8}.

Range(start, stop, step_size) : By default, the value of the step_size=1 and numbers generated with a difference of 1. However, we can generate numbers by specifying the value of step_size according to our requirement.

Example: range(1, 14, 2) is equivalent to [1, 3, 5, 7, 9, 11, 13].

Program 5: To print your name nine times using for loop with range() function.

```
*15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to print your name nine times using for loop with range() function.

name = input("Enter your name: ")
for i in range(0,9):
    print(name)
```

On running the above program, we get the following output:

```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py
Enter your name: Preeti
Preeti
Preeti
Preeti
Preeti
Preeti
Preeti
Preeti
Preeti
Preeti
>>>
```

Program 6: To print multiplication table of an input number.

```
15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to print multiplication table of an input number.

num = int(input("Enter a number: "))
for i in range(1,11):
    print(num, 'X', i, "=", num*i)
```


On running the program, we get the following output:



```
Python 3.11.5 (tags/v3.11.5:ccce5ba9, Aug 24 2023, 14:58:14) [MSC v.1936 64-bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py =====
Enter a number: 10
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
>>>
```

The While Statement

The while statement executes a set of statements repeatedly, until the logical expression evaluates to be true. When the condition becomes false, the control comes out of the loop. The syntax of while statement is given below.

Syntax:

while (test expression) :

Statements

Program 7: To demonstrate the use of a while loop.



```
15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to demonstrate the use of a while loop.

i = 0
a = input('Enter a string: ')
while (i < len(a)):
    if (a[i] == 'a' or a[i] == 'A'):
        i += 1
        continue
    print('Current Letter: ', a[i])
    i += 1

```

You will get the following output:



```
Python 3.11.5 (tags/v3.11.5:ccce5ba9, Aug 24 2023, 14:58:14) [MSC v.1936 64-bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py =====
Enter a string: Hello
Current Letter: H
Current Letter: e
Current Letter: l
Current Letter: l
Current Letter: o
>>>
```

Any non-zero value in the while loop indicates always a true condition, whereas zero indicates always a false condition.

The While Loop Using Else Statement

Python enables the use of else statement with the while loop. The else block is executed when the condition given in the while statement becomes false.

Program 8: To demonstrate the use of while loop with else statement.



```
File Edit Format Run Options Window Help
#Program to demonstrate the use of while loop with else statement.

i = 1
while (i <= 5):
    print(i)
    i+=1
else:
    print('The while loop ends here')
```

On running the above program, we get the following output:



```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:ccc6ba5, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py
1
2
3
4
5
The while loop ends here
>>>
```

Infinite Loop

If the condition given in a loop never becomes false, then the loop will never terminate and run indefinitely. This type of loop is called an infinite loop.

Example:

```
while (1)
print("Hello")
```

Output: Hello will be printed infinite times.

To come out of the infinite loop, we can either close the program window or press Ctrl + C. This will break the running of the program. In fact, at any time during the execution of a

program in Python, you can press the key combination Ctrl + C to stop running the program.

Jumping Statements

Sometimes, there is a situation when the control of the program needs to be transferred out of the loop body, even if all the values of the iterations of the loop have not been completed. For this purpose, jumping statements are used in Python. Python offers two jumping statements—break and continue, which are used within the loop.

The Break Statement

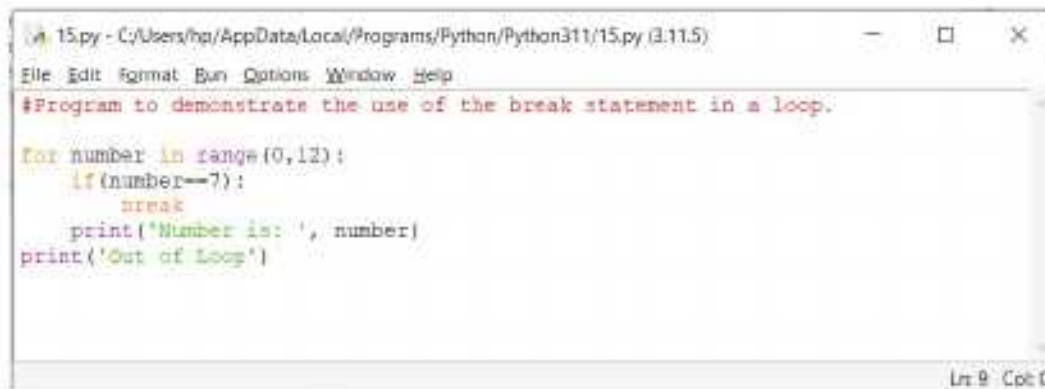
The break is a keyword in Python which is used for bringing the program control out of the loop. The break statement halts the execution of a loop and program flow switches to the statement after the loop. A single break statement will break out of only one loop. The syntax of the break statement is shown below.

Syntax:

```
#loop statement
```

```
break
```

Program 9: To demonstrate the use of the break statement in a loop.



```
15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to demonstrate the use of the break statement in a loop.

for number in range(0,12):
    if(number==7):
        break
    print("Number is: ", number)
print('Out of Loop')
```

On running the above program, we get the following output:



```
IDLE Shell 3.11.5
Python 3.11.5 (tags/v3.11.5:ccc6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py
Number is: 0
Number is: 1
Number is: 2
Number is: 3
Number is: 4
Number is: 5
Number is: 6
Out of Loop
>>>
```


The Continue Statement

The continue statement is used inside loops. When a continue statement is encountered inside a loop, control of the program jumps to the beginning of the loop for next iteration, skipping the execution of rest of the statements of the loop for the current iteration.

Syntax:

#loop statements

continue

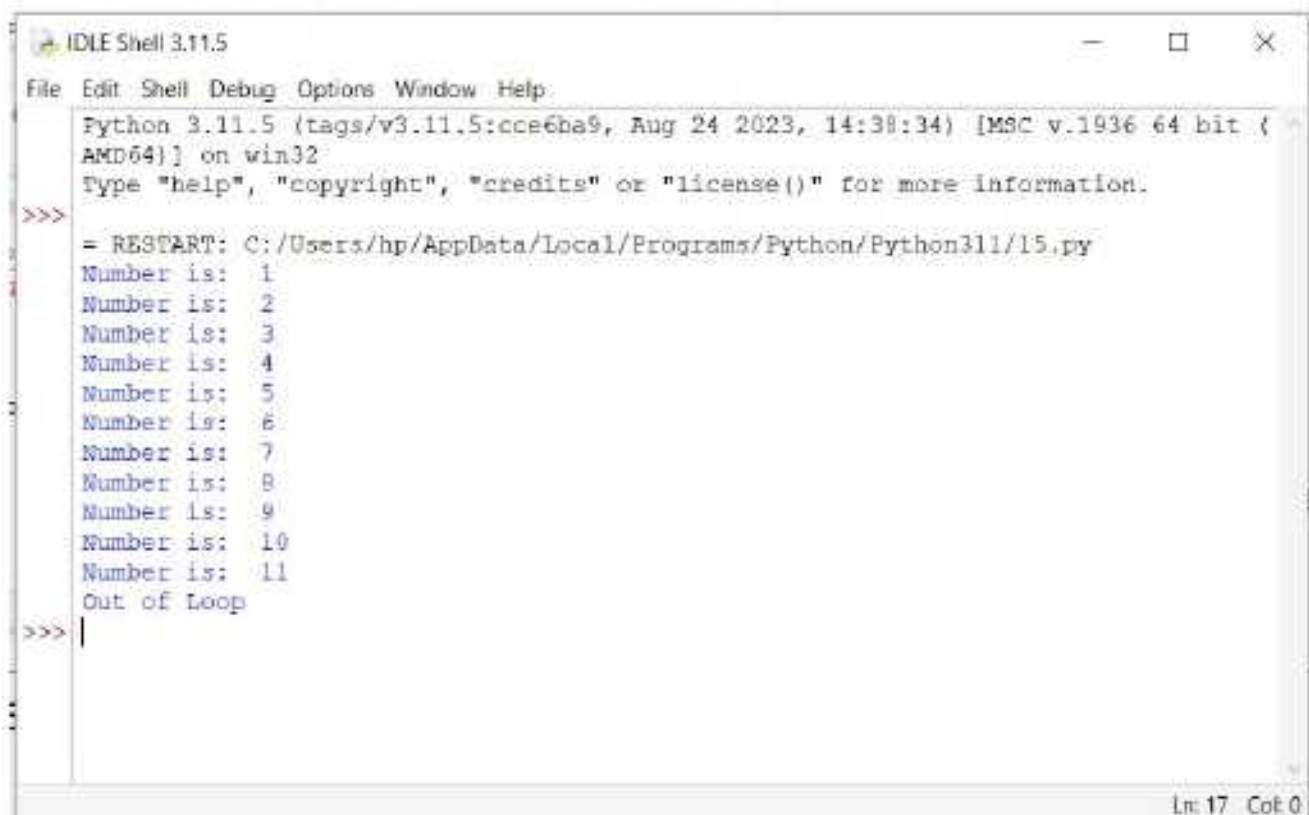
#the code to be skipped

Program 10: To demonstrate the use of the continue statement in a loop.



```
15.py - C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py (3.11.5)
File Edit Format Run Options Window Help
#Program to demonstrate the use of the continue statement in a loop.
for number in range(0,12):
    if(number == 0):
        continue
    print('Number is: ', number)
print('Out of Loop')
```

On running the program, we get the following output:



```
IDLE Shell 3.11.5
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:ccc6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/hp/AppData/Local/Programs/Python/Python311/15.py
Number is: 1
Number is: 2
Number is: 3
Number is: 4
Number is: 5
Number is: 6
Number is: 7
Number is: 8
Number is: 9
Number is: 10
Number is: 11
Out of Loop
>>>
```



COMPUTER ETIQUETTES

Keep your hands and shoulders relaxed while working on a computer. Also, one should not work for too long on a computer.



Let's Implement.

A

Tick (✓) the correct answer.

- The _____ statement allows to repeat a task for a fixed number of times.
a. for ☐ b. while ☐
c. do ☐ d. None of these ☐
- The _____ statement terminates the execution of the loop.
a. break ☐ b. continue ☐
c. Both a & b ☐ d. None of these ☐
- Python offers _____ jumping statements.
a. two ☐ b. three ☐
c. four ☐ d. five ☐
- To come out of the infinite loop, we can either close the program window or press _____.
a. Alt + C ☐ b. Ctrl + C ☐
c. Alt + Ctrl + C ☐ d. Shift + C ☐
- The continue statement is used _____ loops.
a. inside ☐ b. outside ☐
c. Both a & b ☐ d. None of these ☐
- The _____ operator is used to check if a given value exists in the sequence or not.
a. in ☐ b. on ☐
c. + ☐ d. - ☐

B

Fill in the blanks.

- The _____ operator evaluates to be True if it finds a value in the specified sequence, else it returns False.
- The _____ statement executes a set of statements repeatedly, until the logical expression remains true.
- Any _____ value in the while loop indicates an always true condition whereas zero indicates _____ condition.

4. The _____ loop never ends.
5. The _____ and _____ are the jump statements in Python.
6. A _____ is a succession of values bound together by a single name.

C

Write True or False.

1. If the condition given in a loop never becomes false, then the loop will never terminate and run indefinitely.
2. We can use do-while loop in Python.
3. The continue statement breaks the loops one by one.
4. To come out of the infinite loop, we can either close the program window or press Ctrl + C.
5. A sequence is a succession of values bound together by a single name.
6. The while statement is the looping statement.

D

Write the output of the following programs.

1.

```
i=0
while i<=10:
    print(i)
    i=i+1
```
2.

```
x=1
sum=0
while (x<=15):
    sum=sum+x
    x=x+1
    print (sum)
```
3.

```
colours= ["red", "yellow", "pink"]
for x in colours:
    print (x)
```
4.

```
i=10
while i<20:
    print (i*j)
    i+=1
```

E

Answer the following questions.

1. What is looping?

2. Write the syntax to the for loop.
3. What is the use of jump statement?
4. Draw the flowchart of the for loop.
5. Define the use of while statement with the help of example and flowchart.
6. Distinguish between continue and break statements.



► Rewrite the following programs using the while loop.

1.

```
a = int(input("Enter a number="))
f=1
for i in range(1, a+1):
    f=f*i
print("The factorial of the number is: ",f)
```

A large, empty rectangular box with a thin blue border, intended for the student to write the rewritten program using a while loop.

2.

```
n=int(input("Enter the level upto which the series to be printed"))
a=0
b=1
print(a)
print(b)
for i in range(1, n+1):
    c=a+b
    print(c)
    a=b
    b=c
```



3. `n=150`
`s=0`
`for d in range(1, n+1)`
`s=sum+s`
`print("The sum is:",s)`
`print("Game Over")`



Practical Session



Experiential Learning

A Write the Python programs for the following:

1. To print the table of 5 using the while loop
2. To print odd numbers from 1 to 50
3. To print sum of first five natural numbers
4. To print even numbers from 1 to 60
5. To print numbers divisible by 12 from 1 to 200

B Find the output of the following programs:

1.

```
Python 3.11.5
File Edit Format Run Options Window Help
#Program to test whether a pattern using for loop.
for i in range(1, 10):
    print('2' * i)
```

2.

```

File Edit Format Run Options Window Help
#Program to add first five natural numbers.
sum = 0
a = 1
while(a<=5):
    sum+=a
    a+=1
print('The sum of first five natural number is', sum)
Ln: 8 Col: 0

```

3.

```

File Edit Format Run Options Window Help
#Program to print the star pattern.
for i in range(10, 0, -1):
    print(" "*i)
Ln: 4 Col: 0

```

Competency Based Questions

1. Sarthak wants to repeat a block of statements for a given number of times, until the control condition is false. Which type of construct can she use for this? Write its syntax.
2. Samartha wants to write a statement which will cause the program to skip the rest of the statement of the current block and to move to the next iteration of the loop. But he is unable to do it. Suggest the statement he should use for this.

RACKING YOUR BRAIN

Life Skills and Values

A few of your friends have been editing and sharing codes of other classmates without their knowledge. What should you do in this situation?

Just Have a Discussion

Communication

Discuss the different types of loops in the class in detail.



TEACHER'S NOTE

Discuss about all the iterative statements of Python with the students.



AI INTELLIGENCE



Outlines

- Intelligence
- Chatbot
- AI Approach

Artificial intelligence refers to the software technologies that make a robot or computer act and think like humans. It is the simulation of human intelligence processes through machines, especially computer systems. Artificial intelligence imparts cognitive ability to machines, meaning it can make machines perform tasks in ways that are intelligent.

Intelligence

Intelligence refers to the ability to think, to learn from previous experience, to solve complex problems and to adapt new things and environment.

Types of Intelligence

Humans possess cognitive abilities to learn, form concepts, understand, apply logic and reason, including the capability of recognising patterns, comprehend ideas, solve problems, make decisions, retain the information and use the language to communicate.

According to the Harvard psychologist Howard Gardner's theories, humans do not just have an intellectual capacity but also many kinds of intelligences. A person can be strong in a specific area, such as logical intelligence or musical intelligence, but simultaneously can also possess a range of abilities.

There are multiple types of intelligences according to Howard Gardner:

Naturalistic Intelligence

These people possess the following qualities:

- Nature lover
- Love exploring the environment and learning about other species
- Enjoy camping, gardening, hiking and exploring outdoors

Musical Intelligence

These people possess the following qualities:

- Love singing and playing musical instruments
- Good at thinking in patterns, rhythms and sounds
- Recognise musical patterns and tones easily
- Good understanding of musical rhythm

Logical-Mathematical Intelligence

These people possess the following qualities:

- Good at problem-solving
- Love conducting scientific experiments
- Love thinking about abstract ideas
- Solve complex computations easily
- Good at reasoning and recognising patterns

Existential Intelligence

These people possess the following qualities:

- Look at things from different point of view.
- Make connections between what is being learned and the world outside.
- Have a deep and comprehensive understanding not just of the universe, but of the self.

Interpersonal Intelligence

These people possess the following qualities:

- Good at communicating ideas and thoughts
- Good at non-verbal communication
- Have a different perspective on everything
- Good at resolving conflict in groups
- Good at maintaining positive with others

Bodily-Kinesthetic Intelligence

These people possess the following qualities:

- Good at dancing and sports
- Excellent hand-eye and physical coordination
- Love creating things with their hands
- Flexible in different body movements and performing actions
- Tend to remember by doing, rather than hearing and seeing

Linguistic Intelligence

These people possess the following qualities:

- Good at using words
- Good debaters
- Can remember written and spoken information
- Explain things well
- Love reading and writing
- Use humour while telling stories

Intrapersonal Intelligence

These people possess the following qualities:

- Can recognise their strengths and weaknesses
- Aware of their own emotional state and feelings and know how they can motivate themselves.
- Enjoy analysing theories and ideas

Spatial Intelligence

These people possess the following qualities:

- Love reading and writing
- Visualise things easily
- Good at putting puzzles together
- Love drawing and painting
- Recognise patterns easily

AI Approach

The aim of AI is to simulate the human attributes of logical intelligence, reasoning and knowledge acquirement in machines. We have been experiencing how this works in the form of a virtual assistant like Siri, Alexa, and Chatbots.

Generally, AI models have two kinds of approaches:

- Rule based approach
- Learning based approach



Rule Based Approach

A rule-based system uses rules as knowledge representation. These rules are coded in the system in the form of if-then-else statements which help the computer in taking the decisions. The idea behind a rule-based system is to use the knowledge of a human expert in a specialised domain and embody it within the computer system. For example, a system

is developed to identify emails by finding keywords. If this system finds an email containing the word "bill" or "invoice", it forwards the email to the accounts department.

Learning Based Approach

A computer system that achieves AI through a machine learning technique is called a learning-based approach. This approach is often referred to as black boxes as it is not entirely clear how these systems take the decisions. This is a major problem for applications when it comes to making a decision for investment, medical or military reasons as this approach remains unreliable.

Machine Learning is a way of achieving AI. It is a branch of AI and is based on the idea that machines can be built to process data and learn on their own. Machine learning is a subset of AI.

Chatbot

A **chatbot**, also known as abot, is a computer program that allows humans interact with computing devices through voice, text, gesture and touch.

The chatbot works like your virtual assistant, answering your questions based on your inputs and performing the given tasks. For example, you can use the chatbot of a pizza company to order a pizza for you.



The first-ever chatbot was developed by Joseph Weizenbaum, a German American computer scientist at MIT's artificial intelligence lab between 1964 and 1966.

Famous Chatbots

With advanced AI solutions, chatbots can talk like we do. In this section, you will learn about some AI chatbots that can talk like humans. These AI chatbots have won the famous **Loebner** award.



The **Loebner** award is an honour in the field of AI based on the **Turing Test**. The **Turing Test** checks the capability of a chatbot application to think like a human.

Mitsuku



Mitsuku, also known as **Kuki**, is one of the best AI chatbots that acts like your virtual friend. With Mitsuku, you can:

- Chat
- Hear stories or poems or jokes

- Play games
- Surf webpages and pictures



Replika is a friendly chatbot with whom you can connect emotionally and talk like a private friend. You can use the Replika chatbot whenever you feel bored or want to have some fun with it.



Andy chatbot is designed to teach English conversation ally, and can be your best friend if you want to learn English grammar and new words daily. Though it is a paid chatbot, it is considered one of the most effective chatbots to develop English speaking skills.



Using this chatbot, you can:

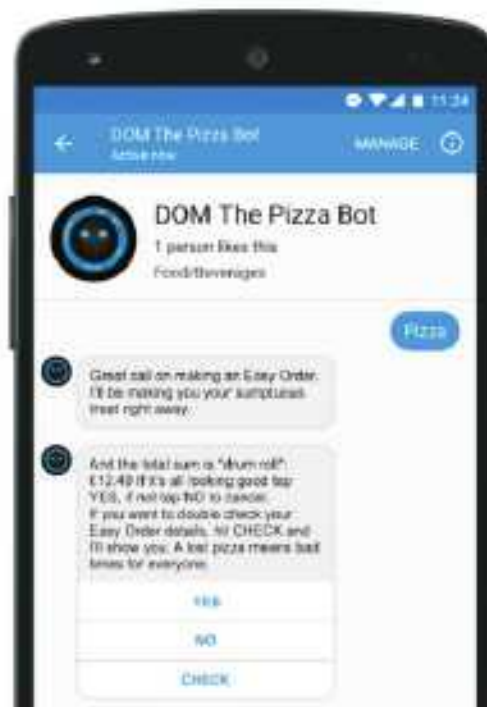
- study the English language in a conversational style
- improve your listening skills with the help of audio messages
- learn new words and grammar rules with the help of fun games

Andy explains the correct answer with a detailed description when you make a mistake in any fun game.

Domino's Pizza Bot



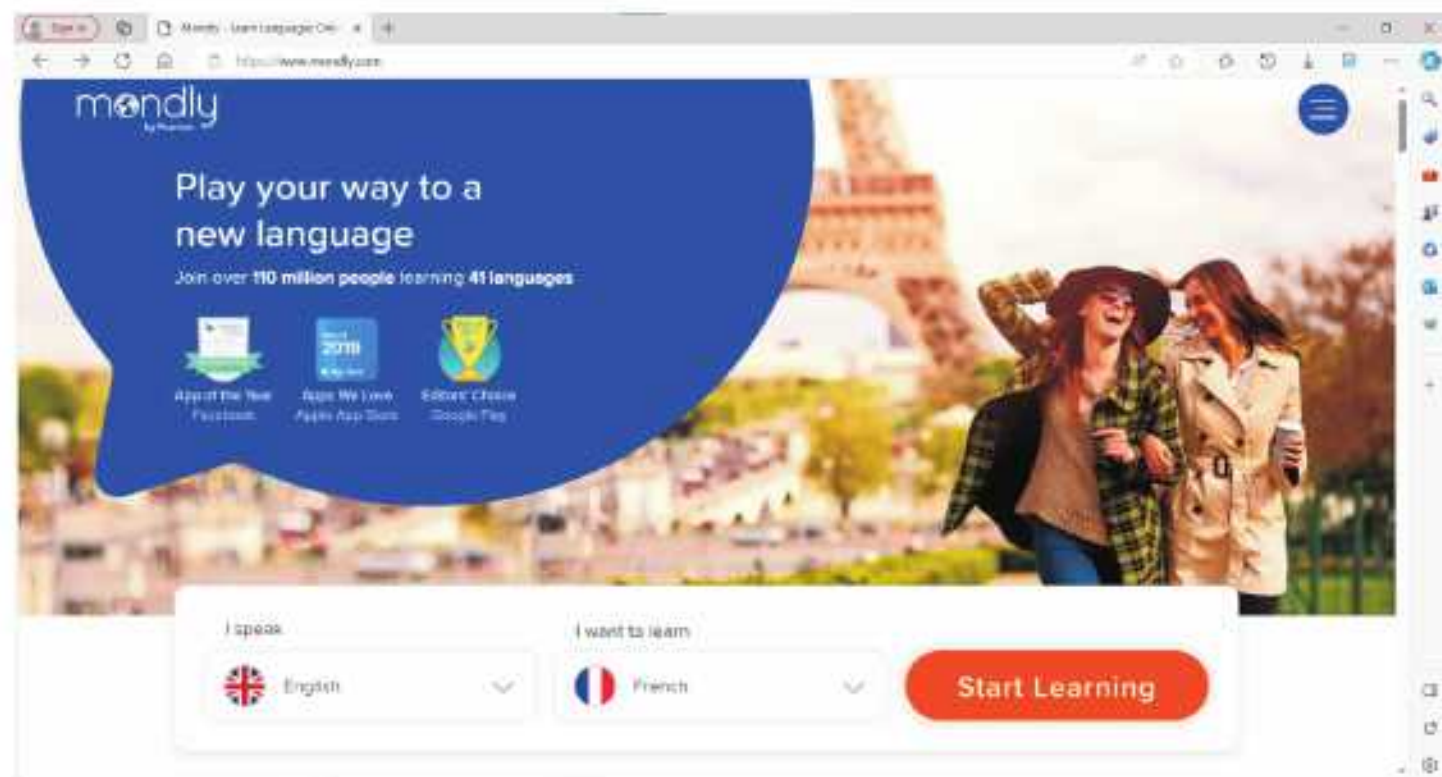
Nowadays, customers can order their food without making a single phonecall. All thanks to customer service chatbots. Domino's has launched a new chatbot called Dom on Facebook Messenger. Dom chatbot lets you order your favourite pizza with a single word.



Mondly: Learning Bot



Mondly is a language learning platform that focuses on interactive online lessons. The chatbot encourages the users to practice English, Spanish, German or French.



COMPUTER ETIQUETTES

Be SMART and SAFE!

- Keep safe by not revealing personal information while using the Internet. It includes e-mail ID, house address, photos, phone number and password.
- Meet online friends only in the presence of elders.
- Accept e-mails and messages only from people you know. Be aware of viruses and spam messages.
- All the information on the Internet may not be correct. Check the reliability of the information with other sources— books, websites or someone who knows.
- Tell your teacher or an adult family member if you or someone you know is uncomfortable about an online communication.



Let's Implement.

A

Tick (✓) the correct answer.

- Which quality do people with visual-spatial intelligence possess?
a. Explaining things well ☐ b. Good at problem-solving ☐
c. Recognising patterns easily ☐ d. Both a & b ☐
- Which AI model approach uses a series of if-else statements that help a computer take the decisions?
a. Learning-based approach ☐ b. Rule-based approach ☐
c. Machine learning ☐ d. None of these ☐
- Which of the following is a way of achieving AI?
a. Machine learning ☐ b. Decision Tree ☐
c. Both a & b ☐ d. None of these ☐
- Which of the following refers to software technologies that make a robot or computer act and think like humans?
a. Real intelligence ☐ b. Artificial intelligence ☐
c. Both a & b ☐ d. None of these ☐
- Which of the following is a language learning platform that focuses on interactive online lessons?
a. Mondly ☐ b. Andy ☐
c. Replika ☐ d. Domino's Pizza Bot ☐

B

Fill in the blanks.

- Siri, Alexa and Chatbots are examples of _____.
- The aim of AI is to simulate the human attributes of logical intelligence, _____ and knowledge acquirement in machines.
- A rule-based system uses rules as _____ representation.
- There are _____ types of intelligences according to Howard Gardner.
- The _____ chatbot encourages the users to practice English, Spanish, German or French.

C

Write True or False.

- The aim of AI is to simulate the human attributes of logical intelligence, reasoning and knowledge acquirement in machines.

2. According to the Harvard psychologist Howard Gardner's theories, humans have an intellectual capacity only.
3. A chatbot can act like your virtual friend.
4. Machine Learning is a way of achieving AI.
5. Machine learning is a subset of AI.

D Answer the following questions.

1. What do you understand by bodily-kinesthetics intelligence?
2. Define rule-based approach.
3. What do you mean by intelligence.
4. Explain the learning based approach in detail.
5. What is a chatbot?



Identify the following chatbots and write their names.



1. _____
2. _____
3. _____
4. _____

Competency Based Questions

1. Sujata loves exploring the environment and also enjoys activities like camping, gardening and hiking. Which type of intelligence can she be associated with?
2. Rajat wants to order his favourite pizza from Domino's. Can you suggest him a chatbot.

RACKING YOUR BRAIN

Life Skills and Values

One of your friends is not able to use Alexa. Then, what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different types of intelligences in the class in detail.



Teacher's Note

Explain the various types of intelligences in detail.



QUICK REVIEW 2

Tick (✓) the correct option.

1. It allow web authors to arrange data, like text, images, links, other tables, etc. into rows and columns of cells.
a. HTML links ☐ b. HTML lists ☐
c. HTML tabs ☐ d. HTML tables ☐
2. _____ controls the sequencing and timing of the graphics and the other elements of animation.
a. Keyframe ☐ b. Frame ☐ c. Layer ☐ d. Timeline ☐
3. This tag will list items using plain bullets.
a. ☐ b. <dl> ☐ c. ☐ d. <dd> ☐
4. To come out of the infinite loop, we can either close the program window or press _____.
a. Alt + C ☐ b. Ctrl + C ☐ c. Alt + Ctrl + C ☐ d. Shift + C ☐
5. Which type of function is append()?
a. Built-in ☐ b. user-defined ☐
c. Not a valid function ☐ d. None of these ☐
6. Which AI model approach uses a series of if-else statements that help a computer take the decisions?
a. Learning-based approach ☐ b. Rule-based approach ☐
c. Machine learning ☐ d. None of these ☐
7. This attribute is used when you want to merge two or more columns into a single column.
a. Caption ☐ b. Colspan ☐
c. Rowspan ☐ d. Hyperlink ☐
8. To hide or show layers, click on the _____ icon.
a. Layers ☐ b. Export ☐ c. Import ☐ d. Eye ☐
9. This Tag is used to indicate the main body of the table.
a. <ttext> ☐ b. <thead> ☐ c. <tbody> ☐ d. <tfoot> ☐
10. Python offers _____ jumping statements.
a. two ☐ b. three ☐
c. four ☐ d. five ☐



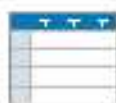
A Fill in the blanks.

1. A network spanned inside a building and operated under single administrative system, is termed as _____.
2. In _____ view of the table, you can enter the field names, their data types and description.
3. Queries, forms and reports are _____ of Access.
4. _____ virus was first detected on ARPANET.
5. In _____, all devices share single communication line or cable.

B Write True or False.

1. The LAN generally expands throughout a city such as cable TV network.
2. Caption property is used to give an alternate name to a field.
3. The Form option is present in the Forms group on the Create tab.
4. All virus programs cause harm to data or programs.
5. Topologies may define both physical and logical aspects of the network.

C Label the pictures given below.



D Answer the following questions.

1. What do you understand by the term 'table'? Name some important components of a table.
2. Differentiate between Form and Report.
3. Define LAN, MAN and WAN.
4. What is a computer virus? Name its types.
5. What is the use of the Print Preview and Design View of a report?



A Fill in the blanks.

1. The _____ tag is used to create table rows and _____ tag is used to create data cells.
2. A _____ is a special frame that enables you define some changes to an object's properties.
3. The _____ loop never ends.
4. HTML and XHTML support a list style which is called _____.
5. The aim of AI is to simulate the human attributes of logical intelligence, _____ and knowledge acquirement in machines.

B Write True or False.

1. The HTML tables are created using the <tt> tag.
2. Krita supports working with multiple layers.
3. By default, image will have a border around it.
4. To come out of the infinite loop, we can either close the program window or press Ctrl + C.
5. Border attribute of <table> tag is used to put a border across all the cells.

C Identify the following chatbots and write their names.



1. _____
2. _____
3. _____
4. _____

D Answer the following questions.

1. Differentiate between Cellpadding and Cellspacing attributes.
2. What are layers used for?
3. Define Ordered Lists with example.
4. Write the syntax to the for loop.
5. What is a chatbot?